National Comprehensive Cancer Network®

NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®)

Multiple Myeloma

NCCN Evidence Blocks™

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NCCN recognizes the importance of clinical trials and encourages participation when applicable and available. Trials should be designed to maximize inclusiveness and broad representative enrollment.

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NCCN Multiple Myeloma Panel Members
NCCN Evidence Blocks Definitions (EB-DEF)

Initial Diagnostic Workup and Clinical Findings (MYEL-1)
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Smoldering Myeloma (Asymptomatic): Primary Treatment and Follow-up/Surveillance (MYEL-3)
Multiple Myeloma (Symptomatic): Primary Treatment and Follow-up/Surveillance (MYEL-4)
Multiple Myeloma (Symptomatic): Response After Primary Therapy and Follow-up Surveillance (MYEL-5)
Multiple Myeloma (Symptomatic): Treatment for Relapse or Progressive Disease (MYEL-6)
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Management of Venous Thromboembolism (VTE) in Multiple Myeloma (MYEL-K)
Management of Renal Disease in Multiple Myeloma (MYEL-L)

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- Monoclonal Gammopathy of Renal Significance (MGRS-1)
- Monoclonal Immunoglobulin Deposition Disease (MIDD-1)
- Monoclonal Gammopathy of Neurological Significance (MGNS-1)

POEMS (Polyneuropathy, Organomegaly, Endocrinopathy, Monoclonal Protein, Skin Changes) (POEMS-1)
Evidence Blocks
Abbreviations (ABBR-1)

Find an NCCN Member Institution: https://www.nccn.org/home/member-institutions

NCCN Categories of Evidence and Consensus: All recommendations are category 2A unless otherwise indicated.

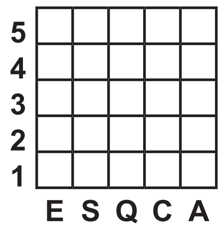
See NCCN Categories of Evidence and Consensus.

NCCN Categories of Preference: All recommendations are considered appropriate.

See NCCN Categories of Preference.

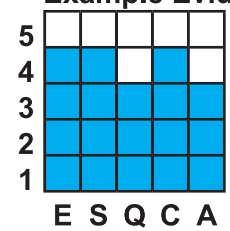
NCCN Guidelines for Patients® available at www.nccn.org/patients

NCCN EVIDENCE BLOCKS CATEGORIES AND DEFINITIONS



E = Efficacy of Regimen/Agent
S = Safety of Regimen/Agent
Q = Quality of Evidence
C = Consistency of Evidence
A = Affordability of Regimen/Agent

Example Evidence Block



E = 4
S = 4
Q = 3
C = 4
A = 3

Efficacy of Regimen/Agent

5	**Highly effective:** Cure likely and often provides long-term survival advantage
4	**Very effective:** Cure unlikely but sometimes provides long-term survival advantage
3	**Moderately effective:** Modest impact on survival, but often provides control of disease
2	**Minimally effective:** No, or unknown impact on survival, but sometimes provides control of disease
1	**Palliative:** Provides symptomatic benefit only

Safety of Regimen/Agent

5	**Usually no meaningful toxicity:** Uncommon or minimal toxicities; no interference with activities of daily living (ADLs)
4	**Occasionally toxic:** Rare significant toxicities or low-grade toxicities only; little interference with ADLs
3	**Mildly toxic:** Mild toxicity that interferes with ADLs
2	**Moderately toxic:** Significant toxicities often occur but life threatening/fatal toxicity is uncommon; interference with ADLs is frequent
1	**Highly toxic:** Significant toxicities or life threatening/fatal toxicity occurs often; interference with ADLs is usual and severe

Note: For significant chronic or long-term toxicities, score decreased by 1

Quality of Evidence

5	**High quality:** Multiple well-designed randomized trials and/or meta-analyses
4	**Good quality:** One or more well-designed randomized trials
3	**Average quality:** Low quality randomized trial(s) or well-designed non-randomized trial(s)
2	**Low quality:** Case reports or extensive clinical experience
1	**Poor quality:** Little or no evidence

Consistency of Evidence

5	**Highly consistent:** Multiple trials with similar outcomes
4	**Mainly consistent:** Multiple trials with some variability in outcome
3	**May be consistent:** Few trials or only trials with few patients, whether randomized or not, with some variability in outcome
2	**Inconsistent:** Meaningful differences in direction of outcome between quality trials
1	**Anecdotal evidence only:** Evidence in humans based upon anecdotal experience

Affordability of Regimen/Agent (includes drug cost, supportive care, infusions, toxicity monitoring, management of toxicity)

5	**Very inexpensive**
4	**Inexpensive**
3	**Moderately expensive**
2	**Expensive**
1	**Very expensive**

INITIAL DIAGNOSTIC WORKUP[a]

- History and physical (H&P) exam
- CBC, differential, and platelet count
- Peripheral blood smear
- Serum BUN/creatinine, electrolytes, liver function tests, albumin,[b] calcium, serum uric acid, serum lactacte dehydrogenase (LDH),[b] and beta-2 microglobulin[b]
- Creatinine clearance (calculated or measured directly)[c]
- Serum quantitative immunoglobulins, serum protein electrophoresis (SPEP), and serum immunofixation electrophoresis (SIFE)
- 24-h urine for total protein, urine protein electrophoresis (UPEP), and urine immunofixation electrophoresis (UIFE)
- Serum free light chain (FLC) assay
- Whole-body FDG-PET/CT (preferred) or low-dose CT[d]
- Unilateral bone marrow aspirate and biopsy, including immunohistochemistry (IHC) and/or multi-parameter flow cytometry
- Plasma cell fluorescence in situ hybridization (FISH)[b] panel on bone marrow[e] [del(13), del (17p13), t(4;14), t(11;14), t(14;16), t(14:20), 1q21 gain/1q21 amplification,[f] 1p deletion]
- Next-generation sequencing (NGS) to assess for TP53 mutation
- N-terminal prohormone B-type natriuretic peptide (NT-proBNP)/BNP[g]

Useful in Certain Circumstances

- If whole-body low-dose CT or FDG-PET/CT is negative, consider whole-body MRI without contrast to discern smoldering myeloma from multiple myeloma (MM)[d]
- Tissue biopsy to confirm suspected plasmacytoma
- Echocardiogram
- Evaluation for light chain amyloidosis, if appropriate (See NCCN Guidelines for Systemic Light Chain Amyloidosis)
- Obtain baseline clonotype identification at diagnosis or store an aspirate sample for future clonotype identification to enable minimal residual disease (MRD) testing by NGS
- Serum viscosity
- Hepatitis B and hepatitis C testing and human immunodeficiency virus (HIV) screening as required
- Assess for circulating plasma cells as clinically indicated
- Renal biopsy if albuminuria or abnormal renal function[h]

CLINICAL FINDINGS

→ Solitary plasmacytoma[i] (MYEL-2)

→ Smoldering myeloma (asymptomatic)[i] (MYEL-3)

→ Multiple myeloma (symptomatic)[i] (MYEL-4)

→ Monoclonal gammopathies of clinical significance
- → For Renal Significance, see MGRS-1
- For Monoclonal Immunoglobulin Deposition Disease (MIDD), see MIDD-1
- → For Neurological Significance, see MGNS-1

→ Polyneuropathy, Organomegaly, Endocrinopathy, Monoclonal Protein, Skin Changes (POEMS-1)

[a] Frailty assessment should be considered in older adults. See NCCN Guidelines for Older Adult Oncology.
[b] These tests are essential for staging. See Disease Staging and Risk Stratification for Multiple Myeloma (MYEL-B).
[c] Management of Renal Disease in Multiple Myeloma (MYEL-L).
[d] Principles of Imaging (MYEL-C).
[e] CD138-positive selected sample is strongly recommended for optimized yield.
[f] 1q21 amplification is defined as ≥4 copies detected by FISH, and a gain is defined as 3 copies of 1q21.
[g] If NT-proBNP is not available, BNP can be performed.
[h] Dimopoulos MA, et al. Lancet Oncol 2023;24:e293-e311.
[i] Definitions of Myeloma and Related Plasma-Cell Disorders (MYEL-A).

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CLINICAL FINDINGS	PRIMARY TREATMENT	FOLLOW-UP/SURVEILLANCE			
Solitary plasmacytoma or Solitary plasmacytoma with minimal marrow involvement[j,k] →	RT[l] ± surgery[m,n] or Consider clinical trial →	Follow-up interval, every 3–6 months[o]: • CBC, differential, and platelet count • Serum chemistry for creatinine and corrected calcium • Serum quantitative immunoglobulins, SPEP, with SIFE • Serum FLC assay • 24-hour urine for total protein and UPEP with UIFE, as needed • Bone marrow aspirate and biopsy as indicated • All plasmacytomas should be imaged yearly, preferably with the same technique used at diagnosis, for at least 5 years[d,p] • See NCCN Guidelines for Survivorship →	Progressive Disease[q] or Response followed by progression[q] →	Restage with myeloma workup →	Multiple Myeloma (symptomatic) (MYEL-4)

[d] Principles of Imaging (MYEL-C).
[j] All criteria must be present for the diagnosis. For diagnostic criteria, see Definitions Myeloma and Related Plasma-Cell Disorders (MYEL-A).
[k] Solitary plasmacytoma with 10% or more bone marrow plasma cells (BMPCs) is regarded as active (symptomatic) MM and systemic therapy should be considered. Consider FISH testing on plasmacytoma biopsy if clonal cells are absent or inadequate in the marrow.
[l] Principles of Radiation Therapy (MYEL-D).
[m] Consider surgery if structurally unstable or if there is neurologic compromise due to mass effect.
[n] Systemic therapy may be indicated in patients with high risk of progression based on the clinical context.
[o] Reassess after at least 3 months following radiation as the assessment of response with imaging may not be accurate if the scans are performed sooner. Patients with soft tissue and head/neck plasmacytoma could be followed less frequently as clinically indicated after initial 3-month follow-up.
[p] Whole-body FDG-PET/CT is the first choice for initial and continued evaluation of solitary extraosseous plasmacytoma. Whole-body MRI (or PET/CT if MRI is not available) is the first choice for initial evaluation of solitary osseous plasmacytoma (If whole-body MRI is not available, then consider MRI of the spine and pelvis, whole-body FDG-PET/CT, or low-dose whole-body CT).
[q] Response Criteria for Multiple Myeloma (MYEL-E).

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CLINICAL FINDINGS		PRIMARY TREATMENT	FOLLOW-UP/SURVEILLANCE		
Smoldering myeloma (asymptomatic)[i]	Low risk[r] →	Clinical trial[t] or Observe at 3- to 6-month intervals[s] (category 1)	→ • Every 3–6 months: ‣ CBC, differential, platelet count ‣ Creatinine, corrected calcium ‣ Serum quantitative immunoglobulins, SPEP, SIFE ‣ Serum FLC assay ‣ 24-hour urine for total protein, UPEP, and UIFE as clinically indicated • Bone marrow aspirate and biopsy with FISH, NGS, or multi-parameter flow cytometry as clinically indicated • Whole-body MRI without contrast, low-dose CT, FDG-PET/CT annually or as needed, ideally with the same technique used at diagnosis[d] • See NCCN Guidelines for Survivorship	→ Progression to symptomatic myeloma	→ Multiple Myeloma (symptomatic) (MYEL-4)
	High risk[r,s] →	Clinical trial[t] (preferred) or Observe at 3-month intervals as clinically indicated[s] or Select patients: • Daratumumab (category 1) • Lenalidomide (category 2B)[u]	→ (same follow-up/surveillance)		

[d] Principles of Imaging (MYEL-C).
[i] Definitions of Myeloma and Related Plasma-Cell Disorders (MYEL-A).
[r] For risk criteria, refer to Lakshman A, et al. Blood Cancer J 2018;8:59 and Mateos MV, et al. Blood Cancer J 2020;10:102.
[s] Patients with rising parameters are considered high risk and should be closely monitored.
[t] The NCCN Panel strongly recommends enrolling eligible patients with smoldering myeloma in clinical trials.
[u] Consider collection of hematopoietic stem cells for future use.

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CLINICAL FINDINGS	PRIMARY TREATMENT	FOLLOW-UP/SURVEILLANCE		
MM (symptomatic)[i,v] → (For MM with CNS disease, see CNSM-1)	Initiate myeloma therapy[w,x] and bone-targeting treatment[y] + Supportive care treatment as indicated for symptom management[c,y] (also see NCCN Guidelines for Supportive Care) Assess for candidacy for transplant after starting therapy and reassess for transplant as performance status improves[z,aa] • Refer to hematopoietic cell transplant (HCT) center • Harvest hematopoietic stem cells (consider for 2 transplants if appropriate) →	• Laboratory assessments appropriate for monitoring treatment toxicities may include: complete blood count (CBC) with differential and metabolic panel • Serum quantitative immunoglobulins, SPEP, and SIFE[bb] • 24-hour urine for total protein, UPEP, and UIFE[bb] as clinically indicated • Serum FLC assay • Whole-body MRI without contrast, low-dose CT, FDG-PET/CT annually or as clinically indicated, ideally with the same technique used at diagnosis[d] • Bone marrow aspirate and biopsy at relapse with FISH as clinically indicated • Consider MRD testing as indicated for prognostication after shared decision with patient • See NCCN Guidelines for Survivorship	Response[q] → Progression[q] →	Response after primary therapy (MYEL-5) Additional Treatment (MYEL-6)

[c] Management of Renal Disease in Multiple Myeloma (MYEL-L).
[d] Principles of Imaging (MYEL-C).
[i] Definitions Myeloma and Related Plasma-Cell Disorders (MYEL-A).
[q] Response Criteria for Multiple Myeloma (MYEL-E).
[v] Disease Staging and Risk Stratification for Multiple Myeloma (MYEL-B).
[w] Myeloma Therapy (MYEL-G).
[x] See MYEL-F for considerations for myeloma therapy and for specific populations (eg, Black/African American individuals).
[y] Supportive Care Treatment for Multiple Myeloma (MYEL-I).
[z] Autologous transplantation: Category 1 evidence supports proceeding directly after primary therapy to high-dose therapy and HCT. Collecting stem cells and delaying HCT is also an option. See Discussion.
[aa] Renal dysfunction and advanced age are not contraindications to transplant.
[bb] Needed only if protein electrophoresis is negative during follow-up.

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MULTIPLE MYELOMA (SYMPTOMATIC)

FOLLOW-UP/SURVEILLANCE

Response after primary therapy[q,cc] →

- **Autologous**[z,aa,cc,dd,ee] **HCT (category 1)** → **Response or stable disease**[q] → **Maintenance therapy (category 1)**[w,x,dd,ff] → **Relapse or progression**[q] **(MYEL-6)**
- **Continuous myeloma therapy or maintenance therapy**[w,x,dd,ff] → **Relapse or progression**[q] **(MYEL-6)**
- **Useful in Certain Circumstances for patients with high risk of progression/relapse:** **Tandem autologous or allogeneic HCT**[dd,ee] → **Response or stable disease**[q] → **Maintenance therapy (category 1)**[w,x,dd,ff] → **Relapse or progression**[q] **(MYEL-6)**

[q] Response Criteria for Multiple Myeloma (MYEL-E).
[w] Myeloma Therapy (MYEL-G).
[x] See MYEL-F for considerations for myeloma therapy and for specific populations (eg, Black/African American individuals).
[z] Autologous transplantation: Category 1 evidence supports proceeding directly after primary therapy to high-dose therapy and HCT. Collecting stem cells and delaying HCT is also an option. See Discussion.
[aa] Renal dysfunction and advanced age are not contraindications to transplant.
[cc] Patients with stable disease can be considered for autologous HCT.
[dd] Follow up with the tests listed on MYEL-4 under Follow-up/Surveillance.
[ee] Whole-body FDG-PET/CT is recommended around day 100 after autologous HCT.
[ff] The length of therapy should be balanced with toxicity and depth of response and disease status.

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MULTIPLE MYELOMA (SYMPTOMATIC)

TREATMENT

Relapse[q]

or

Progressive disease[q]

→

- Discuss the patient's preferences and goals of care
- Consider referral to:
  - Chimeric antigen receptor (CAR) T-cell therapy specialist
  - Palliative care specialist and supportive care[y] for symptom management (See NCCN Guidelines for Palliative Care and NCCN Guidelines for Supportive Care)

→

Clinical trial, if eligible

or

Therapy for previously treated myeloma[w,x,dd]

or

Autologous HCT[dd,gg]

or

Allogeneic HCT[dd,gg,hh]

→

Refractory disease[q] and lack of treatment options

→

Continue palliative care; consider re-evaluation of goals of care and hospice initiation (See NCCN Guidelines for Supportive Care and NCCN Guidelines for Palliative Care)

(For MM with CNS disease, see CNSM-1)

[q] Response Criteria for Multiple Myeloma (MYEL-E).
[w] Myeloma Therapy (MYEL-G).
[x] See MYEL-F for considerations for myeloma therapy and for specific populations (eg, Black/African American individuals).
[y] Supportive Care Treatment for Multiple Myeloma (MYEL-H).
[dd] Follow up with the tests listed on MYEL-4 under Follow-up/Surveillance.
[gg] Assess for HCT candidacy.
[hh] Donor lymphocyte infusion can be considered in patients with disease relapse after allogeneic HCT.

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CLINICAL PRESENTATION OF CNS INVOLVEMENT[a]	WORKUP[b]	DIAGNOSIS[b]	MANAGEMENT	RESPONSE EVALUATION
Confusion, headache, visual symptoms, weakness, cranial nerve palsies →	<u>Essential</u> • Brain and spine MRI with and without contrast • Lumbar puncture with flow cytometry, cytopathology, cell count, total protein • Workup for infections <u>Useful in Certain Circumstances</u> • Biopsy of affected tissue if above test results are negative and suspicion is high →	<u>Definitive Diagnosis</u> • Clonal plasma cells observed in CSF by immunophenotyping • Tissue diagnosis of plasma cell myeloma in affected CNS tissue <u>Probable Diagnosis</u> • Leptomeningeal enhancement or parenchymal lesions on MRI without definitive clonal plasma cells demonstrated *<u>Note:</u>* Extension of skull or spinal plasmacytoma without meningeal infiltration causing compressive neurologic disease is not considered central nervous system (CNS) disease and should be treated per treatment guidelines for MM (MYEL-G) →	• Multimodality therapy (MYEL-H) and • Consider referral to palliative care specialist and supportive care for symptom management (See NCCN Guidelines for Palliative Care and NCCN Guidelines for Supportive Care) →	• Lumbar puncture (LP) with flow cytometry, cytopathology, cell count and/or • Repeat MRI imaging for CNS evaluation (most often brain and spine)

[a] Hyperammonemic encephalopathy is a rare presentation of myeloma–this is not considered CNS involvement and resolves with systemic plasma cell-directed therapy. In addition, hyperviscosity and hypercalcemia may also cause neurologic symptoms but is not considered CNS involvement. CNS involvement may occur as part of initial myeloma presentation or relapse with or without systemic involvement.
[b] Cerebrospinal fluid (CSF) SPEP is not appropriate for diagnosis of CNS disease in patients with MM.

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DEFINITIONS OF MYELOMA AND RELATED PLASMA-CELL DISORDERS

Disease	Definition
Smoldering Myeloma[a,b] (Asymptomatic)	• Serum monoclonal protein ≥3 g/dL or • Bence-Jones protein ≥500 mg/24 h and/or • Clonal bone marrow plasma cells (BMPCs) 10%–59% and • Absence of myeloma-defining events or amyloidosis
Multiple Myeloma[a,c] (Symptomatic)	• Clonal BMPCs ≥10% or biopsy-proven bony or extramedullary plasmacytoma and any one or more of the following myeloma-defining events: • Myeloma-defining events: ‣ Evidence of end organ damage that can be attributed to the underlying plasma cell proliferative disorder, specifically: ◊ Hypercalcaemia: serum calcium >0·25 mmol/L (>1 mg/dL) higher than the upper limit of normal or >2·75 mmol/L (>11 mg/dL) ◊ Renal insufficiency: creatinine clearance <40 mL per min or serum creatinine >177 µmol/L (>2 mg/dL) ◊ Anaemia: haemoglobin value of >20 g/L below the lower limit of normal, or a haemoglobin value <100 g/L ◊ Bone lesions: one or more osteolytic lesions on skeletal radiography, CT, or PET-CT ‣ Any one or more of the following biomarkers of malignancy: ◊ Clonal bone marrow plasma cell percentage ≥60% ◊ Involved:uninvolved serum free light chain ratio ≥100 ◊ >1 focal lesions on MRI studies
Solitary Plasmacytoma[a]	• Biopsy-proven solitary lesion of bone or soft tissue with evidence of clonal plasma cells • Normal skeletal survey and MRI (or CT) of spine and pelvis (except for the primary solitary lesion) • Absence of myeloma defining events • Normal bone marrow with no evidence of clonal plasma cells
Solitary Plasmacytoma with minimal marrow involvement[a]	• Biopsy-proven solitary lesion of bone or soft tissue with evidence of clonal plasma cells • Normal skeletal survey and MRI (or CT) of spine and pelvis (except for the primary solitary lesion) • Absence of myeloma defining events • Clonal bone marrow plasma cells <10%
Plasma Cell Leukemia	• Presence of ≥5% of plasma cells in circulation

[a] Adapted with permission from Rajkumar SV, Dimopoulos MA, Palumbo A, et al. International Myeloma Working Group updated criteria for the diagnosis of multiple myeloma. Lancet Oncol 2014;15:e538-e548.
[b] BMPCs >20%, M-protein >2 g/dL, and FLCr >20 are variables used to risk stratify patients at diagnosis. Patients with two or more of these risk factors are considered to have a high risk of progression to MM. Lakshman A, Rajkumar SV, Buadi FK, et al. Risk stratification of smoldering multiple myeloma incorporating revised IMWG diagnostic criteria. Blood Cancer J 2018;8:59.
[c] Other examples of active disease include: repeated infections, amyloidosis, light chain deposition disease, or hyperviscosity.

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DISEASE STAGING AND RISK STRATIFICATION SYSTEMS FOR MULTIPLE MYELOMA

International Staging System (ISS)	Revised-ISS (R-ISS)[1]	R2-ISS[2,a]	IMS-IMWG[3]
ISS Stage I • Serum beta-2 microglobulin <3.5 mg/L • Serum albumin ≥3.5 g/dL	**R-ISS Stage I** • ISS stage I and standard risk chromosomal abnormalities by FISH • Serum LDH ≤ the ULN	**Low-risk: 0 points[b]** • Not ISS stage II or III • Serum LDH ≤ the ULN • del(17p), t(4;14), 1q+: Not detected	**Standard Risk** • All patients not meeting criteria for HR
ISS Stage II • Not ISS stage I or III	**R-ISS Stage II** • Not R-ISS stage I or III	**Low-intermediate risk: 0.5–1 points[b]** • ISS stage II or • Serum LDH > the ULN • or • del(17p) or t(4;14) or 1q+: Detected	
ISS Stage III • Serum beta-2 microglobulin ≥5.5 mg/L	**R-ISS Stage III** • ISS stage III • And either high-risk chromosomal abnormalities [del(17p) or translocation (4;14) or translocation (14;16)] by FISH or Serum LDH > the ULN	**Intermediate-high risk: 1.5–2.5 points[b]** • Any combination of high-risk features that equals a score of 1.5–2.5	**High-Risk IMS-IMWG** • Del(17p) (>20% of plasma cells) and/or TP53 mutation • One of these translocations co-occurring with 1q+ and/or del(1p32): ▸ t(4;14) ▸ t(14;16) ▸ t(14;20) • Monoallelic del(1p32) along with 1q+ or biallelic del(1p32) • High β2M (>5.5 mg/dL) with normal creatinine (<1.2 mg/dL)
		High-risk: 3–5 points[b] • Any combination of high-risk features that equals a score of 3–5	

[1] Palumbo A, Avet-Loiseau H, Oliva S, et al. Revised International Staging System for Multiple Myeloma: A Report from International Myeloma Working Group. J Clin Oncol 2015;33:2863-2869.
[2] D'Agostino M, Cairns DA, Lahuerta JJ, et al. Second revision of the international staging system (R2-ISS) for overall survival in multiple myeloma: A European Myeloma Network (EMN) report within the harmony project. J Clin Oncol 2022;40:3406-3418.
[3] Avet-Loiseau H, Davies FE, Samur MK, et al. International Myeloma Society/ International Myeloma Working Group Consensus Recommendations on the Definition of High-Risk Multiple Myeloma. J Clin Oncol 2025:Jco2401893

[a] R2-ISS is only validated for newly diagnosed Multiple Myeloma.
[b] For R2-ISS classification a numerical value is assigned to each risk factor based on their influence on OS: ISS-III is 1.5 points, ISS-II is 1 point, del(17p) is 1 point, t(4;14) is 1 point, 1q+ is 0.5 points, and serum LDH > the ULN is 1 point.

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DISEASE STAGING AND RISK STRATIFICATION SYSTEMS FOR MULTIPLE MYELOMA

Factors Considered as High Risk for Progression/Relapse	
For Those with Newly Diagnosed MM	**For Those with Relapsed MM**

• R-ISS III (MYEL-B 1 of 2)
• Extramedullary disease
• Circulating plasma cells
• Cytogenetic abnormalities[c]
  ▸ Del(1p32)
  ▸ t(4;14)
  ▸ t(14;16)
  ▸ t(14;20)
  ▸ Del(17p)/monosomy 17/*TP53* mutation
  ▸ 1q21 gain/1q21 amplification[d,4]
  ▸ MYC translocation[5]
• High-risk gene expression profile
• Markers of high proliferation rate
	• Disease relapse within 2 years of initial therapy when transplant and maintenance are used.
	
	• Relapse within 18 months in case of non-transplant–based treatment.
	
	• Acquisition of 1q gain/amplification and/or del(17p)/*TP53* mutation
	
	• Extramedullary disease at relapse and/or circulating plasma cells

[4] Schmidt TM, Fonseca R, Usmani SZ. Chromosome 1q21 abnormalities in multiple myeloma. Blood Canc J 2021;11:83.
[5] Abdullah N, Baughn LB, Rajkumar SV, et al. Implications of MYC Rearrangements in Newly Diagnosed Multiple Myeloma. Clin Cancer Res 2020;26:6581-6588.

[c] Patients presenting with two or more of these cytogenetic abnormalities are considered to have very high risk of disease progression/relapse.
[d] Sole abnormality in 1q21 is not considered a marker for high risk of progression/relapse.

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PRINCIPLES OF IMAGING

Imaging for Initial Diagnostic Workup (for patients suspected of having myeloma/solitary plasmacytoma)

- **Whole-body FDG-PET/CT (preferred) or whole-body low-dose CT is recommended for initial diagnostic workup of patients suspected of having MM or solitary plasmacytoma. Skeletal survey is acceptable in certain circumstances.**
- **If FDG-PET/CT or whole-body low-dose CT is negative, whole-body MRI without contrast may be considered to discern smoldering myeloma from MM.**

Imaging of Solitary Plasmacytoma

- **Whole-body FDG-PET/CT is the first choice for initial and continued evaluation of solitary extraosseous plasmacytoma. For solitary osseous plasmacytoma, whole-body MRI (or FDG-PET/CT if MRI is not available) is the first choice for initial evaluation.**
- **Since the risk of progression of solitary plasmacytoma into MM or relapse is relatively high (14%–38% within the first 3 years of diagnosis), yearly, follow-up with the imaging technique used at first diagnosis for the first 5 years and subsequently only in case of clinical or laboratory signs or symptoms.[1]**

Imaging for Follow-up of Smoldering Myeloma

- **Advanced whole-body imaging (ie, FDG-PET/CT, low-dose CT, MRI without contrast) is recommended annually or as clinically indicated. If imaging findings are the only parameters indicating initiation of treatment and if findings are doubtful, the same imaging technique should be repeated after 3 to 6 months. If only an MRI had been performed, whole-body low-dose CT should be done to exclude lytic lesions.**

Imaging for Follow-up of MM

- **Advanced whole-body imaging (ie, FDG-PET/CT, low-dose CT, MRI without contrast) is recommended as needed. Residual focal lesions detected by either FDG-PET/CT or MRI have been shown to be of adverse prognostic significance.[2-5]**
- **Patients who do not have measurable levels of M protein or free light chain should be followed using imaging at regular intervals.**

References

[1] Paiva B, Chandia M, Vidriales MB, et al. Multiparameter flow cytometry for staging of solitary bone plasmacytoma: new criteria for risk of progression to myeloma. Blood 2014;124:1300-1303.
[2] Walker R, Barlogie B, Haessler J, et al. Magnetic resonance imaging in multiple myeloma: diagnostic and clinical implications. J Clin Oncol 2007;25:1121-1128.
[3] Bartel TB, Haessler J, Brown TL, et al. F18-fluorodeoxyglucose positron emission tomography in the context of other imaging techniques and prognostic factors in multiple myeloma. Blood 2009;114:2068-2076.
[4] Zamagni E, Nanni C, Mancuso K, et al. PET/CT improves the definition of complete response and allows to detect otherwise unidentifiable skeletal progression in multiple myeloma. Clin Cancer Res 2015;21:4384-4390.
[5] Moreau P, Attal M, Caillot D, et al. Prospective evaluation of magnetic resonance imaging and [(18)F]fluorodeoxyglucose positron emission tomography-computed tomography at diagnosis and before maintenance therapy in symptomatic patients with multiple myeloma included in the IFM/DFCI 2009 trial: Results of the IMAJEM study. J Clin Oncol 2017;35:2911-2918.

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PRINCIPLES OF RADIATION THERAPY

Solitary Plasmacytoma

General Principles:
- Radiation therapy (RT) is the intervention of choice for solitary plasmacytoma.

- Treatment of solitary plasmacytomas should be performed using modern treatment principles including imaging-based delineation (MRI, CT with contrast, and/or FDG-PET/CT) of a gross tumor volume (GTV), clinical target volume (CTV), and planning target volume (PTV) and adjacent organs at risk (OARs). CTV expansions should generally include at least 0.5 cm of margin for microscopic extent, and up to 2–3 cm for involvement of long bones. PTV margins should be minimized using modern daily image guidance. Treatment of adjacent vertebral bodies for spine lesions is not required if there is no suspicion of clinical involvement.

- RT should be utilized with advanced technology (ie, intensity-modulated RT [IMRT], VMAT, protons) when these modalities can help limit radiation doses to surrounding OARs. Principles of involved-site RT (ISRT) should be used to avoid large radiation fields and inappropriately including uninvolved sites, which will increase the risk of toxicity.

Treatment Information/Dosing:
- Solitary plasmacytoma (MYEL-2)
  - RT (40–50 Gy in 1.8–2.0 Gy fractions [20–25 total fractions]) to involved site.
  - Treatment with 35–40 Gy is an acceptable alternative for solitary plasmacytomas <5 cm in size, due to the high rates of local control reported for smaller tumors.

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PRINCIPLES OF RADIATION THERAPY

Multiple Myeloma

General Principles:
- **Treatment recommendations should be made after joint consultation and/or discussion by a multidisciplinary team including surgical, radiation, and medical oncologists, and radiologists.**
- **RT is primarily used for palliation in patients with MM.**
- **Careful planning in defining the radiation field and radiation technique must be utilized to minimize toxicity to the spinal cord, brain, bone marrow, and adjacent OARs as patients may be treated multiple times during their disease course.**
- **Careful planning of radiation fields for cord compression in the thoracic area at the level of the heart should be utilized to avoid radiation dose exiting into the heart structures, which could lead to cardiac toxicity.**
- **If surgical intervention is not indicated for impending fracture, structural instability, or emergent decompression, it should be avoided to circumvent potential complications and delays in systemic therapy. RT can result in functional re-ossification in a large proportion of patients.**
- **For principles of RT for CNS disease in patients with MM, see Principles of Management of CNS Disease in Patients with MM (MYEL-H).**

Timing and Sequencing of Therapy:
- **Systemic therapy should not be delayed for RT. Data suggest that systemic therapy and palliative RT can be used concurrently without evidence of increased toxicity, but that patients should be carefully monitored for toxicities.**
- **If urgent surgical intervention is indicated, RT should be delivered postoperatively to improve pain control and prevent local recurrence. Patients in a resource-limited setting with access to systemic therapy may consider forgoing postoperative RT.**

Palliative RT Dosing for MM:
- **Low-dose RT (8 Gy x 1 fraction) or 20–30 Gy in 5–15 total fractions can be used as palliative treatment for indications such as uncontrolled pain, for impending pathologic fracture, or for impending cord compression. Moderately fractionated courses of 20–25 Gy in 8–10 fractions are generally preferred over higher doses (30 Gy) absent extenuating circumstances (eg, severe symptomatic cord compression) to limit toxicity risk and reduce future toxicity risk in the event additional irradiation is needed to adjacent or overlapping sites (eg, overlapping sites in the spine/spinal cord.**
- **Limited involved sites should be used to limit the impact of irradiation on hematopoietic stem cell harvest or impact on potential future treatments.**
- **For RT dose constraint suggestions regarding bone marrow and other OARs, see NCCN Guidelines for Hodgkin Lymphoma.**

References
Elhammali A, Amini B, Ludmir EB, et al. New paradigm for radiation in multiple myeloma: lower yet effective dose to avoid radiation toxicity. Haematologica 2020;105:e355-e357.
Guerini AE, Tucci A, Alongi F, et al. RR Myelo POINT: A Retrospective Single-Center Study Assessing the Role of Radiotherapy in the Management of Multiple Myeloma and Possible Interactions with Concurrent Systemic Treatment. Cancers (Basel). 2022;14:2273.
Resende Salgado L, Wang S, Adler A, et al. The Safety Profile of Concurrent Therapy for Multiple Myeloma in the Modern Era. Adv Radiat Oncol 2018;4:112-117.
Tsang RW, Campbell BA, Goda JS, et al. Radiation Therapy for Solitary Plasmacytoma and Multiple Myeloma: Guidelines From the International Lymphoma Radiation Oncology Group Int J Radiat Oncol Biol Phys 2018;101:794-808.
Tsang RW, Gospodarowicz MK, Pintilie M, et al. Solitary plasmacytoma treated with radiotherapy: impact of tumor size on outcome. Int J Radiat Oncol Biol Phys 2001;50:113-120.

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RESPONSE CRITERIA FOR MULTIPLE MYELOMA
(Revised based on the new criteria by International Myeloma Working Group [IMWG])

IMWG criteria for response assessment including criteria for minimal residual disease (MRD)	
Response Category[a]	**Response Criteria**
IMWG MRD criteria (requires a complete response as defined below)	
Sustained MRD-negative	MRD negativity in the marrow (next-generation flow [NGF], next-generation sequencing [NGS], or both) and by imaging as defined below, confirmed minimum of 1 year apart. Subsequent evaluations can be used to further specify the duration of negativity (eg, MRD-negative at 5 years).[b]
Flow MRD-negative	Absence of phenotypically aberrant clonal plasma cells by NGF[c] on bone marrow aspirates using the EuroFlow standard operation procedure for MRD detection in multiple myeloma (or validated equivalent method) with a minimum sensitivity of 1 in 10^5 nucleated cells or higher.
Sequencing MRD-negative	Absence of clonal plasma cells by NGS on bone marrow aspirate in which presence of a clone is defined as less than two identical sequencing reads obtained after DNA sequencing of bone marrow aspirates using a validated equivalent method with a minimum sensitivity of 1 in 10^5 nucleated cells[d] or higher.
Imaging plus MRD-negative	MRD negativity as defined by NGF or NGS plus disappearance of every area of increased tracer uptake found at baseline or a preceding FDG-PET/CT or decrease to less mediastinal blood pool standardized uptake value (SUV) or decrease to less than that of surrounding normal tissue.[e]
Standard IMWG response criteria[f]	
Stringent complete response	Complete response as defined below plus normal FLC ratio[g] and absence of clonal cells in bone marrow biopsy by immunohistochemistry (κ/λ ratio ≤4:1 or ≥1:2 for κ and λ patients, respectively, after counting ≥100 plasma cells).[h]
Complete response[i]	Negative immunofixation on the serum and urine and disappearance of any soft tissue plasmacytomas and <5% plasma cells in bone marrow aspirates.
Very good partial response	Serum and urine M-protein detectable by immunofixation but not on electrophoresis or ≥90% reduction in serum M-protein plus urine M-protein level <100 mg per 24 h.
Partial response	≥50% reduction of serum M-protein plus reduction in 24-h urinary M-protein by ≥90% or to <200 mg per 24 h. If the serum and urine M-protein are unmeasurable, a ≥50% decrease in the difference between involved and uninvolved FLC levels is required in place of the M-protein criteria. If serum and urine M-protein are unmeasurable, and serum-free light assay is also unmeasurable, ≥50% reduction in plasma cells is required in place of M-protein, provided baseline bone marrow plasma-cell percentage was ≥30%. In addition to these criteria, if present at baseline, a ≥50% reduction in the size (sum of the products of the maximal perpendicular diameters [SPD] of measured lesions)[j] of soft tissue plasmacytomas is also required.
Minimal response	≥25% but ≤49% reduction of serum M-protein and reduction in 24-h urine M-protein by 50%–89%. In addition to the above listed criteria, if present at baseline, a 25%–49% reduction in SPD[j] of soft tissue plasmacytomas is also required.

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RESPONSE CRITERIA FOR MULTIPLE MYELOMA
(Revised based on the new criteria by International Myeloma Working Group [IMWG])

Standard IMWG response criteria[f] (continued from previous page)	
Stable disease	• Not recommended for use as an indicator of response; stability of disease is best described by providing the time-to-progression estimates. Not meeting criteria for complete response, very good partial response, partial response, minimal response, or progressive disease.
Progressive disease[k,l]	Any one or more of the following criteria: • Increase of 25% from lowest confirmed response value in one or more of the following criteria: • Serum M-protein (absolute increase must be ≥0.5 g/dL); • Serum M-protein increase ≥1 g/dL, if the lowest M component was ≥5 g/dL; • Urine M-protein (absolute increase must be ≥200 mg/24 h); • In patients without measurable serum and urine M-protein levels, the difference between involved and uninvolved FLC levels (absolute increase must be >10 mg/dL); • In patients without measurable serum and urine M-protein levels and without measurable involved FLC levels, bone marrow plasma-cell percentage irrespective of baseline status (absolute increase must be ≥10%); • Appearance of a new lesion(s), ≥50% increase from nadir in SPD[j] of >1 lesion, or ≥50% increase in the longest diameter of a previous lesion >1 cm in short axis; • ≥50% increase in circulating plasma cells (minimum of 200 cells per μL) if this is the only measure of disease.
Clinical relapse	Clinical relapse requires one or more of the following criteria: • Direct indicators of increasing disease and/or end organ dysfunction (calcium elevation, renal failure, anemia, lytic bone lesions [CRAB features]) related to the underlying clonal plasma cell proliferative disorder. It is not used in calculation of time to progression or progression-free survival but is listed as something that can be reported optionally or for use in clinical practice; • Development of new soft tissue plasmacytomas or bone lesions (osteoporotic fractures do not constitute progression); • Definite increase in the size of existing plasmacytomas or bone lesions. A definite increase is defined as a 50% (and ≥1 cm) increase as measured serially by the SPD[j] of the measurable lesion; • Hypercalcemia (>11 mg/dL); • Decrease in hemoglobin of ≥2 g/dL not related to therapy or other non–myeloma-related conditions; • Rise in serum creatinine by 2 mg/dL or more from the start of the therapy and attributable to myeloma; • Hyperviscosity related to serum paraprotein.
Relapse from complete response (to be used only if the endpoint is disease-free survival)	Any one or more of the following criteria: • Reappearance of serum or urine M-protein by immunofixation or electrophoresis[i]; • Development of ≥5% plasma cells in the bone marrow; • Appearance of any other sign of progression (ie, new plasmacytoma, lytic bone lesion, or hypercalcemia) (see above).
Relapse from MRD negative (to be used only if the endpoint is disease-free survival)	Any one or more of the following criteria: • Loss of MRD negative state (evidence of clonal plasma cells on NGF or NGS, or positive imaging study for recurrence of myeloma); • Reappearance of serum or urine M-protein by immunofixation or electrophoresis; • Development of ≥5% clonal plasma cells in the bone marrow; • Appearance of any other sign of progression (ie, new plasmacytoma, lytic bone lesion, or hypercalcemia).

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RESPONSE CRITERIA FOR MULTIPLE MYELOMA
FOOTNOTES

[a] All response categories require two consecutive assessments made any time before starting any new therapy; for MRD there is no need for two consecutive assessments, but information on MRD after each treatment stage is recommended (eg, after induction, high-dose therapy/autologous HCT, consolidation, maintenance). MRD tests should be initiated only at the time of suspected complete response. All categories of response and MRD require no known evidence of progressive or new bone lesions if radiographic studies were performed. However, radiographic studies are not required to satisfy these response requirements except for the requirement of FDG-PET if imaging MRD-negative status is reported.

[b] Sustained MRD negativity when reported should also annotate the method used (eg, sustained flow MRD-negative, sustained sequencing MRD-negative).

[c] Bone marrow MFC should follow NGF guidelines. The reference NGF method is an eight-color two-tube approach, which has been extensively validated. The two-tube approach improves reliability, consistency, and sensitivity because of the acquisition of a greater number of cells. The eight-color technology is widely available globally and the NGF method has already been adopted in many flow laboratories worldwide. The complete eight-color method is most efficient using a lyophilised mixture of antibodies, which reduces errors, time, and costs. Five million cells should be assessed. The Flow Cytometry Method (FCM) method employed should have a sensitivity of detection of at least 1 in 10^5 plasma cells. Paiva B, Gutierrez NC, Rosinol L, et al, for the GEM (Grupo Españolde MM)/PETHEMA (Programa para el Estudio de la Terapéutica en Hemopatías Malignas) Cooperative Study Groups. High-risk cytogenetics and persistent minimal residual disease by multiparameter flow cytometry predict unsustained complete response after autologous stem cell transplantation in multiple myeloma. Blood 2012;119:687-691.

[d] DNA sequencing assay on bone marrow aspirate should use a validated assay.

[e] Criteria used by Zamagni and colleagues, and expert panel (IMPetUs; Italian Myeloma Criteria for PET Use). Baseline positive lesions were identified by presence of focal areas of increased uptake within bones, with or without any underlying lesion identified by CT and present on at least two consecutive slices. Alternatively, an SUV_{max} = 2.5 within osteolytic CT areas >1 cm in size, or SUV_{max} = 1.5 within osteolytic CT areas ≤1 cm in size were considered positive. Imaging should be performed once MRD negativity is determined by MFC or NGS. Zamagni E, Nanni C, Mancuso K, et al. PET/CT improves the definition of complete response and allows to detect otherwise unidentifiable skeletal progression in multiple myeloma. Clin Cancer Res 2015;21:4384-4390.

[f] Derived from international uniform response criteria for multiple myeloma. Minor response definition and clarifications derived from Rajkumar and colleagues. When the only method to measure disease is by serum FLC levels: complete response can be defined as a normal FLC ratio of 0.26 to 1.65 in addition to the complete response criteria listed previously. Very good partial response in such patients requires a ≥90% decrease in the difference between involved and uninvolved FLC levels. All response categories require two consecutive assessments made at any time before the institution of any new therapy; all categories also require no known evidence of progressive or new bone lesions or extramedullary plasmacytomas if radiographic studies were performed. Radiographic studies are not required to satisfy these response requirements. Bone marrow assessments do not need to be confirmed. Each category, except for stable disease, will be considered unconfirmed until the confirmatory test is performed. The date of the initial test is considered as the date of response for evaluation of time dependent outcomes such as duration of response. Durie BG, Harousseau JL, Miguel JS, et al, for the International Myeloma Working Group. International uniform response criteria for multiple myeloma. Leukemia 2006;20:1467-1473.

[g] All recommendations regarding clinical uses relating to serum FLC levels or FLC ratio are based on results obtained with the validated serum FLC assay.

[h] Presence/absence of clonal cells on immunohistochemistry is based upon the κ/λ/L ratio. An abnormal κ/λ ratio by immunohistochemistry requires a minimum of 100 plasma cells for analysis. An abnormal ratio reflecting presence of an abnormal clone is κ/λ of >4:1 or <1:2.

[i] Special attention should be given to the emergence of a different monoclonal protein following treatment, especially in the setting of patients having achieved a conventional complete response, often related to oligoclonal reconstitution of the immune system. These bands typically disappear over time and in some studies have been associated with a better outcome. Also, appearance of monoclonal IgG κ in patients receiving monoclonal antibodies should be differentiated from the therapeutic antibody.

[j] Plasmacytoma measurements should be taken from the CT portion of the PET/CT, or MRI scans, or dedicated CT scans where applicable. For patients with only skin involvement, skin lesions should be measured with a ruler. Measurement of tumor size will be determined by the SPD.

[k] Positive immunofixation alone in a patient previously classified as achieving a complete response will not be considered progression. For purposes of calculating time to progression and progression-free survival, patients who have achieved a complete response and are MRD-negative should be evaluated using criteria listed for progressive disease. Criteria for relapse from a complete response or relapse from MRD should be used only when calculating disease-free survival.

[l] In the case where a value is felt to be a spurious result per physician discretion (eg, a possible laboratory error), that value will not be considered when determining the lowest value.

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CONSIDERATIONS FOR MYELOMA THERAPY

General Principles:
- Systemic therapy should not be delayed for advanced imaging if diagnosis of active myeloma is otherwise clear.
- Quadruplet regimen is preferred. Based on functional status, patients may be started with a 2- or 3-drug regimen with additional drugs added as performance status improves.
- Consider dose modifications based on functional status and age.
- Consider discontinuing dexamethasone at earliest possible timepoint, based on response, frailty, and toxicity considerations.
- For additional supportive care while on myeloma therapy, see Supportive Care Treatment for Multiple Myeloma (MYEL-I).

Considerations for Special Populations:
- Older Adults
  - Frailty assessment should be considered in older adults. See NCCN Guidelines for Older Adult Oncology.
  - For the Myeloma Frailty Score Calculator developed by IMWG for the prognosis, see http://www.myelomafrailtyscorecalculator.net/[1]
- Black/African American Individuals
  - Black/African American individuals have the highest rate of myeloma of any race/ethnic group and twice the incidence of all other races and ethnicities, earlier age at diagnosis, and present with more severe symptoms at diagnosis including increased anemia, higher calcium levels, worse renal dysfunction, and more extramedullary disease at presentation.
  - The biology of myeloma in these individuals can also differ. Black/African American individuals are more likely to have chromosome 14 translocations such as t(11;14), t(14;16), and t(14;20), and are less likely to have high-risk mutations such as del17p and gain/amp 1q compared to white individuals.
  - Decreased participation in clinical trials exists based on race, ethnicity and age, which leads to limited understanding of whether results of the studies apply to all patients.
  - A study of outcomes of patients with MM showed that without disparities in the overall use of novel agents or autologous HCT, Black/African American individuals and age <65 years had a superior median overall survival (OS) (7.07 years; 95% CI, 6.36–7.70 years) compared with white patients (5.83 years; 95% CI, 5.44–6.09 years; $P < .001$) (Fillmore NR, et al. Blood 2019;133:2615-2618).
  - Toxicity of drugs can vary in Black/African American individuals, including hyperpigmentation of skin with use of IMiDs and higher risk for cytokine release syndromes (CRS) during CAR T-cell therapy (Blevins F, et al. Blood 2020;136:23-24; Peres LC, et al. Blood Adv 2024;8:251-259).
  - Due to these differences, in order to reflect populations most affected, it is recommended to conduct equitable and inclusive clinical trials.

[1] Palumbo A, Bringhen S, Mateos MV, et al. Geriatric assessment predicts survival and toxicities in elderly myeloma patients: An International Myeloma Working Group report. Blood 2015;125:2068-2074.

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CONSIDERATIONS FOR MYELOMA THERAPY

Additional Consideration for Relapsed/Refractory Disease:
- **Consideration for appropriate regimen in previously treated myeloma should be based on the context of clinical relapse.**
- **In order to maximize benefit of systemic therapy, agents/regimens may be reconsidered or repeated if relapse is after at least 6 months after stopping therapy.**
- **A new triplet regimen should preferably include drugs or drug classes in patients who have not been exposed to, or not exposed to for at least 6 months.**
- **Clinical trials with these triplet regimens primarily included patients who were naïve or sensitive to the novel drug in the doublet comparator arm. Patients with disease refractory to the novel drug in the doublet backbone should be considered for triplet therapy that does not contain the drug they are progressing on.**
- **Immunoglobulin replacement should be considered for patients with an IgG <400 mg/dL[a] or at a higher level for supportive care as clinically indicated. (See MYEL-J)**
- **Patients may be candidates for TCR Therapy. (See MYEL-G).**

For HCT and Stem Cell Storage:
- **Exposure to myelotoxic agents (including alkylating agents and nitrosoureas) should be limited to avoid compromising stem cell reserve prior to stem cell harvest in patients who may be candidates for HCT.**
- **Consider harvesting autologous peripheral blood stem cells within the first 6 cycles of therapy initiation prior to prolonged exposure to lenalidomide and/or daratumumab in patients for whom HCT is being considered.**
- **If delayed HCT is considered, then autologous stem cells should be collected and stored.**

Dosing and Administration:
- **An FDA-approved biosimilar is an appropriate substitute for any recommended systemic biologic therapy in the NCCN Guidelines.**
- **Subcutaneous bortezomib is the preferred method of administration.**
- **Both weekly (preferred) and twice-weekly dosing schemas of bortezomib may be appropriate.**
- **Carfilzomib may be used weekly (preferred) or twice-weekly and at different doses.**
- **For any regimen that includes daratumumab, this could be daratumumab for intravenous infusion or daratumumab and hyaluronidase-fihj for subcutaneous injection. Daratumumab and hyaluronidase-fihj for subcutaneous injection has different dosing and administration instructions compared to daratumumab for intravenous infusion.**
- **Steroids should be reduced to 20 mg weekly in patients who are older and should be decreased or discontinued with treatment response plateau and/or toxicity.**

Side Effects and Lab Interference:
- **Daratumumab and isatuximab-irfc may interfere with serologic testing and cause false-positive indirect Coombs test.**
- **Type and screen should be performed before using daratumumab or isatuximab-irfc.**
- **Monoclonal antibodies can produce a false positive serum immunofixation if the monoclonal protein is IgG kappa and special interference testing or mass spectrometry based assessment can differentiate between the two.**
- **Carfilzomib can potentially cause cardiac, renal, and pulmonary toxicity, especially in patients who are older.**
- **Agents such as bendamustine can impact the ability to collect T cells for CAR T-cell therapy. See NCCN Guidelines for Management of Immunotherapy-Related Toxicities.**

[a] In assessing the need for gammaglobulin replacement in patients with IgG myeloma, it is important to take into the account the portion of IgG that is clonal. To estimate the normal IgG levels, subtract the M spike value from the IgG.

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PRIMARY THERAPY FOR HCT CANDIDATES[a-d]
Preferred • Daratumumab/Lenalidomide/Bortezomib/Dexamethasone (category 1) • Isatuximab-irfc/Bortezomib/Lenalidomide/Dexamethasone (category 1)
Other Recommended • Bortezomib/Lenalidomide/Dexamethasone (category 1) • Carfilzomib/Lenalidomide/Dexamethasone • Daratumumab/Carfilzomib/Lenalidomide/Dexamethasone • Isatuximab-irfc/Carfilzomib/Lenalidomide/Dexamethasone
Useful in Certain Circumstances • Bortezomib/Cyclophosphamide/Dexamethasone • Carfilzomib/Cyclophosphamide/Dexamethasone[e] • Daratumumab/Bortezomib/Cyclophosphamide/Dexamethasone • Dexamethasone/Thalidomide/Cisplatin/Doxorubicin/Cyclophosphamide/Etoposide/Bortezomib[f] (VTD-PACE)

MAINTENANCE THERAPY[g]
Preferred • Lenalidomide[h] (category 1)
Other Recommended • Carfilzomib/Lenalidomide[h] • Daratumumab/Lenalidomide[h]
Useful in Certain Circumstances • Bortezomib ± Lenalidomide[h] • Daratumumab • Ixazomib (category 2B)

See Evidence Blocks on EB-1

[a] Selected, but not inclusive of all regimens. The regimens under each preference category are listed by order of NCCN Category of Evidence and Consensus alphabetically.
[b] Supportive Care Treatment for Multiple Myeloma (MYEL-I).
[c] See MYEL-F for considerations for myeloma therapy and for specific populations (eg, Black/African American individuals).
[d] Management of Renal Disease in Multiple Myeloma (MYEL-L).
[e] Treatment option for patients with renal insufficiency and/or peripheral neuropathy.
[f] Generally reserved for the treatment of aggressive MM.
[g] Two-drug maintenance recommended for high-risk MM.
[h] There appears to be an increased risk for secondary cancers, especially with lenalidomide maintenance following transplant. The benefits and risks of maintenance therapy versus secondary cancers should be discussed with patients.

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PRIMARY THERAPY IF HCT-DEFERRED[i] OR IF HCT IS NOT INDICATED[a-d,j]	
In general, continue primary therapy until progression with de-escalation of therapy (modification of dose and duration) as needed.	
Preferred • Daratumumab/Lenalidomide/Dexamethasone (category 1) • Daratumumab/Bortezomib/Lenalidomide/Dexamethasone (for patients <80 years old who are not frail) (category 1) • Isatuximab-irfc/Bortezomib/Lenalidomide/Dexamethasone (for patients <80 years old who are not frail) (category 1)	
Other Recommended • Carfilzomib/Lenalidomide/Dexamethasone • Bortezomib/Lenalidomide/Dexamethasone (category 1) • Isatuximab-irfc/Lenalidomide/Dexamethasone	
Useful In Certain Circumstances • Lenalidomide/Low-Dose Dexamethasone (category 1) • Bortezomib/Cyclophosphamide/Dexamethasone • Bortezomib/Dexamethasone • Bortezomib/Lenalidomide/Dexamethasone (VRD-lite) for patients assessed as being frail	• Carfilzomib/Cyclophosphamide/Dexamethasone[e] • Daratumumab/Cyclophosphamide/Bortezomib/Dexamethasone[k] • Isatuximab-irfc/Carfilzomib/Lenalidomide/Dexamethasone (category 2B) • Ixazomib/Lenalidomide/Dexamethasone • Lenalidomide/Cyclophosphamide/Dexamethasone

See Evidence Blocks on EB-2

[a] Selected, but not inclusive of all regimens. The regimens under each preference category are listed by order of NCCN Category of Evidence and Consensus alphabetically.
[b] Supportive Care Treatment for Multiple Myeloma (MYEL-I).
[c] See MYEL-F for considerations for myeloma therapy and for specific populations (eg, Black/African American individuals).
[d] Management of Renal Disease in Multiple Myeloma (MYEL-L).
[e] Treatment option for patients with renal insufficiency and/or peripheral neuropathy.
[i] Strongly consider collecting and storing stem cells for future HCT.
[j] Bortezomib or carfilzomib may be substituted with ixazomib in select patients in case of intolerance/logistical reasons.
[k] Treatment option for patients with renal insufficiency.

Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page EB-DEF
All recommendations are category 2A unless otherwise indicated.

THERAPY FOR PREVIOUSLY TREATED MULTIPLE MYELOMA[a-d,l-o]
Relapsed/Refractory Disease After 1–3 Prior Therapies

Preferred*
Order of regimens does not indicate comparative efficacy

Anti-CD38 Refractory	Bortezomib-Refractory	Lenalidomide-Refractory
• Carfilzomib/Lenalidomide/Dexamethasone (category 1) • Carfilzomib/Pomalidomide/Dexamethasone • Pomalidomide/Bortezomib/Dexamethasone (category 1) ***After two prior therapies including lenalidomide and a proteasome inhibitor (PI)*** ‣ Elotuzumab/Pomalidomide/Dexamethasone ***After two prior therapies including an IMiD and a PI and with disease progression on/within 60 days of completion of last therapy*** ‣ Ixazomib/Pomalidomide/Dexamethasone	• Carfilzomib/Lenalidomide/Dexamethasone (category 1) • Daratumumab/Carfilzomib/Dexamethasone (category 1) • Daratumumab/Lenalidomide/Dexamethasone (category 1) • Isatuximab-irfc/Carfilzomib/Dexamethasone (category 1) • Carfilzomib/Pomalidomide/Dexamethasone ***After one prior therapy including Lenalidomide and a PI*** ‣ Daratumumab/Pomalidomide/Dexamethasone (category 1) ‣ Daratumumab/Teclistamab-cqyv (category 1) ***After two prior therapies including Lenalidomide and a PI*** ‣ Isatuximab-irfc/Pomalidomide/Dexamethasone (category 1) ‣ Elotuzumab/Pomalidomide/Dexamethasone	• Daratumumab/Bortezomib/Dexamethasone (category 1) • Daratumumab/Carfilzomib/Dexamethasone (category 1) • Isatuximab-irfc/Carfilzomib/Dexamethasone (category 1) • Pomalidomide/Bortezomib/Dexamethasone (category 1) • Carfilzomib/Pomalidomide/Dexamethasone ***After one prior therapy including Lenalidomide and a PI*** ‣ Daratumumab/Pomalidomide/Dexamethasone (category 1) ‣ Daratumumab/Teclistamab-cqyv (category 1) ***After two prior therapies including Lenalidomide and a PI*** ‣ Isatuximab-irfc/Pomalidomide/Dexamethasone (category 1) ‣ Elotuzumab/Pomalidomide/Dexamethasone ***After two prior therapies including an IMiD and a PI and with disease progression on/within 60 days of completion of last therapy*** ‣ Ixazomib/Pomalidomide/Dexamethasone

CAR T-Cell Therapy

See Evidence Blocks on EB-3

After one prior line of therapy including IMiD and a PI, and refractory to lenalidomide
‣ Ciltacabtagene autoleucel (category 1)

After two prior lines of therapies including an IMiD, an anti-CD38 monoclonal antibody and a PI
‣ Idecabtagene vicleucel (category 1)

* For Other Recommended regimens and for regimens Useful in Certain Circumstances for Relapsed/Refractory Disease After 1–3 Prior Therapies, see MYEL-G 4 of 5

[a] Selected, but not inclusive of all regimens. The regimens under each preference category are listed by order of NCCN Category of Evidence and Consensus alphabetically.
[b] Supportive Care Treatment for Multiple Myeloma (MYEL-I).
[c] See MYEL-F for considerations for myeloma therapy and for specific populations (eg, Black/African American individuals).
[d] Management of Renal Disease in Multiple Myeloma (MYEL-L).
[l] Regimens included under 1–3 prior therapies can also be used later in the disease course. Attempt should be made to use drugs/drug classes the patients have not been exposed to or exposed to >1 line prior.
[m] Autologous HCT should be considered in patients who are eligible and have not previously received HCT or had a prolonged response to initial HCT.
[n] In order to maximize benefit of systemic therapy, agents/regimens may be reconsidered or repeated if relapse is after at least 6 months of stopping therapy.
[o] Alkylating agents can impact the ability to collect T cells for CAR T-cell therapy. See NCCN Guideline for Management of Immunotherapy-Related Toxicities.

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All recommendations are category 2A unless otherwise indicated.

THERAPY FOR PREVIOUSLY TREATED MULTIPLE MYELOMA[a-d,l-p]
Relapsed/Refractory Disease After 1–3 Prior Therapies

Other Recommended

- Elotuzumab/Lenalidomide/Dexamethasone (category 1)
- Ixazomib/Lenalidomide/Dexamethasone (category 1)
- Selinexor/Bortezomib/Dexamethasone (category 1)
- Bortezomib/Cyclophosphamide/Dexamethasone
- Bortezomib/Lenalidomide/Dexamethasone
- Carfilzomib/Cyclophosphamide/Dexamethasone
- Daratumumab/Cyclophosphamide/Bortezomib/Dexamethasone
- Daratumumab/Carfilzomib/Pomalidomide/Dexamethasone
- Elotuzumab/Bortezomib/Dexamethasone
- Ixazomib/Cyclophosphamide/Dexamethasone
- Lenalidomide/Cyclophosphamide/Dexamethasone

After two prior therapies including an IMiD and a PI
- Belantamab mafodotin-blmf/Bortezomib/Dexamethasone (category 1)

After two prior therapies including an IMiD and a PI and disease progression on/within 60 days of completion of last therapy
- Pomalidomide/Cyclophosphamide/Dexamethasone (category 1)

See Evidence Blocks on EB-4

Useful in Certain Circumstances

- Bortezomib/Dexamethasone (category 1)
- Bortezomib/Liposomal Doxorubicin/Dexamethasone (category 1)
- Lenalidomide/Dexamethasone (category 1)
- Carfilzomib/Cyclophosphamide/Thalidomide/Dexamethasone
- Carfilzomib/Dexamethasone (category 1)
- Selinexor/Carfilzomib/Dexamethasone
- Selinexor/Daratumumab/Dexamethasone
- Venetoclax/Dexamethasone ± Daratumumab or PI only for t(11;14) patients

After two prior therapies including IMiD and a PI and with disease progression on/within 60 days of completion of last therapy
- Pomalidomide/Dexamethasone (category 1)
- Selinexor/Pomalidomide/Dexamethasone

For treatment of aggressive MM
- Dexamethasone/Cyclophosphamide/Etoposide/Cisplatin (DCEP)
- Dexamethasone/Thalidomide/Cisplatin/Doxorubicin/Cyclophosphamide/Etoposide (DT-PACE) ± Bortezomib (VTD-PACE)

After at least three prior therapies including a PI and an IMiD or are double-refractory to a PI and an IMiD
- Daratumumab

See Evidence Blocks on EB-5

[a] Selected, but not inclusive of all regimens. The regimens under each preference category are listed by order of NCCN Category of Evidence and Consensus alphabetically.
[b] Supportive Care Treatment for Multiple Myeloma (MYEL-I).
[c] See MYEL-F for considerations for myeloma therapy and for specific populations (eg, Black/African American individuals).
[d] Management of Renal Disease in Multiple Myeloma (MYEL-L).
[l] Regimens included under 1–3 prior therapies can also be used later in the disease course. Attempt should be made to use drugs/drug classes the patients have not been exposed to or exposed to >1 line prior.
[m] Autologous HCT should be considered in patients who are eligible and have not previously received HCT or had a prolonged response to initial HCT.
[n] In order to maximize benefit of systemic therapy, agents/regimens may be reconsidered or repeated if relapse is after at least 6 months of stopping therapy.
[o] Alkylating agents can impact the ability to collect T cells for CAR T-cell therapy. See NCCN Guideline for Management of Immunotherapy-Related Toxicities.
[p] Consider single-agent lenalidomide or pomalidomide for patients with steroid intolerance.

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All recommendations are category 2A unless otherwise indicated.

THERAPY FOR PREVIOUSLY TREATED MULTIPLE MYELOMA[a-d,l-o] Relapsed/Refractory Disease After 3 Prior Lines of Therapy
Preferred[r]
‣ ***CAR T-cell Therapy[q]:*** ◊ Ciltacabtagene autoleucel ◊ Idecabtagene vicleucel ***After at least four prior therapies, including an anti-CD38 monoclonal antibody, a PI, and an IMiD*** ‣ *Bispecific Antibodies (BsAb)*[s]: ◊ Elranatamab-bcmm ◊ Linvoseltamab-gcpt ◊ Talquetamab-tgvs ◊ Teclistamab-cqyv
Other Recommended
• High-Dose or Fractionated Cyclophosphamide ***After at least four prior therapies and whose disease is refractory to at least two PIs, at least two immunomodulatory agents, and an anti-CD38 monoclonal antibody*** • Selinexor/Dexamethasone
Useful in Certain Circumstances[r]
• Belantamab mafodotin-blmf • Bendamustine[o] • Bendamustine/Bortezomib/Dexamethasone[o] • Bendamustine/Carfilzomib/Dexamethasone[o] • Bendamustine/Lenalidomide/Dexamethasone[o] • Talquetamab + Teclistamab[s] See Evidence Blocks on EB-6

[a] Selected, but not inclusive of all regimens. The regimens under each preference category are listed by order of NCCN Category of Evidence and Consensus alphabetically.
[b] Supportive Care Treatment for Multiple Myeloma (MYEL-I).
[c] See MYEL-F for considerations for myeloma therapy and for specific populations (eg, Black/African American individuals).
[d] Management of Renal Disease in Multiple Myeloma (MYEL-L).
[l] Regimens included under 1–3 prior therapies can also be used later in the disease course. Attempt should be made to use drugs/drug classes the patients have not been exposed to or exposed to >1 line prior.
[m] Autologous HCT should be considered in patients who are eligible and have not previously received HCT or had a prolonged response to initial HCT.
[n] In order to maximize benefit of systemic therapy, agents/regimens may be reconsidered or repeated if relapse is after at least 6 months of stopping therapy.
[o] Alkylating agents can impact the ability to collect T cells for CAR T-cell therapy. See NCCN Guideline for Management of Immunotherapy-Related Toxicities.
[q] Talquetamab-tgvs may be considered as a bridge to BCMA CAR-T therapy in relapsed and/or refractory multiple myeloma.
[r] Patients can receive more than one B-cell maturation antigen (BCMA) targeted therapy. Optimal sequencing of sequential BCMA targeted therapies is not known; however accumulated data suggests immediate follow on with second BCMA directed therapy after relapse may be associated with lower response rates
[s] Prophylactic tocilizumab may be considered prior to first dose to reduce the risk of CRS.

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All recommendations are category 2A unless otherwise indicated.

PRINCIPLES OF MANAGEMENT OF CNS DISEASE IN PATIENTS WITH MM

General

- Multimodality therapy (radiation and systemic therapy) has been associated with improved outcomes.
- Incorporation of palliative care along with multimodal therapy is strongly recommended.

Radiation Therapy (RT)

- RT can be part of multimodality therapy.
- Focal RT can be used to rapidly reverse or prevent progression of focal neurologic deficits resulting from localized disease within the brain, spine, or cranial nerves while awaiting further workup or if alternative therapies are not immediately available.
- Whole brain radiation therapy (WBRT) can be considered for patients with poorly localized or cranial nerve disease.
  - Doses of WBRT ≤ 23.4 Gy in 13 fractions are associated with low toxicity.
  - Consider adding craniospinal irradiation (CSI) for patients with negative or otherwise well-controlled extra-CNS involvement.
    - If using CSI: conformal volumetric modulated arc therapy (VMAT) or protons can spare bone marrow.
    - CSI serves as an effective bridge to consolidation therapy like CAR T-cell therapy or autologous HCT.

Intrathecal (IT) Chemotherapy

- Consider as part of multimodality therapy.
- Concurrent IT Methotrexate and radiation can increase risk of myelopathy and should be given no less than 2 weeks apart.
  - Other concurrent combinations such as T-Cell Redirecting (TCR) therapies and systemic therapy are considered safe.
- Recommended IT regimens:
  - Thiotepa/Hydrocortisone
  - Methotrexate/Cytarabine/Hydrocortisone

Systemic Chemotherapy

- Systemic chemotherapy can be considered in combination with immunomodulatory drug (IMiD) as part of multimodal therapy if there are no other alternatives.
- Limited data on efficacy supports the use of systemic chemotherapy (high-dose methotrexate, cytarabine, or bendamustine). Bendamustine and other alkylating agents can impact the ability to collect T cells for CAR T-cell therapy. See NCCN Guidelines for Management of Immunotherapy-Related Toxicities.

Novel Agents

- Consider as part of multimodality therapy (with IT chemotherapy and RT).
  - IMiDs
    - Blood brain barrier penetration has been observed clinically with pomalidomide and lenalidomide.
    - Retrospective studies incorporating IMiDs have demonstrated improved outcomes.
  - Selinexor
    - Can penetrate blood brain barrier but has not been tested in management of CNS disease in patients with MM.
  - Venetoclax
    - Preclinical data supports blood brain barrier penetration.
    - Consider in t(11;14) if no other options are available.
  - Immunotherapy: CAR T-cell and BsAb Therapy
    - A history of treated CNS disease should not preclude use of CAR T-cell and BsAb therapy.
    - CAR T-cell and BsAb therapies may be useful as consolidative therapy to treat extra-CNS disease in patients receiving multi-modality CNS-directed therapy, including RT.
    - In general, there is limited data available regarding the ability of current immunotherapies to penetrate the CNS.
  - Monoclonal antibody (mAbs) therapy
    - In general there is limited data of mAbs to penetrate blood brain barrier
    - Can use if systemic therapy required, but no clear data to support use in management of CNS disease in patients with myeloma.
- Proteasome inhibitors (PI)
  - Commercially available agents exhibit limited penetration of the blood brain barrier.
  - No clear signal for activity of PI in CNS disease.

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All recommendations are category 2A unless otherwise indicated.

SUPPORTIVE CARE FOR MULTIPLE MYELOMA

Bone Disease

- All patients receiving primary myeloma therapy should be given bone-targeting treatment (bisphosphonates [category 1][a] or denosumab[b]).
  - A baseline dental exam is strongly recommended.
  - Assess vitamin D status.
  - Monitor for renal dysfunction with use of bisphosphonate therapy.
  - Monitor for osteonecrosis of the jaw.
  - Continue bone-targeting treatment (bisphosphonates or denosumab) for up to 2 years. The frequency of dosing (monthly vs. every 3 months) would depend on the individual patient criteria, response to therapy, and agent used. Continuing beyond 2 years should be based on clinical judgment.
  - Patients receiving denosumab for bone disease who subsequently discontinue therapy should be given maintenance denosumab every 6 months or a single dose of bisphosphonate to mitigate risk of rebound osteoporosis.[c]
- DEXA scans are not useful to assess myeloma bone disease.
- For RT recommendations see Principles of Radiation Therapy (MYEL-D).
- Orthopedic consultation should be sought for impending or actual long-bone fractures or bony compression of spinal cord or vertebral column instability.
- Consider vertebroplasty or kyphoplasty for symptomatic vertebral compression fractures.

Hypercalcemia

- Hydration, bisphosphonates (zoledronic acid preferred), denosumab, steroids, and/or calcitonin are recommended.

Hyperviscosity

- Plasmapheresis should be used as adjunctive therapy for symptomatic hyperviscosity.

Anemia

- See NCCN Guidelines for Hematopoietic Growth Factors.
- Consider erythropoietin for anemic patients.[d]

Infections

- See NCCN Guidelines for Prevention and Treatment of Cancer-Related Infections.
- See CDC for Use of COVID-19 Vaccines in the US.
- For prophylaxis and management of infections in patients with multiple myeloma, see MYEL-J.

Renal Dysfunction

- See Management of Renal Disease in Multiple Myeloma (MYEL-L).

Venous Thromboembolism (VTE)

- For management of VTE, risk stratification, and VTE prophylaxis, see MYEL-K.

[a] Both pamidronate and zoledronic acid have shown equivalence in terms of reducing risk of skeletal-related events in randomized trials.
[b] Denosumab is preferred in patients with renal insufficiency, caution for risk of hypocalcemia.
[c] This is based on observations with denosumab discontinuation in non-myeloma settings. Cummings SR, Ferrari S, Eastell R, et al. Vertebral fractures after discontinuation of denosumab: A post hoc analysis of the randomized placebo-controlled FREEDOM trial and its extension. J Bone Miner Res 2018;33:190-198.
[d] Increased risk of VTE has been reported in patients receiving erythropoiesis-stimulating agents (ESAs).

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All recommendations are category 2A unless otherwise indicated.

MANAGEMENT OF INFECTIONS IN PATIENTS WITH MULTIPLE MYELOMA[1]

For additional recommendations on infection management, see NCCN Guidelines for Prevention and Treatment of Cancer-Related Infections

Factors associated with increased infection risk in patients with MM

Infection Risk	Increased General Infection Risk	Increased Bacterial Infection Risk	Increased Viral Infection Risk	Increased Fungal Infection Risk
Contributing Factors	• Autologous HCT • BsAb therapy • CAR T-cell therapy • CD38 antibodies • Cytotoxic chemotherapy	• Diagnosis of MM • BsAb therapy • CAR T-cell therapy	• Proteasome inhibitors • Monoclonal antibodies • BsAb therapy • CAR T-cell therapy	• High-dose steroids • Prolonged Neutropenia • BsAb therapy • CAR T-cell therapy

Common Bacterial Infections				
Interventions & Prophylaxis	**Indication & Duration**			
	CAR-T	**BsAb**	**AutoHCT**	**Other Indications**
Levofloxacin: 500 mg PO daily **Alternatives:** Cefdinir 300 mg PO twice a day or augmentin 875 mg PO twice a day	Start when ANC <500 or per clinician discretion and continue until neutrophil recovery.	Start when ANC <500 or per clinician discretion and continue until neutrophil recovery.		Newly diagnosed MM: Consider levofloxacin for 12 weeks after diagnosis.[2]
Immunoglobulin replacement	After CAR T-cell therapy, regular IVIG infusions based on clinical context.	Consider for the duration of BsAb therapy		For IgG <400 mg/dL and/or recurrent life-threatening infections.[a]
Pneumococcal vaccination	Revaccination starting 3–6 months after treatment.	Update vaccination status prior to starting BsAb treatment.	Revaccination starting 3-6 months after treatment.	The CDC recommends 1 dose of PCV20 or 1 dose of PCV15 followed by 1 dose of PPSV23 at least 1 year later.

[a] In assessing the need for gammaglobulin replacement in patients with IgG myeloma, it is important to take into the account the total portion of IgG that is derived from clonal plasma cells. To estimate the normal IgG levels, subtract the M spike value for the IgG. If IgG <400 mg/dL, is not polyclonal, than replacement can be considered.

[1] Mohan M, et al. Blood Adv. 2022;6:2466–2470.
[2] Drayson MT, et al. Lancet Oncol 2019;20:1760-1772.

Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page EB-DEF
All recommendations are category 2A unless otherwise indicated.

Continued

MANAGEMENT OF INFECTIONS IN PATIENTS WITH MULTIPLE MYELOMA[1]

For additional recommendations on infection management, see NCCN Guidelines for Prevention and Treatment of Cancer-Related Infections

Common Viral and Fungal Infections				
Interventions & Prophylaxis	**Indication & Duration**			
	CAR-T[b]	**BsAb[b]**	**AutoHCT**	**Other Indications**
Herpes simplex virus or Varicella-zoster virus: Acyclovir 400–800 mg PO twice/day or valacyclovir 500 mg PO once or twice/ day.	A minimum of 1 year; indefinite (preferred), irrespective of vaccination status.	Indefinite, irrespective of vaccination status	For 1 year post-HCT or as clinically indicated.	While receiving a regimen with PIs or monoclonal antibody and for at least 3 months beyond end of therapy or per institutional practice.
Hepatitis B virus and HIV: Screening	• HBsAg-positive • HBsAg-negative • HBcAb-IgG positive • See NCCN Guidelines for Prevention and Treatment of Cancer-Related Infections for HBV and HIV treatment		See NCCN Guidelines for Prevention and Treatment of Cancer-Related Infections for HBV and HIV treatment	
Pneumocystis jirovecii pneumonia (PJP): Trimethoprim-sulfamethoxazole (TMP-SMX) therapy or pentamidine or atovaquone	Start with therapy and continue for at least 6 months after the infusion or until CD4 ≥200/mm³ (whichever is longer).		When equivalent dexamethasone dosing is >40 mg/day for 4 days per week or as clinically indicated per institutional practice.	
SARS-CoV-2: COVID-19 vaccination	See The CDC's Staying Up to Date with COVID-19 Vaccines for specific guidance and see NCCN Guidelines for Prevention and Treatment of Cancer-Related Infections for treatment.			
Influenza virus: Vaccination	Per CDC for patients who are immunosuppressed.			
RSV: Bivalent vaccine	• Single dose of bivalent vaccine for patients with MM aged ≥60 years. • See CDC guidance for all other patient populations.			
CMV: Prophylaxis	• Monitor viral load (by PCR) only in patients with suspected CMV-related disease (eg, colitis, pneumonitis, hepatitis) or otherwise unexplained fever and/or cytopenias or in patients with high risk of infections.[b] • See NCCN Guidelines for Prevention and Treatment of Cancer-Related Infections for treatment indication and treatment duration.			
Adenovirus, EBV, JCV, parvovirus: No prophylaxis	Routine monitoring of viral load is not recommended.			
Yeast: Fluconazole 400 mg PO daily	Start when ANC <500 or per clinician discretion and continue until neutrophil recovery.			
Mold: Azole	In patients with high risk of infections[b]: Consider ongoing prophylaxis with anti-mold azole.			

[b] Patients receiving CAR T-cell and BsAb treatment, recipients of >1 dose of tocilizumab, use of second line agents such as anakinra or siltuximab for management of CRS and ICANS, prolonged and/or high-dose steroid use (requiring >3 days of 10 mg dexamethasone per day with a 7-day period or receiving higher doses of methylprednisolone >1 g per day) are at high risk of infection.

[1] Mohan M, et al. Blood Adv 2022;6:2466–2470.

Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page EB-DEF
All recommendations are category 2A unless otherwise indicated.

MANAGEMENT OF VENOUS THROMBOEMBOLISM (VTE) IN MULTIPLE MYELOMA

VTE RISK STRATIFICATION USING IMPEDE OR SAVED SCORING SYSTEM

IMPEDE Score[1] for Risk Stratification (points assigned)			
Individual Risk Factors	Points	Myeloma Risk Factors	Points
Positive Factors			
Central venous catheter/Tunneled central line	+2	Immunomodulatory drug (IMiD)	+4
Pelvic, hip, or femur fracture	+4	Erythropoiesis-stimulating agent	+1
Obesity (body mass index ≥25)	+1	Dexamethasone ≤160 mg/month	+2
Previous VTE	+5	Dexamethasone >160 mg/month	+4
		Doxorubicin or multiagent chemotherapy	+3
Negative Factors			
Ethnicity/Race = Asian/Pacific Islander	-3		
Existing thromboprophylaxis: prophylactic LMWH (low-molecular-weight heparin) or aspirin	-3		
Existing thromboprophylaxis: therapeutic LMWH or warfarin	-4		

SAVED Score[2] for Risk Stratification	
Variable	Points
Surgery within 90 days	+2
Asian race	-3
VTE history	+3
Age ≥80 years	+1
Dexamethasone (regimen dose) • Standard dose (120–160 mg/cycle) • High dose (>160 mg/cycle)	 +1 +2

[1] Adapted with permission from Sanfilippo KM, et al. Am J Hematol 2019;94:1176-1184.
[2] Adapted with permission from Li A, et al. J Natl Compr Canc Netw 2019;17:840-847.

Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page EB-DEF
All recommendations are category 2A unless otherwise indicated.

Continued

MANAGEMENT OF VENOUS THROMBOEMBOLISM (VTE) IN MULTIPLE MYELOMA

General Principles:
- The highest risk for VTE is in the first 6 months following new diagnosis of MM.
- VTE prophylaxis is administered assuming there are no contraindications to anticoagulation agents or anti-platelets (see VTE-A within the NCCN Guidelines for Cancer-Associated Venous Thromboembolic Disease).
- All anticoagulants carry increased risk of bleeding; careful consideration needs to be made regarding risks and benefits for each patient.
- Warfarin at international normalized ratio (INR) 2–3 is not directly comparable to the other agents listed at prophylactic doses with respect to bleeding and thrombotic risks.
- Patients already on therapeutic anticoagulants for other reasons (eg, atrial fibrillation) should continue anticoagulation therapy.
- If no other coagulopathy, full-dose anticoagulation is contraindicated with thrombocytopenia <50,000/μL; in patients with high risk for VTE, prophylactic anticoagulation may be appropriate even if platelet count is as low as 25,000/μL.
- Indications for long-term anticoagulation include unprovoked VTE or provoked VTE in the presence of a risk factor that is still present.
- For any patients who develop VTE on IMiD-based therapy, continue using therapeutic dose anticoagulants for as long as IMiD-based therapy is indicated.

Factors Impacting the Choice of Optimal VTE Prophylaxis Agent:
- Bleeding risk (eg, concurrent coagulopathy, disseminated intravascular coagulation, hyperviscosity)
- Cytopenias (eg, platelet count ± hemoglobin)
- Concurrent medications (eg, strong cytochrome P inducers/inhibitors, single/dual anti-platelets)
- Current renal function (eg, creatinine clearance)
- Patient choice (eg, preference for mode of administration, dietary restrictions)
- Insurance coverage/restrictions (including cost of therapy)
- Availability of reversal agents in case of emergency bleeding
- History of heparin-induced thrombocytopenia
- Extremes of body weight
- Carfilzomib + IMiD therapy

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Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page EB-DEF
All recommendations are category 2A unless otherwise indicated.

Continued

MANAGEMENT OF VENOUS THROMBOEMBOLISM (VTE) IN MULTIPLE MYELOMA

RECOMMENDATIONS FOR VTE PROPHYLAXIS

VTE Prophylaxis Recommendations	
≤3 Points by IMPEDE Score or <2 Points by SAVED Score	≥4 Points by IMPEDE Score or ≥2 SAVED Score[a]
• Aspirin 81–325 mg once daily	• LMWH (equivalent to enoxaparin 40 mg daily) OR • Rivaroxaban 10 mg daily OR • Apixaban 2.5 mg twice daily OR • Fondaparinux 2.5 mg daily OR • Warfarin (target INR, 2.0–3.0)

Duration of VTE Prophylaxis
• Indefinite while on myeloma therapy • 3–6 months followed by aspirin (longer periods of anticoagulation may be considered in the presence of additional patient, treatment-specific, or transient VTE risk factors)

[a] A less common choice of agent includes dalteparin 5000 units subcutaneously (SC) daily (category 2B).

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All recommendations are category 2A unless otherwise indicated.

MANAGEMENT OF RENAL DISEASE IN MULTIPLE MYELOMA[a]

Tests[b]
- Serum creatinine, electrolytes, and uric acid
- Urinalysis, electrolytes, and sediment
- 24-hour urine collection for protein and UPEP/UIFE
- SPEP/SIFE and serum FLCs
- Consider renal ultrasound and renal biopsy.

Treatment Options
- Pulse dexamethasone
- Regimens containing bortezomib and/or daratumumab
- Can switch to other regimen once renal function has improved or stabilized
- See Response Criteria for Multiple Myeloma (MYEL-E)
- See Myeloma Therapy (MYEL-G)

Supportive Care
- Provide hydration to dilute tubular light chains; goal urine output is 100–150 cc/h
- Monitor fluid status
- Treat hypercalcemia, hyperuricemia, and other metabolic abnormalities
- Discontinue nephrotoxic medications
- Dialysis
  - Refractory electrolyte disturbances, uremia, and fluid overload
- Mechanical removal of serum FLCs with high cutoff dialysis filters or plasmapheresis may have a limited role. Systemic therapy should not be delayed if performing this procedure.
- Renal dosing of all medications

Recommendations for Lenalidomide Dosing in Patients with Multiple Myeloma who Have Renal Impairment

Category	Renal Function (Cockcroft-Gault CrCL)	Lenalidomide Dosing in Multiple Myeloma
Moderate renal impairment	CrCL ≥30 mL/min to <60 mL/min	10 mg every 24 h
Severe renal impairment	CrCL <30 mL/min (not requiring dialysis)	15 mg every 48 h
End-stage renal disease	CrCL<30 mL/min (requiring dialysis)	5 mg once daily; on dialysis days, dose should be administered after dialysis

Bone-Modifying Agent Dosing in Patients with Multiple Myeloma Who Have Renal Impairment

Degree of Renal Impairment	Pamidronate (focal segmental glomerulosclerosis)	Zoledronic Acid (tubular cell toxicity)	Denosumab
None	90 mg IV every 3–4 wks	4 mg IV every 1–3 mo	120 mg SQ Q 4 wks
Mild/moderate renal impairment	Use standard dose	Reduce dose	120 mg SQ Q 4 wks
Severe renal impairment	60–90 mg	Not recommended	120 mg SQ Q 4 wks[c]

[a] Defined as serum creatinine >2 mg/dL or established glomerular filtration rate (eGFR) <60 mL/min/1.73 sqm.
[b] Consider other diagnosis such as amyloid and light chain disease for patients with significant proteinuria. Dimopoulos MA, et al. Lancet Oncol 2023;24:e293-e311.
[c] Patients with creatinine clearance <30 cc/min can experience severe hypocalcemia and should be monitored.

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MONOCLONAL GAMMOPATHY OF RENAL SIGNIFICANCE

CLINICAL FINDINGS

Monoclonal gammopathy of renal significance (MGRS) suspected →

INITIAL WORKUP

Evaluate for kidney disease
- Kidney function: established GFR (eGFR)
- Urinalysis
- 24-h urine for total protein, UPEP, and UIFE
- Metabolic panel

→

Renal biopsy recommended if:
- Acute kidney injury (AKI) stage 3
- eGFR <60 mL/min and >2 mL/min per year decline
- Proteinuria (>1 g/day) Albumin:creatinine >30 mg/mmol
- Fanconi syndrome

Consider renal biopsy if:
- AKI stage 1 or 2
- eGFR <60 mL/min and <2 mL/min per year decline
- Albumin:creatinine 3–30 mg/mmol and glomerular filtration rate (GFR) <60 mL/min
- Evidence of light chain proteinuria

Defer renal biopsy if:
- Stable eGFR
- Normal urinalysis
- No evidence of light chain proteinuria

→

ADDITIONAL WORKUP

To confirm diagnosis of MGRS:
- Light microscopy
- Immunofluorescence staining for IgG subclasses, IgA and IgM, and kappa and lambda
  - Note: M protein detected in serum and/or urine must match the one found in the renal biopsy
- Electron microscopy
- FDG-PET/CT, low-dose CT, or whole-body MRI without contrast as clinically indicated
- Bone marrow biopsy if suspected to have MM or Waldenström macroglobulinemia/ lymphoplasmacytic lymphoma (WM/ LPL)

Additional workup as clinically indicated:
- FISH panel for myeloma and polymerase chain reaction (PCR) assay for *MYD88* L265P
- Excisional lymph node biopsy, if other B-cell lymphomas are suspected
- Peripheral blood flow cytometry for diagnosis of chronic lymphocytic leukemia (See NCCN Guidelines for Chronic Lymphocytic Leukemia/ Small Lymphocytic Lymphoma)
- Evaluate for light chain amyloidosis (See NCCN Guidelines for Systemic Light Chain Amyloidosis)

→ For management, see MGRS-2

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MONOCLONAL GAMMOPATHY OF RENAL SIGNIFICANCE

TREATMENT

- For plasma cell-related MGRS, use the management algorithm for MM (MYEL-4)
- For lymphoplasmacytic-related MGRS, see NCCN Guidelines for Waldenström Macroglobulinemia/Lymphoplasmacytic Lymphoma[a]
- For any MGRS with monoclonal B-cell lymphocytosis (MBL) features, see NCCN Guidelines for Chronic Lymphocytic Leukemia/Small Lymphocytic Lymphoma

→

RESPONSE ASSESSMENT

- For IgG- or IgA-associated MGRS, use the Response Criteria for MM (MYEL-E)
- For IgM-associated MGRS, use the response criteria for WM (See NCCN Guidelines for Waldenström Macroglobulinemia/Lymphoplasmacytic Lymphoma)
- For FLC-associated MGRS, use the response criteria for amyloidosis (See NCCN Guidelines for Systemic Light Chain Amyloidosis)
- For cases in which the causal monoclonal paraprotein is not detectable or is difficult to measure:
  - evaluate renal function
  - evaluate bone marrow involvement or radiologic findings

→ Relapse → Individualize treatment based on response and toxicity of prior therapy, patient's performance status, and renal function at the time of relapse

[a] Systemic agents associated with neurotoxicity should be used with caution.

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MONOCLONAL IMMUNOGLOBULIN DEPOSITION DISEASE

CLINICAL FINDINGS

Monoclonal Immunoglobulin Deposition Diseases (MIDD)
- Light chain deposition disease (LCDD) ~80% (of which 80%–90% are Kappa LC)
- Heavy chain deposition disease (HCDD) ~10%
- Light and heavy chain deposition disease (LHCDD) ~10%

→

WORKUP

Tests outlined on MYEL-1 to identify the plasma cell abnormality, if not previously done

Evaluate for kidney or other organ damage
- For evaluating kidney damage, see MGRS-1
- For evaluating other organ damage, see NCCN Guidelines for Systemic Light Chain Amyloidosis

Additional Evaluation as Clinically Indicated
- PCR assay for MYD88 L265P
- Excisional lymph node biopsy, if other B-cell lymphomas are suspected
- Peripheral blood flow cytometry for diagnosis of CLL/SLL or LPL (see NCCN Guidelines for Chronic Lymphocytic Leukemia/Small Lymphocytic Lymphoma and NCCN Guidelines for Waldenström Macroglobulinemia/Lymphoplasmacytic Lymphoma)

→ Single or multi-organ damage[a] →

TREATMENT

Treat if evidence of organ damage by underlying pathogenic clone:
- For plasma cell-related MIDD, see MM (MYEL-4 and MYEL-G)
- For lymphoplasmacytic-related MIDD, see NCCN Guidelines for Waldenström Macroglobulinemia/Lymphoplasmacytic Lymphoma
- For MIDD with monoclonal B-cell lymphocytosis features, see NCCN Guidelines for Chronic Lymphocytic Leukemia/Small Lymphocytic Lymphoma

→

TRANSPLANT CONSIDERATIONS

- Autologous HCT consolidation preferable in eligible patients (by clone subtype)
- If patient has deep and sustained clonal response after induction therapy (with or without autologous HCT consolidation)
  - Consider solid organ transplant to treat the affected organ/s in eligible patients
  - Candidacy for solid organ transplant is acceptable without consolidative autologous HCT if patients attained deep and durable response

[a] Organ involvement by frequency: kidney (~96%), heart (~21%), liver (~19%), peripheral nervous system (~8%).

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MONOCLONAL GAMMOPATHY OF NEUROLOGICAL SIGNIFICANCE

IgM[a] MGNS (Monoclonal gammopathy of neurological significance) suspected →

INITIAL WORKUP

- Rule out other causes of neuropathy
  - Diabetes
  - Cobalamin deficiency
  - Thyroid dysfunction
  - Lyme disease
  - HIV infection
  - Syphilis
  - Autoimmune disease
  - Cryoglobulinemia
  - Evaluate for the following as appropriate:
    - ◊ Light chain amyloidosis (See NCCN Guidelines for Systemic Light Chain Amyloidosis)
    - ◊ WM (See NCCN Guidelines for WM/LPL)
    - ◊ POEMS (POEMS-1)
- Anti-MAG antibodies[a]
- Ganglioside antibody panel
- Nerve conduction study (NCS)/electromyogram (EMG)[a]
- Neurology consult
- *MYD88*[b] L265P allele-specific PCR (AS-PCR) testing of bone marrow
- Chest/abdomen/pelvis CT with contrast when possible

Useful in certain circumstances
- Sural nerve biopsy
- *CXCR4* gene mutation testing

→

CLINICAL FINDINGS

High suspicion
- Sensory predominant
- Length dependent
- Slow progression (years)
- Bilateral and symmetrical
- Antibodies present
- Demyelination by EMG/NCS OR intermediate suspicion (not high or low suspicion) AND affecting activities of daily living (ADLs)

Low suspicion
- Motor/pain predominant
- Non-length dependent
- Rapid progression (weeks to months)
- Unilateral/asymmetrical
- Antibodies not present
- No demyelination by EMG/NCS OR intermediate/high suspicion AND not affecting ADLs

→ See NCCN Guidelines for Waldenström Macroglobulinemia/Lymphoplasmacytic Lymphoma

OR

Observation

[a] In patients presenting with suspected disease related to peripheral/axonal neuropathy, rule out amyloidosis in patients presenting with nephrotic syndrome or unexplained cardiac problems.
[b] *MYD88* wild-type occurs in <10% of patients and should not be used to exclude diagnosis of WM if other criteria are met.

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POLYNEUROPATHY, ORGANOMEGALY, ENDOCRINOPATHY, MONOCLONAL PROTEIN, SKIN CHANGES (POEMS)

POEMS suspected →

INITIAL WORKUP

- Complete H&P examination
- Evaluate for organomegaly
- Fundoscopic exam
- Hyperhidrosis
- Diarrhea
- Weight loss
- Menstrual and sexual function
- Skin examination for hyperpigmentation, hypertrichosis, acrocyanosis, glomeruloid hemangiomata, plethora, flushing, clubbing, etc
- Detailed neurologic history (numbness, pain, weakness, balance, orthostasis) and exam (sensation and motor function)

→

RECOMMENDED INITIAL TESTING

- Electrophysiologic (nerve conduction) studies
- CT chest/abdomen/pelvis to document lymphadenopathy, organomegaly, ascites, pleural effusion, edema
- Testosterone, estradiol, fasting glucose, thyroid-stimulating hormone, parathyroid hormone, prolactin, serum cortisol, luteinizing hormone
- CBC, complete metabolic panel, serum immunoglobulins (IgG, IgA, IgM), electrophoresis and immunofixation, serum FLC, 24-h urine for total protein, UPEP, and UIFE, vascular endothelial growth factor (VEGF), and interleukin 6 (IL-6)

Bone marrow aspirate and biopsy with FISH panel for myeloma

- Echocardiography to assess right ventricular systolic and pulmonary artery pressures
- CT body bone windows and or FDG-PET/CT for sclerotic bone lesions

→

ADDITIONAL TESTING AS INDICATED

- For criteria for diagnosis, see POEMS-3
- Sural nerve biopsy
- Follicle-stimulating hormone, adrenocorticotropin hormone, cosyntropin stimulation test
- Biopsy of bone lesion if needed

→ For management of POEMS syndrome, see POEMS-2

- Excisional lymph node biopsy, if Castleman disease or other B-cell lymphomas are suspected
- FISH panel for myeloma
- Evaluate for light chain amyloidosis, if appropriate (see NCCN Guidelines for Systemic Light Chain Amyloidosis)

→

DIAGNOSIS

If diagnosis is MM, follow MM algorithm

If diagnosis is WM, see NCCN Guidelines for WM/LPL

If diagnosis is Castleman disease, see NCCN Guidelines for Castleman Disease

If diagnosis is Systemic Light Chain Amyloidosis, see NCCN Guidelines for Systemic Light Chain Amyloidosis

Adapted with permission: Dispenzieri A. Am J Hematol 2019;94:812-827.

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POLYNEUROPATHY, ORGANOMEGALY, ENDOCRINOPATHY, MONOCLONAL PROTEIN, SKIN CHANGES (POEMS)

TREATMENT

- RT alone to isolated bone lesion (<3 sites) in patients without clonal BMPC
- Autologous HCT in patients who are eligible as sole therapy or as consolidation after induction therapy
  - Induction therapy options include:
    - ◊ Lenalidomide/Dexamethasone ± Daratumumab
    - ◊ Bortezomib[a,b]/Dexamethasone
    - ◊ Cyclophosphamide/Dexamethasone
    - ◊ Pomalidomide/Dexamethasone
- In patients who are not candidates for HCT, options include:
  - All regimen listed above
  - Melphalan/Dexamethasone

Useful in Certain Circumstances
- Myeloma systemic therapy options can be used in certain circumstances (MYEL-G)

RESPONSE ASSESSMENT

→ See POEMS-4 for Response Criteria → Progression → Individualize treatment based on response and toxicity of prior therapy and patient's performance status at the time of progression

[a] Bortezomib may cause exacerbation of neuropathy.
[b] Bortezomib may be replaced with carfilzomib in select patients.

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POLYNEUROPATHY, ORGANOMEGALY, ENDOCRINOPATHY, MONOCLONAL PROTEIN, SKIN CHANGES (POEMS)

Table 1. Criteria for the Diagnosis of POEMS syndrome[a]

Mandatory major criteria	1. Polyneuropathy (typical demyelinating)
	2. Monoclonal plasma cell-proliferative disorder (almost always λ)
Other major criteria (one required)	3. Castleman disease[b]
	4. Sclerotic bone lesions
	5. Vascular endothelial growth factor elevation
Minor criteria	6. Organomegaly (splenomegaly, hepatomegaly, lymphadenopathy)
	7. Extravascular volume overload (edema, pleural effusion, or ascites)
	8. Endocrinopathy (adrenal, thyroid, pituitary, gonadal, parathyroid, pancreatic)
	9. Skin changes (hyperpigmentation, hypertrichosis, glomeruloid hemangiomata, plethora, acrocyanosis, flushing, white nails)
	10. Papilledema
	11. Thrombocytosis/polycythemia[c]
Other signs and symptoms	Clubbing, weight loss, hyperhidrosis, pulmonary hypertension, restrictive lung disease, thrombotic diatheses, diarrhea, low vitamin B_{12} levels

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The diagnosis of POEMS syndrome is confirmed when both of the mandatory major criteria, one of the other three major criteria, and one of the six minor criteria are present.

[a] There is a Castleman disease variant of POEMS that occurs without evidence of a clonal plasma cell disorder that is not accounted for in this table. This entity should be considered separately.
[b] Because of the high prevalence of diabetes mellitus and thyroid abnormalities, this diagnosis alone is not sufficient to meet this minor criterion.
[c] Approximately 50% of patients will have bone marrow changes that distinguish it from a typical MGUS or myeloma bone marrow. Anemia and/or thrombocytopenia are distinctively unusual in this syndrome unless Castleman disease is present.

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POLYNEUROPATHY, ORGANOMEGALY, ENDOCRINOPATHY, MONOCLONAL PROTEIN, SKIN CHANGES (POEMS)

Table 2. Response Criteria for POEMS Syndrome

Parameter	Evaluable	Complete Response	Improvement	Progression[a]
Plasma VEGF	2x ULN	Normal[b]	50% reduction from baseline[b]	50% increase from lowest level
Hematologic	M-spike 0.5 g/dL,[c] 1.0 g/dL[d,e]	Negative serum and urine IFE and bone marrow[b]	50% reduction of M-spike from baseline[f]	25% increase from lowest level, which must be >0.5 g/dL
PET/CT	At least one lesion with FDG SUV_{max}[g]	No FDG uptake	50% reduction in sum of SUV_{max}[g]	30% increase in sum of SUV_{max}[g] from lowest level which must be at least 4 SUV_{max}[g] OR appearance of new FDG avid lesion
mNIS $+7_{POEMS}$	All patients	...	15% decrease from baseline (a minimum of 10 points)	15% increase from lowest value (a minimum of 10 points)
Ascites/effusion/edema	Present	Absent	Improved by 1 CTCAE grade from baseline	Worsened by 1 CTCAE grade from lowest grade
ECHO RVSP	≥40 mm Hg	...	<40 mm Hg	
Papilledema	Present		Absent	Worsening by 1 CTCAE grade
DLCO	<70% predicted	≥70% predicted	...	Worsening by 1 CTCAE grade

Abbreviations: CTCAE, common terminology criteria for adverse events, IFE, immunofixation electrophoresis, ECHO RVSP, echocardiogram right ventricular systolic pressure, DLCO, diffusing capacity of carbon monoxide.

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[a] Any progression event (VEGF, hematologic, or clinical will be considered progression, assuming change is attributable to disease and not an adverse event). To document progression, option exists for repeating value. If confirmed, progression date is first date of suspected progression.
[b] For VEGF, M-spike, and IFE response documentation, blood values need to be repeated for verification.
[c] For VGPR evaluable.
[d] For PR evaluable.
[e] Quantitative IgA is acceptable surrogate for M-spike for proteins migrating in the beta region.
[f] VGPR is defined as no measurable monoclonal protein on serum or urine electrophoresis, but positive IFE.
[g] By body weight.

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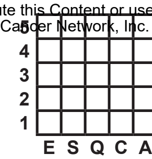

E = Efficacy of Regimen/Agent
S = Safety of Regimen/Agent
Q = Quality of Evidence
C = Consistency of Evidence
A = Affordability of Regimen/Agent

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Isatuximab-irfc/Bortezomib/Lenalidomide/Dexamethasone	
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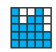
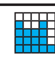
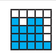
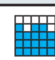
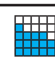
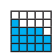
Bortezomib/Lenalidomide/Dexamethasone	
Carfilzomib/Lenalidomide/Dexamethasone	
Daratumumab/Carfilzomib/Lenalidomide/Dexamethasone	
Isatuximab-irfc/Carfilzomib/Lenalidomide/Dexamethasone	
Useful In Certain Circumstances	
Bortezomib/Cyclophosphamide/Dexamethasone	
Carfilzomib/Cyclosphosphamide/Dexamethasone	
Daratumumab/Bortezomib/Cyclophosphamide/ Dexamethasone	
Dexamethasone/Thalidomide/Cisplatin/Doxorubicin/ Cyclophosphamide/Etoposide/Bortezomib (VTD-PACE)	

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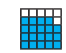
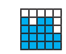
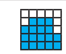
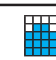
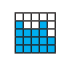
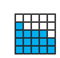
Lenalidomide	
Other Recommended Regimens	
Carfilzomib/Lenalidomide	
Daratumumab/Lenalidomide	
Useful In Certain Circumstances	
Bortezomib	
Bortezomib/Lenalidomide	
Daratumumab	
Ixazomib	

Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page EB-DEF

EVIDENCE BLOCKS FOR MULTIPLE MYELOMA THERAPY

PRIMARY THERAPY IF HCT-DEFERRED OR IF HCT IS NOT INDICATED	
Preferred Regimens	
Daratumumab/Lenalidomide/Dexamethasone	
Daratumumab/Bortezomib/Lenalidomide/Dexamethasone (for patients <80 years old who are not frail)	*
Isatuximab-irfc/Bortezomib/Lenalidomide/Dexamethasone (for patients <80 years old who are not frail)	
Other Recommended Regimen	
Carfilzomib/Lenalidomide/Dexamethasone	
Bortezomib/Lenalidomide/Dexamethasone	
Isatuximab-irfc/Lenalidomide/Dexamethasone	*
Useful In Certain Circumstances	
Lenalidomide/low-dose Dexamethasone	
Bortezomib/Dexamethasone	
Bortezomib/Cyclophosphamide/Dexamethasone	
Bortezomib/Lenalidomide/Dexamethasone (VRD-lite) for patients assessed as being frail	
Carfilzomib/Cyclophosphamide/Dexamethasone	
Daratumumab/Cyclophosphamide/Bortezomib/Dexamethasone	
Isatuximab-irfc/Carfilzomib/Lenalidomide/Dexamethasone	
Ixazomib/Lenalidomide/Dexamethasone	*
Lenalidomide/Cyclophosphamide/Dexamethasone	

*Evidence Block development in progress

Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page EB-DEF

EVIDENCE BLOCKS FOR MULTIPLE MYELOMA THERAPY

THERAPY FOR PREVIOUSLY TREATED MULTIPLE MYELOMA
Relapsed/Refractory Disease After 1–3 Prior Therapies
Preferred Regimens
Order of regimens do not indicate comparative efficacy

Anti-CD-38 Refractory	
Carfilzomib/Lenalidomide/ Dexamethasone	
Carfilzomib/Pomalidomide/ Dexamethasone	
Pomalidomide/Bortezomib/ Dexamethasone	
After two prior therapies including lenalidomide and a PI	
Elotuzumab/Pomalidomide/ Dexamethasone	
After two prior therapies including an IMiD and a PI and with disease progression on/ within 60 days of completion of last therapy	
Ixazomib/pomalidomide/ Dexamethasone	

CAR T-Cell Therapy	
After one prior line of therapy including IMiD and a PI, and refractory to lenalidomide	
Ciltacabtagene autoleucel	
After two prior lines of therapy including an IMiD, an anti-CD38 monoclonal antibody and a PI	
Idecabtagene vicleucel	

*Evidence Block development in progress

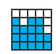
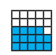
Bortezomib-Refractory *Order of regimens do not indicate comparative efficacy*	
Carfilzomib/Lenalidomide/ Dexamethasone	
Daratumumab/Carfilzomib/ Dexamethasone	
Daratumumab/Lenalidomide/ Dexamethasone	
Isatuximab-irfc/Carfilzomib/ Dexamethasone	
Carfilzomib/Pomalidomide/ Dexamethasone	
After one prior therapy including lenalidomide and a PI	
Daratumumab/Pomalidomide/ Dexamethasone	
Daratumumab/Teclistamab-cqyv	*
After two prior therapies including lenalidomide and a PI	
Isatuximab-irfc/Pomalidomide/ Dexamethasone	
Elotuzumab/Pomalidomide/ Dexamethasone	

Lenalidomide-Refractory *Order of regimens do not indicate comparative efficacy*	
Daratumumab/Bortezomib/ Dexamethasone	
Daratumumab/Carfilzomib/ Dexamethasone	
Isatuximab-irfc/Carfilzomib/ Dexamethasone	
Pomalidomide/Bortezomib/ Dexamethasone	
Carfilzomib/Pomalidomide/ Dexamethasone	
After one prior therapy including lenalidomide and a PI	
Daratumumab/Pomalidomide/ Dexamethasone	
Daratumumab/Teclistamab-cqyv	*
After two prior therapies including lenalidomide and a PI	
Isatuximab-irfc/Pomalidomide/ Dexamethasone	
Elotuzumab/Pomalidomide/ Dexamethasone	
After two prior therapies including an IMiD and a PI and with disease progression on/within 60 days of completion of last therapy	
Ixazomib/Pomalidomide/ Dexamethasone	

Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page EB-DEF

EVIDENCE BLOCKS FOR MULTIPLE MYELOMA THERAPY

THERAPY FOR PREVIOUSLY TREATED MULTIPLE MYELOMA
Other recommended therapy for Relapsed/Refractory Disease After 1–3 Prior Therapies

Elotuzumab/Lenalidomide/ Dexamethasone	
Ixazomib/Lenalidomide/ Dexamethasone	*
Selinexor/Bortezomib/ Dexamethasone	
Selinexor/Bortezomib/ Dexamethasone (once weekly)	
Bortezomib/Cyclophosphamide/ Dexamethasone	
Bortezomib/Lenalidomide/ Dexamethasone	
Carfilzomib/Cyclophosphamide/ Dexamethasone	
Daratumumab/ Cyclophosphamide/Bortezomib/ Dexamethasone	
Daratumumab/Carfilzomib/ Pomalidomide/Dexamethasone	
Elotuzumab/Bortezomib/ Dexamethasone	
Ixazomib/Cyclophosphamide/ Dexamethasone	
Lenalidomide/ Cyclophosphamide/ Dexamethasone	

After two prior therapies including an IMiD and a PI	
Belantamab mafodotin-blmf/ Bortezomib/Dexamethasone	*

After two prior therapies including an IMiD and a PI and disease progression on/within 60 days of completion of last therapy	
Pomalidomide/Cyclophosphamide/ Dexamethasone	

*Evidence Block development in progress

Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page EB-DEF

EVIDENCE BLOCKS FOR MULTIPLE MYELOMA THERAPY

THERAPY FOR PREVIOUSLY TREATED MULTIPLE MYELOMA

Useful in Certain Circumstances for

Relapsed/Refractory Disease After 1–3 Prior Therapies

Bortezomib/Dexamethasone	
Bortezomib/Liposomal Doxorubicin/ Dexamethasone	
Lenalidomide/Dexamethasone	
Carfilzomib/Cyclophosphamide/ Thalidomide/Dexamethasone	
Carfilzomib (weekly)/Dexamethasone	
Selinexor/Carfilzomib/Dexamethasone	
Selinexor/Daratumumab/ Dexamethasone	
Venetoclax/Dexamethasone only for t(11;14) patients	
Venetoclax/Dexamethasone/ Daratumumab	
Venetoclax/Dexamethasone/PI	

After two prior therapies including IMiD and a PI and with disease progression on/within 60 days of completion of last therapy	
Pomalidomide/Dexamethasone	
Selinexor/Pomalidomide/Dexamethasone	

For treatment of aggressive MM	
Dexamethasone/Cyclophosphamide/ Etoposide/Cisplatin (DCEP)	
Dexamethasone/Thalidomide/Cisplatin/ Doxorubicin/Cyclophosphamide/Etoposide (DT-PACE) ± bortezomib (VTD-PACE)	

After at least three prior therapies including a PI and an IMiD or are double-refractory to a PI and an IMiD	
Daratumumab	

Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page EB-DEF

EVIDENCE BLOCKS FOR MULTIPLE MYELOMA THERAPY

Relapsed/Refractory Disease After 3 Prior Lines of Therapy

Preferred Regimens	
CAR T-cell Therapy	
Ciltacabtagene autoleucel	
Idecabtagene vicleucel	
Bispecific Antibodies	
Elranatamab-bcmm	
Linvoseltamab-gcpt	*
Talquetamab-tgvs	
Teclistamab-cqyv	

Other Recommended Regimens	
High-dose or fractionated Cyclophosphamide	
After at least four prior therapies and whose disease is refractory to at least two PIs, at least two immunomodulatory agents, and an antiCD38 monoclonal antibody	
Selinexor/Dexamethasone	

Useful in Certain Circumstances	
Bendamustine	
Bendamustine/Bortezomib/ Dexamethasone	
Bendamustine/Carfilzomib/ Dexamethasone	
Bendamustine/Lenalidomide/ Dexamethasone	
High-dose or fractionated Cyclophosphamide	
Talquetamab-tgvs + teclistamab-cqyv	*
After at least four prior therapies, including an anti-CD38 monoclonal antibody, a PI, and an IMiD	
Belantamab mafodotin-blmf	

*Evidence Block development in progress

Note: For more information regarding the categories and definitions used for the NCCN Evidence Blocks™, see page EB-DEF

ABBREVIATIONS

ADL	activities of daily living
AKI	acute kidney injury
ANC	absolute neutrophil count
AS-PCR	allele-specific polymerase chain reaction
BCMA	B-cell maturation antigen
BMPC	bone marrow plasma cell
BNP	b-type natriuretic peptide
BUN	blood urea nitrogen
BsAb	bispecific antibody
CAR	chimeric antigen receptor
CBC	complete blood count
CLL	chronic lymphocytic leukemia
CMV	cytomegalovirus
CNS	central nervous system
CrCl	creatinine clearance
CRS	cytokine release syndromes
CSF	cerebrospinal fluid
CSI	craniospinal irradiation
CTCAE	common terminology criteria for adverse events
CTV	clinical target volume
DLCO	diffusing capacity of the lung for carbon monoxide
EBV	Epstein-Barr virus
ECHO	echocardiogram
eGFR	estimated glomerular filtration rate
EMG	electromyogram
ESA	erythropoiesis-stimulating agent
FCM	flow cytometry
FDG	fluorodeoxyglucose
FISH	fluorescence in situ hybridization
FLC	free light chain
FLCr	serum free light chain ratio
GFR	glomerular filtration rate
GTV	gross tumor volume
H&P	history and physical
HBcAb	hepatitis B core antibody
HBsAg	hepatitis B surface antigen
HCDD	heavy chain deposition disease
HCT	hematopoietic cell transplant
HIV	human immunodeficiency virus
ICANS	immune effector cell-associated neurotoxicity syndrome
IFE	immunofixation electrophoresis
Ig	immunoglobulin
IgG	immunoglobulin G
IHC	immunohistochemistry
IL-6	interleukin-6
IMiD	immunomodulatory drug
IMPEDE	IMiD, BMI, Pathologic fracture, ESA (erythropoietin stimulating agent), Dexamethasone/Doxorubicin, Ethnicity
IMRT	intensity-modulated radiation therapy
IMS	International Myeloma Society
IMWG	International Myeloma Working Group
INR	international normalized ratio
ISS	International Staging System
ISRT	involved-site radiation therapy
IT	intrathecal
IVIG	itravenous immunoglobulin
JCV	John Cunningham virus

ABBREVIATIONS

LCDD	light chain deposition disease
LDH	lactate dehydrogenase
LHCDD	light and heavy chain deposition disease
LMWH	low-molecular-weight heparin
LP	lumbar puncture
mAb	monoclonal antibody
MBL	monoclonal B-cell lymphocytosis
MFC	multicolor flow cytometry
MGNS	monoclonal gammopathy of neurological significance
MGRS	monoclonal gammopathy of renal significance
MGUS	monoclonal gammopathy of undetermined significance
MIDD	monoclonal immunoglobulin deposition disease
MM	multiple myeloma
MRD	minimal residual disease
NCS	nerve conduction study
NGF	next-generation flow
NGS	next-generation sequencing
NT-proBNP	N-terminal prohormone B-type natriuretic peptide
OAR	organs at risk
OS	overall survival
PCR	polymerase chain reaction
PCV	pneumococcal conjugate vaccine
PI	proteasome inhibitor
PJP	pneumocystis jirovecii pneumonia
POEMS	polyneuropathy, organomegaly, endocrinopathy, monoclonal protein, skin changes
PR	partial response
PTV	planning target volume
R-ISS	Revised International Staging System
RSV	respiratory syncytial virus
RSVP	right ventricular systolic pressure
SARS-CoV-2	severe acute respiratory syndrome coronavirus 2
SAVED	Surgery within 90 days, Asian race, Venous thromboembolism history, age over Eighty (80), dexamethasone
SIFE	serum immunofixation electrophoresis
SLL	small lymphocytic lymphoma
SNP	single nucleotide polymorphism
SPD	sum of product of greatest perpendicular diameters
SPEP	serum protein electrophoresis
SUV	standardized uptake value
TCR	T-cell redirecting therapy
UIFE	urine immunofixation electrophoresis
ULN	upper limit of normal
UPEP	urine protein electrophoresis
VEGF	vascular endothelial growth factor
VGPR	very good partial response
VMAT	volumetric modulated arc therapy
VTE	venous thromboembolism
WBRT	whole brain radiation therapy
WM/LPL	Waldenström macroglobulinemia/ lymphoplasmacytic lymphoma

NCCN Evidence Blocks™

NCCN Categories of Evidence and Consensus	
Category 1	Based upon high-level evidence (≥1 randomized phase 3 trials or high-quality, robust meta-analyses), there is uniform NCCN consensus (≥85% support of the Panel) that the intervention is appropriate.
Category 2A	Based upon lower-level evidence, there is uniform NCCN consensus (≥85% support of the Panel) that the intervention is appropriate.
Category 2B	Based upon lower-level evidence, there is NCCN consensus (≥50%, but <85% support of the Panel) that the intervention is appropriate.
Category 3	Based upon any level of evidence, there is major NCCN disagreement that the intervention is appropriate.

All recommendations are category 2A unless otherwise indicated.

NCCN Categories of Preference	
Preferred	Interventions that are based on superior efficacy, safety, and evidence; and, when appropriate, affordability.
Other recommended	Other interventions that may be somewhat less efficacious, more toxic, or based on less mature data; or significantly less affordable for similar outcomes.
Useful in certain circumstances	Other interventions that may be used for selected patient populations (defined with recommendation).

All recommendations are considered appropriate.

Discussion

This discussion corresponds to the NCCN Guidelines for Multiple Myeloma. Last updated: November 26, 2025.

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Overview

Multiple myeloma (MM) is a malignant neoplasm of plasma cells that typically accumulate in bone marrow, leading to bone destruction, elevated blood calcium, and anemia as well as renal damage secondary to the secreted monoclonal protein (M-protein). MM is most frequently diagnosed among people aged 65 to 74 years, with the median age being 69 years.[1] The American Cancer Society has estimated 36,100 new MM cases and an estimated 12,030 deaths in the United States in 2025.[2] Globally, age-standardized prevalence and mortality rates, as well as disability-adjusted life years have all increased between 1990 and 2021, with the number of MM cases, deaths, and disability-adjusted life-years higher in males than in females.[3]

Guidelines Update Methodology

The complete details of the development and update of the NCCN Guidelines are available at www.NCCN.org.

Literature Search Criteria

Prior to the annual update of the NCCN Clinical Practice Guidelines in Oncology (NCCN Guidelines®) for Multiple Myeloma, an electronic search of the PubMed database was performed to obtain key literature published since the previous update, using the following search terms: Smoldering Multiple Myeloma, Solitary Plasmacytoma, Multiple Myeloma, Monoclonal Gammopathy of Undetermined Significance, and POEMS syndrome. The PubMed database was chosen because it remains the most widely used resource for medical literature and indexes peer-reviewed biomedical literature.

The search results were narrowed by selecting studies in humans published in English. Results were confined to the following article types: Clinical Trial, Phase II; Clinical Trial, Phase III; Clinical Trial, Phase IV; Guideline; Practice Guideline; Randomized Controlled Trial; Meta-Analysis; Systematic Reviews; and Validation Studies. The data from key PubMed articles as well as articles from additional sources deemed as relevant to these guidelines as discussed by the Panel have been included in this version of the Discussion section. Recommendations for which high-level evidence is lacking are based on the Panel's review of lower-level evidence and expert opinion.

Sensitive/Inclusive Language

NCCN Guidelines strive to use language that advances the goals of equity, inclusion, and representation. NCCN Guidelines endeavor to use language that is person-first; not stigmatizing; anti-racist, anti-classist, anti-misogynist, anti-ageist, anti-ableist, and anti-weight-biased; and inclusive of individuals of all sexual orientations and gender identities. NCCN Guidelines incorporate non-gendered language, focusing on organ-specific recommendations. This language is accurate and inclusive and can help fully address the needs of individuals of all sexual orientations and gender identities. NCCN Guidelines will continue to use the terms men, women, female, and male when citing statistics, recommendations, or data from organizations or sources that do not use inclusive terms. Most studies do not report how sex and gender data are collected and use these terms interchangeably or inconsistently. If sources do not differentiate gender from sex assigned at birth or organs present, the information is presumed to represent cisgender individuals. NCCN encourages researchers to collect more specific data in future studies and organizations to use more inclusive and accurate language in their future analyses.

Diagnosis and Workup

It is important to distinguish MM from other plasma cell neoplasms/dyscrasias to determine prognosis and provide appropriate treatment.

The initial diagnostic workup in all patients should include a history and physical examination. To differentiate symptomatic from asymptomatic MM the following baseline laboratory studies are needed: a complete blood count (CBC) with differential; examination of peripheral blood smear; blood urea nitrogen (BUN); serum creatinine; creatinine clearance (calculated or measured directly) and serum electrolytes; liver function tests; serum calcium; serum uric acid; albumin; lactate dehydrogenase (LDH); and beta-2 microglobulin.

Peripheral smear may show abnormal distribution of red blood cells (RBCs) such as the Rouleaux formation (red cells taking on the appearance of a stack of coins) due to elevated serum proteins.[4] Smear may also show myeloma cells in the circulation. Increased BUN and creatinine indicate decreased kidney function, whereas LDH and beta-2 microglobulin levels reflect tumor burden. N-terminal prohormone of brain natriuretic peptide (NT-proBNP) is also recommended, and if NT-proBNP is not available BNP can be performed.

Serum and Urine Analysis: Serum analysis includes quantitative immunoglobulin levels (IgG, IgA, and IgM); serum protein electrophoresis (SPEP) for quantitation of M protein; and serum immunofixation electrophoresis (SIFE) to obtain more specific information about the type of M protein present. Assessing changes in levels of various proteins, particularly the M protein, helps track disease progression and response to treatment. Urine analysis as a part of the initial diagnostic workup includes evaluating 24-hour urine for total protein, urine protein electrophoresis (UPEP), and urine immunofixation electrophoresis (UIFE).

Free Light Chain Assay: The serum free light chain (FLC) assay along with serum M protein analyses (SPEP and SIFE) is an important screening tool for MM and related plasma cell disorders.[5-7] It is also helpful in prognostication of monoclonal gammopathy of undetermined significance (MGUS), smoldering myeloma, active MM, immunoglobulin light chain amyloidosis, and solitary plasmacytoma.[7,8] The serum FLC assay also allows for disease monitoring in patients with light chain amyloidosis and light chain myeloma. In addition to all of the above, the FLC ratio (FLCr) is required for documenting stringent complete response (sCR) according to the International Myeloma Working Group (IMWG) Uniform Response Criteria.[9] The serum FLC assay cannot replace the 24-hour UPEP for monitoring patients with measurable urinary M protein and can also be affected by renal function Once individuals are found to have an M protein, light chain abnormalities, or both, the same studies should be used serially. Patients with a negative UPEP do not need this study repeated, except to confirm remission or if the clinical situation changes.

Bone Marrow Evaluation: The percentage of clonal bone marrow plasma cells (BMPCs; ≥10%) is a major criterion for the diagnosis of MM. The percentage of plasma cells in bone marrow is estimated by unilateral bone marrow aspiration and biopsy. Immunohistochemistry and/or flow cytometry can be used to confirm presence of monoclonal plasma cells, and to more accurately quantify plasma cell involvement.[10] The cytoplasm of abnormal plasma cells contains either kappa or lambda light chains, and predominance of one or the other light chain expressing plasma cells indicates clonality. Specific immunophenotypic profiles of the myeloma cells may have prognostic implications.[11] Bone marrow biopsy is indicated even if the diagnosis of myeloma has been made through biopsy of a plasmacytoma since it is important to obtain cytogenetic information and should not be omitted.

Cytogenetic and Molecular Studies: Although MM may be morphologically similar, several subtypes of the disease have been identified at the genetic and molecular levels. Bone marrow studies at initial diagnosis should include chromosome analysis by fluorescence in situ hybridization (FISH) performed on the plasma cells obtained from bone marrow aspiration. Specific chromosomal abnormalities have been identified in patients with

MM involving translocations, deletions, or amplifications. According to the NCCN Multiple Myeloma Panel members, the FISH panel (performed on CD138-positive/purified cells and *not* whole bone marrow) for prognostic estimation of plasma cells should be examined for del(17p13), t(4;14), t(11;14), t(14;16), t(14:20), 1q21 gain/amplification, and 1p deletion. The utility of this information is to determine biological subtypes and for prognostic recommendations as well as candidacy for clinical trials.

Deletion of 17p13 (the locus for the tumor-suppressor gene, *p53*) leads to loss of heterozygosity of *TP53* and is considered a high-risk feature in MM.[12-14] Mutation involving the *TP53* gene is also a poor prognostic marker but this requires DNA sequencing for identification. The Panel recommends next-generation sequencing (NGS) using CD138-positive/purified cells to assess for *TP53* mutation.

Several studies have confirmed that patients with MM with t(4;14), t(14;16), and t(14;20) have a poor prognosis, while t(11;14) is believed to impart less risk.[15-18]

Abnormalities of chromosome 1 are also among the frequent chromosomal alterations in MM.[19] The short arm is most often associated with deletions and the long arm with amplifications.[20] Gains/amplification of 1q21 increases the risk of MM progression and incidence of the amplification is higher in relapsed MM compared with newly diagnosed MM (NDMM).[19,21]

Risk stratification based on chromosomal aberrations is being utilized for prognostic counseling, selection, and sequencing of therapy.[22-25]

Imaging: A skeletal survey has been the standard for assessing bone disease for any individual with suspected MM for decades.[26] However, this technique has significant limitations related to lower sensitivity compared to advanced imaging. Whole body low-dose CT (WB LD-CT) alone or CT in combination with FDG-PET has been shown to be significantly superior regarding the sensitivity to detect osteolytic lesions in patients with monoclonal plasma cell disorders. A multicenter analysis by the IMWG compared conventional skeletal survey with WB LD-CT from 212 patients with monoclonal plasma cell disorders. WB LD-CT was positive in 25.5% of patients with negative skeletal survey. The sensitivity of the skeletal survey and WB LD-CT in the long bones is not significantly different; the difference is mainly in detection of abnormalities in the spine and pelvis.[26] In a study of 29 patients, 5 (17%) showed osteolytic lesions in CT while skeletal survey results were negative.[27] Furthermore, studies have shown WB LD-CT is superior to skeletal survey radiographs in areas that are difficult to visualize with skeletal surveys such as the skull and ribs.[28]

FDG-PET/CT too has been shown to identify more lesions than plain x-rays and detect lesions in patients with negative skeletal surveys.[29-31] It is important to note that if FDG-PET/CT is chosen instead of WB LD-CT, the imaging quality of the CT part of the FDG-PET/CT should be equivalent to a WB LD-CT. Usually the CT part is used only for attenuation correction, which may not be sufficient to assess bone disease due to MM and stability of the spine. Whole-body PET/CT is also useful in detecting extramedullary disease outside of the spine.

For initial diagnostic workup of patients suspected of having MM, the NCCN Panel recommends either whole-body FDG-PET/CT (preferred) or WB LD-CT. The Panel has also noted that skeletal survey is acceptable in certain circumstances. Skeletal surveys are much less sensitive and should only be used as the sole imaging modality in circumstances such as when FDG-PET/CT or WB LD-CT is not accessible or when it is the only available option.

Additional Diagnostic Tests

The NCCN Multiple Myeloma Panel recommends additional tests that may be useful in some circumstances. MRI and FDG-PET/CT are useful for discerning smoldering myeloma from MM. Since the disease burden in patients with smoldering myeloma is lower than those with MM, imaging techniques with high sensitivity need to be used and MRI is a sensitive technique for detecting marrow infiltration by myeloma.[32,33] PET is used specifically to rule out lytic bone disease as well as extramedullary involvement. According to the NCCN Panel, if WB LD-CT or FDG-PET/ CT is negative, consider whole-body or spine MRI without contrast to discern smoldering myeloma from MM.

A tissue biopsy may also be necessary to confirm the diagnosis of suspected plasmacytomas, particularly extramedullary deposits. Also, if amyloidosis is suspected, the diagnosis is established by following the recommendations outlined in the NCCN Guidelines for Systemic Light Chain Amyloidosis (available at www.NCCN.org).

Serum viscosity should be evaluated when clinical symptoms of hyperviscosity are suspected, particularly in those with very high M-protein levels.

Hepatitis B and C testing and human immunodeficiency virus (HIV) screening should be done as required. Reactivation of hepatitis B has been reported with the use of anti-CD 38 antibodies.

The Panel also suggests baseline clonotype identification at diagnosis or storage of a bone marrow aspirate sample for clonotype identification to enable minimal residual disease (MRD) testing by NGS in the future. Risk assessment by gene expression profiling can also be considered.[34]

If abnormal renal function or non-selective proteinuria (mainly albuminuria) is found in the absence of other known causes of renal impairment, then a renal biopsy should be performed.[35]

Clinical Findings

Based on the results of the clinical and laboratory evaluation, patients are initially classified as having solitary plasmacytoma, MGUS, smoldering myeloma (asymptomatic disease), or multiple myeloma (symptomatic disease). Patients with MGUS who have organ dysfunction related to monoclonal gammopathy have been variably classified as having monoclonal gammopathy of clinical significance (MGCS). The algorithm divides MGCS into different categories based on the organ system involved: eg, monoclonal gammopathy of renal significance (MGRS) or monoclonal gammopathy of neurological significance (MGNS). Patients may also be classified as having POEMS (polyneuropathy, organomegaly, endocrinopathy, monoclonal protein, and skin changes).

Staging and Risk Stratification Systems for MM

Definitions: The IMWG definition of MM includes biomarkers in addition to requirements of CRAB features.[36] The CRAB criteria that define MM include: increased **c**alcium levels (>11 mg/dL), **r**enal insufficiency (creatinine >2 mg/dL or creatinine clearance <40 mL/min), **a**nemia (hemoglobin <10 g/dL or 2 g/dL less than normal), and presence of **b**one lesions. The IMWG has also clarified that presence of one or more osteolytic lesions seen on skeletal radiography, whole-body MRI, or whole-body FDG-PET/CT fulfills the criteria for bone disease.[36] The MM-defining biomarkers identified by the IMWG SLiM features (SLiM stands for **S**ixty, **Li**ght chain ratio, **M**RI) include one or more of the following: greater than or equal to **s**ixty percent clonal plasma cells in the bone marrow; involved/uninvolved free **l**ight chain ratio of 100 or more with the involved FLC being greater than or equal 100 mg/L; or **M**RI with more than one focal marrow (non-osteolytic) lesion.[36] All of these myeloma-defining events are referred to as SLiM-CRAB. Patients who fulfill one of these criteria are considered as having active MM and should be offered treatment.

The criteria by the IMWG for patients with smoldering (asymptomatic) MM include serum M protein (IgG or IgA) ≥30 g/L and/or clonal BMPCs 10% to 59% *and* absence of CRAB features, myeloma-defining events, or amyloidosis.[36] The updated IMWG diagnostic criteria for MM allow initiation of therapy before end-organ damage on the basis of specific biomarkers, and also allow the use of sensitive imaging criteria to diagnose MM, including whole-body FDG-PET/CT and MRI.[36] A study analyzed clinical and laboratory information from 421 patients with smoldering myeloma and identified M protein >2 g/dL, FLCr >20, and plasma cells >20% as important risk factors for progression. Patients with 2 or more of these features had a median time to progression (TTP) of 29 months.[37] Mateos et al in an analysis of 2004 patients found that the presence of t(4;14), t(14;16), 1q amp/gain, or del(13q) was a further risk factor and patients with one of these findings, along with all 3 of the 20/2/20 model, had a risk of progression of 67% at 2 years.[38]

For patients with solitary plasmacytoma, the criteria by the IMWG includes biopsy-proven solitary lesion of bone or soft tissue with evidence of clonal plasma cells.

Risk Stratification: The NCCN Panel has provided a list of factors that put patients at elevated risk of disease progression and/or relapse. For NDMM, these factors include being diagnosed as R-ISS III, having extramedullary disease,[39] having circulating plasma cells,[40] having certain cytogenetic abnormalities [eg, del(1p32), t(4;14), t(14;14), t (14;20), del(17p)/monosomy 17; 1q21 gain/1q21 amplification; MYC translocation][25]; or high-risk gene expression profile.[34]

The NCCN Panel has also listed high risk factors for those with relapsed MM. These include disease relapse within 2 years of initial therapy with hematopoietic cell transplant (HCT) and maintenance regimens or within 18 months of primary induction therapy without HCT; acquisition of 1q gain and/or del(17p)[25]; or presence of extramedullary disease at relapse.[39]

These factors are useful for counseling patients regarding prognosis and for selection and sequencing of subsequent therapy.

Staging Systems: Those with active MM can be staged using the International Staging System (ISS).[41] The ISS identifies three stages based on serum beta-2 microglobulin and serum albumin.

The revised ISS (R-ISS) includes serum beta-2 microglobulin, serum albumin, and prognostic information from the LDH and high-risk chromosomal abnormalities [t(4;14), t(14;16), 17p13 deletion] detected by FISH and is the preferred staging approach.[42] Having del(17p) and/or translocation t(4;14) and/or translocation t(14;16) are considered as high risk. Those with no high-risk chromosomal abnormality are considered standard risk. The R-ISS also identifies three stages: 1) R-ISS I, which includes ISS I with standard risk chromosomal abnormalities and normal LDH levels; 2) R-ISS III, which includes ISS III and either high-risk chromosomal abnormalities or elevated LDH levels; and 3) R-ISS II, which includes all the other possible combinations and is neither R-ISS I nor R-ISS II.

The main limitation of the R-ISS is that most of the patients were classified as R-ISS II or the intermediate-risk group; it therefore included patients with large variations in risk of progression/death.

In the second revision of the R-ISS (R2-ISS), the intermediate-risk group has been further divided into low- and high-risk groups.[24] Therefore, the R2-ISS identifies four risk groups by assigning a numerical value to each risk factor based on their influence on overall survival (OS): ISS-III is 1.5

points, ISS-II is 1 point, del(17p) is 1 point, t(4;14) is 1 point, 1q+ is 0.5 points, and serum LDH > the upper limit of normal is 1 point. The low-risk group is 0 points, low-intermediate-risk group is 0.5–1 points; intermediate high-risk group is 1.5–2.5 points, and high-risk group is 3–5 points.[24] The limitation of R2-ISS is that it has only been validated in NDMM.

The IMS/IMWG uses genomic data and serum beta-2 microglobulin to identify patients with high-risk MM. IMS/IMWG consensus staging defines high-risk MM by the presence of any one or more of the following: 1) Del(17p) and/or TP53 mutation; 2) t(4;14), t(14;16), or t(14;20) co-occurring with 1q+ and/or del(1p32); 3) Monoallelic del(1p32) co-occurring with 1q+, or biallelic del(1p32); 4) High B2M (≥5.5 mg/dL) with normal creatinine (<1.2 mg/dL).[25] MM without any high-risk criteria is classified as standard risk MM.

Solitary Plasmacytoma

The diagnosis of solitary plasmacytoma or solitary plasmacytoma with minimal marrow involvement (<10% plasma cells in bone marrow) requires a thorough evaluation with advanced imaging studies to rule out the presence of additional lesions or systemic disease, because many patients presumed to have solitary plasmacytomas are found to have additional sites.[43,36,44] FISH testing on plasmacytoma biopsy should be considered if clonal cells are absent or inadequate in the bone marrow aspirate to identify the cytogenetic abnormality.[45]

Whole-body imaging with low-dose CT or FDG-PET/CT is recommended for initial diagnostic workup of patients suspected of having MM or solitary plasmacytoma. Skeletal survey is acceptable in certain circumstances. However, skeletal survey is significantly less sensitive than whole-body low-dose CT and FDG-PET/CT in detecting osteolytic lesions in patients with monoclonal plasma cell disorders.[26]

Whole-body imaging with MRI (or FDG-PET/CT, if MRI is not available) is the first choice for initial evaluation of solitary osseous plasmacytoma, and whole-body FDG-PET/CT is the first choice for initial evaluation of solitary extraosseous plasmacytoma. The sensitivity of FDG-PET/CT for areas of increased metabolism and the high soft-tissue resolution of MRI enable both techniques to provide information on the presence or absence of plasmacytomas. While the sensitivity of both techniques for the detection of focal lesions is similar, MRI provides a higher sensitivity for a diffuse infiltration.[46,47] No data exist on the comparison of FDG-PET/CT and MRI in solitary plasmacytoma. In retrospective analyses, the risk of progression to MM within 2 years of diagnosis has been shown to be higher with osseous plasmacytoma (35%) compared with extramedullary lesions (7%).[48] This might, at least in part, be due to undetected diffuse infiltration reflecting systemic disease, which makes the superior sensitivity of MRI significant in this regard. Since the risk of progression of solitary plasmacytoma into MM or relapse is relatively high (14%–38% within the first 3 years of diagnosis), annual follow-up with the same imaging technique used at first diagnosis should be performed for the first 5 years and subsequently only in case of clinical or laboratory signs or symptoms.[49]

Primary Therapy for Solitary Plasmacytoma

The treatment and follow-up options for solitary plasmacytoma or solitary plasmacytoma with minimal marrow involvement are similar. RT has been shown to provide excellent local control of solitary plasmacytomas.[50-56] The largest retrospective study (N = 258) included patients with solitary plasmacytoma (n = 206) or extramedullary plasmacytoma (n = 52).[57] Treatments included RT alone (n = 214), RT plus chemotherapy (n = 34), and surgery alone (n = 8). Five-year OS was 74%, disease-free survival was 50%, and local control was 85%.[57]

Patients with solitary plasmacytoma (n = 206) who received localized RT (varying doses) had a lower rate of local relapse (12% in those who received between 40–50 Gy) than those who did not receive RT and were treated with surgery (80%).[56]

According to the NCCN Panel, treatment planning of solitary plasmacytomas should be performed using modern treatment principles including imaging-based delineation (MRI, CT with contrast, and/or FDG-PET/CT) of a gross tumor volume (GTV), clinical target volume (CTV), and planning target volume (PTV) and adjacent organs at risk (OARs). CTV expansions should include at least 0.5 cm of margin for microscopic extent, and up to 2 to 3 cm for involvement of long bones. PTV margins should be minimized using modern daily image guidance. Treatment of adjacent vertebral bodies for spine lesions is not required if there is no suspicion of clinical involvement. In addition, advanced technology (ie, intensity-modulated RT [IMRT], volumetric modulated arc therapy [VMAT], protons) should be used when feasible to limit radiation doses to surrounding OARs. Principles of involved-site RT (ISRT) should be used to avoid large radiation fields and inappropriately including uninvolved sites that would increase the risk of toxicity.

The dose used in most published papers ranges from 30 to 60 Gy.[55,56,58]

The recommended RT dose by the NCCN Panel is 40 to 50 Gy in 1.8 to 2.0 Gy fractions (20–25 total fractions) to the involved site. The Panel also recommends treatment with 35 to 40 Gy as an alternative for solitary plasmacytomas <5 cm in size, due to the high rates of local control reported for smaller tumors.[53,59-61] The response to RT should be assessed after at least 3 months of completion of RT.

Surveillance/Follow-up Tests for Solitary Plasmacytoma

Follow-up and surveillance tests for solitary plasmacytoma consist of blood and urine tests and imaging. Serial measurements to check for re-emergence or appearance of M protein are required to confirm disease sensitivity to RT. The recommended follow-up interval for patients with plasmacytoma after RT is every 3 to 6 months. The imaging results may not be accurate if the scans are performed sooner than 3 months following the conclusion of RT. However, patients with soft tissue and head/neck plasmacytoma could be followed less frequently after an initial 3-month follow-up. All plasmacytomas should be imaged annually, preferably with the same technique used at diagnosis, for at least 5 years.

The blood tests include CBC with differential and platelet count, serum chemistry for creatinine, and corrected calcium. Serum FLC assay, serum quantitative immunoglobulins, and SPEP with SIFE may be performed as needed.

The urine tests as needed include 24-hour urine assay for total protein, UPEP, and UIFE. Bone marrow aspirate and biopsy are recommended as clinically indicated.

If progression to MM occurs, then the patient should be re-evaluated as described in *Diagnosis and Workup*, and systemic therapy must be administered as clinically indicated.

Smoldering (Asymptomatic) Myeloma

Smoldering (asymptomatic) myeloma describes a stage of disease with no symptoms and no related organ or tissue impairment.[62] Smoldering myeloma is a precursor to MM. All patients with smoldering myeloma have a risk of progression to MM.[63] However, the rate of progression varies from months to several years based on certain risk features.[63]

The Mayo 2018 20/2/20 criteria stratify patients with smoldering MM based on risk of disease progression to MM. The criteria take into consideration the following risk factors: percentage of BMPCs >20%, M protein >2 g/dL, and FLCr g >20. Patients with 2 to 3 of the above risk factors are considered to have elevated risk.

These risk factors were developed from a retrospective study of patients with smoldering myeloma (n = 417). In those with high risk (≥2 factors present), the estimated median TTP was 29 months, in those with intermediate risk (1 factor present), the estimated median TTP was 68 months, and for those with low risk (none of the risk factors present), the estimated median TTP was 110 months.[37] The Mayo 2018 20/2/20 criteria were validated in a large retrospective analysis of 2004 patients with smoldering myeloma.[64] The estimated progression rates at 2 years among those with low-, intermediate-, and high-risk disease were 5%, 17%, and 46%, respectively.[64]

The IMWG has also confirmed the Mayo 20/2/20 model 2-year progression risk: 6% for low risk (0 risk factors; 18% for intermediate risk (1 risk factor), and 44% for high risk (2–3 factors).[38] In addition, the IMWG has also proposed a risk stratification model, building upon the above Mayo criteria to incorporates genetic abnormalities like t(4;14), t(14;16), 1q gain, or del(13q) to further risk stratification into 4-risk groups: low risk with 0, low intermediate risk with 1, intermediate risk with 2, and high risk with ≥3 risk factors. The 2-year progression risks of each of the above 4 groups are 6%, 23%, 46%, and 63%, respectively, depending on the number of risk factors present.[38]

Primary Therapy for Smoldering (Asymptomatic) Myeloma

While the historic approach for management of all smoldering myeloma has been close observation, recently there has been mounting evidence for early intervention for those with high-risk features.

Daratumumab: A phase III trial (AQUILA) of patients with high-risk smoldering myeloma (n = 390) compared the PFS between patients receiving daratumumab (n = 194) and patients undergoing active monitoring (n = 196).[65] High-risk status in this trial was defined as at least 10% marrow PCs and the presence of at least one of the following risk factors: serum M-protein level of ≥30g/L, IgA SMM, immunoparesis with reduced levels of two uninvolved immunoglobulin isotypes, a ratio of involved free light chains to uninvolved free light chains (FLC ratio) in serum of 8 to <100, or a percentage of clonal plasma cells in the bone marrow of >50% to <60%.[37,38] Three years of subcutaneous daratumumab monotherapy improved 5-year PFS compared to active monitoring (63.1% vs. 40.8%).[65] In subgroup analysis by 20/2/20 criteria, the PFS benefit was clear in patients with high-risk SMM (hazard ratio [HR], 0.36; 95% CI, 0.23–0.58) but did not reach statistical significance in those with intermediate-risk SMM (HR, 0.70; 95% CI, 0.43–1.14) or low-risk SMM (HR, 0.59; 95% CI, 0.24–1.45). The 5-year OS with daratumumab in the entire study population was 93% versus 86.9% (HR, 0.52; 95% CI, 0.27–0.98).

Lenalidomide: A relatively small, randomized, prospective, phase III study by the PETHEMA group investigated whether early treatment with lenalidomide and dexamethasone in patients (n = 119) with smoldering myeloma, at high risk of progression to active MM, prolongs the TTP.[66] The high-risk group in the study was defined using the following criteria: plasma cell bone marrow infiltration of at least 10% and/or a monoclonal component (defined as an IgG level of ≥3 g/dL, an IgA level of ≥2 g/dL, or a urinary Bence Jones protein level of >1 g per 24 hours); and at least 95% phenotypically aberrant plasma cells in the bone marrow infiltrate. The OS reported in the trial at 3 years was higher in the group treated with the lenalidomide and dexamethasone arm (94% vs. 80%; HR, 0.31; 95% CI, 0.10–0.91; P = .03).[66] At a median follow-up of 75 months (range, 27–57 months), treatment with lenalidomide and dexamethasone delayed median TTP to symptomatic disease compared to no treatment (TTP was not reached in the treatment arm compared to 23 months in the observation arm; HR, 0.24; 95% CI, 0.14–0.41).[67] The high OS rate seen after 3 years was also maintained (HR, 0.43; 95% CI, 0.20–0.90). According to the NCCN Panel, the flow cytometry-based high-risk criteria specified in the study is not uniformly available and participants did not receive advanced imaging. Based on the criteria used in the trial, some patients with active myeloma were classified as having high-risk smoldering myeloma.

In a larger, multicenter, phase III, randomized trial, patients with smoldering myeloma (n = 182) were either treated with single-agent lenalidomide or observed. The lenalidomide group experienced improved PFS) and decreased end organ damage (eg, renal failure, bone lesions) when compared with those who were observed.[68] Grade 3 or 4 adverse events were reported in 41% of patients treated with lenalidomide.[68] On subgroup analysis, the PFS benefit was seen in those with high-risk smoldering myeloma but was less clear in those with low- or intermediate-risk disease.[68]

The NCCN Panel suggests using the Mayo 2018/IMWG 20/2/20 criteria to stratify patients based on risk. According to the NCCN Panel, individuals with no high-risk criteria, should be enrolled in a clinical trial or observed at 3- to 6-month intervals (category 1). It is important to note that patients can evolve from having low-risk to high-risk SMM over time, requiring recalibration of follow-up strategies.

For the high-risk group, the NCCN Panel prefers enrollment in an ongoing clinical trial (strongly recommended and preferred option) or close observation at 3-month intervals, as clinically indicated. Those with rising markers or high-risk factors should be monitored closely. The Panel recommends intervention with systemic therapy in carefully selected patients based on thorough discussion and guided by a shared decision making process. Treatment should be with single agent daratumumab (category 1)[65] or single-agent lenalidomide (category 2B).[66,68]

Surveillance/Follow-up Tests for Smoldering (Asymptomatic) Myeloma

The surveillance/follow-up tests for smoldering myeloma include CBC with differential and platelet count; serum chemistry for creatinine, albumin, corrected calcium, serum quantitative immunoglobulins, SPEP, and SIFE; and serum FLC assay as clinically indicated. The urine tests, as needed, include 24-hour urine assay for total protein, UPEP, and UIFE.

Bone marrow aspirate and biopsy with FISH, NGS, or multiparameter flow cytometry may be used as clinically indicated.

Imaging studies with MRI without contrast, whole-body low-dose CT and/or CT, and/or whole-body FDG-PET/CT are recommended annually or as clinically indicated. The NCCN Panel recommends using the same imaging modality used during the initial workup for the follow-up assessments.

If the disease progresses to symptomatic myeloma, then patients should be treated according to the guidelines for symptomatic MM.

Active (Symptomatic) Multiple Myeloma

NDMM is typically sensitive to a variety of classes of drugs: immunomodulatory drugs (IMiDs), proteasome inhibitors (PIs), and monoclonal antibodies.

Primary Therapy for Active (Symptomatic) Multiple Myeloma

Patients presenting with active (symptomatic) myeloma are initially treated with primary therapy and followed by autologous HCT in those who are eligible.

Early referral to an HCT center to assess whether a patient is eligible for HCT is important. Stem cell toxins, such as nitrosoureas or alkylating agents, compromise stem cell reserve. If delaying HCT, then stem cells should be collected and stored. The Panel recommends harvesting peripheral blood stem cells within the first six cycles of therapy initiation prior to prolonged exposure to lenalidomide and/or daratumumab if HCT is being considered.

The algorithm has a list of primary therapy regimens recommended by the NCCN Multiple Myeloma Panel members for HCT-eligible and non-HCT candidates and lists drugs recommended for maintenance therapy in each setting. The list is selected and is not inclusive of all regimens.

The NCCN Multiple Myeloma Panel has categorized all myeloma therapy regimens as: "preferred," "other recommended," or "useful in certain circumstances." The purpose of classifying regimens as such is to convey the sense of the Panel regarding the relative efficacy and toxicity of the regimens. Factors considered by the Panel include evidence, efficacy, toxicity, pre-existing comorbidities such as renal insufficiency, and in some cases access to certain agents.

The NCCN Panel prefers 4-drug regimens as the standard for primary treatment of all patients who are HCT eligible. This is based on improved response rates, depth of response, and rates of PFS or OS seen in clinical trials. Frailty assessment should be performed in older adults, as well as consideration of dose modifications based on functional status and age.

It is also important to consider supportive care for all patients at diagnosis. For example, 80% of patients have bone disease and up to 33% have renal compromise. In all patients, careful attention to supportive care is critical to avoid early complications that may compromise therapeutic outcome.

Considerations for Special Populations

Older Adults: An assessment for frailty is considered for older adult patients. The NCCN Guidelines for Older Adult Oncology contain specific considerations by treatment type for components of frailty including comorbidities, cognition, mobility, functional status, and nutrition. Additionally, the Myeloma Frailty score calculator developed by IMWG can be referenced for the prognosis of older adult patients (see http://www.myelomafrailtyscorecalculator.net).

Disparities in specific populations: Black/African American individuals have the highest myeloma rates and twice the incidences of any race/ethnic groups.[69,70]

The biology of myeloma in these individuals can also differ. For example, Black/African American individuals are more likely to have chromosome 14 translocations such as t(11;14), t(14;16), and t(14;20), and are less likely to have high-risk mutations such as del17p and gain/amp 1q compared to white individuals. Biological differences in Black/African American individuals may lead to more severe symptoms at diagnosis such as increased anemias, higher calcium levels, worse renal dysfunction, and more extramedullary disease at presentation. Toxicity of drugs too can vary in Black/African American individuals, including hyperpigmentation of skin with use of IMIDs and higher risk for cytokine release syndromes (CRS)[71] during chimeric antigen receptor (CAR) T-cell therapy.

Duffy blood group antigen (Ag) patterns in Black/African American individuals have historically resulted in trial exclusions and unnecessary dose reductions or delays during therapy, contributing to unmet needs in underserved populations where the incidences and risks of myeloma are highest. The most common Duffy phenotype in Black/African American individuals, the Duffy-null phenotype is associated with lower absolute neutrophil counts but remains physiologically insignificant regarding myeloma treatment.[72] In a recent survey, the Duffy-null phenotype was found in approximately 67% of healthy individuals self-identifying as Black/African American.[72] Duffy-null status has been inappropriately compared with a rigid reference range that may lead to treatment disparities via dose delays/reductions.[73-75]

A decreased participation in clinical trials exists based on race, ethnicity, and age, which leads to limited understanding of whether results of the studies apply to all patients.[76] To reflect populations most affected, the NCCN Panel recommends conducting equitable and inclusive clinical trials.

General Considerations for Dosage and Administration of Commonly Used Agents:

While weekly and twice-weekly dosing schemas of bortezomib are acceptable and supported by data, weekly dosing is preferred. Twice-weekly bortezomib can be associated with neuropathy that may limit efficacy due to treatment delays or discontinuation. Therefore, Reeder et al modified the CyBorD regimen to a once-weekly schedule of bortezomib.[77] In the study, patients treated with weekly bortezomib achieved responses similar to the twice-weekly schedule (overall response rate [ORR], 93% vs. 88%; very good partial response [VGPR], 60% vs. 61%). In addition, they experienced fewer grade 3 or 4 adverse events (37%/3% vs. 48%/12%). Fewer dose reductions of bortezomib/dexamethasone were required in the modified schedule and neuropathy rates were the same in both cohorts, even though the total bortezomib dose per cycle was higher in the weekly versus the twice-weekly schedule (6.0 vs. 5.2 mg/m^2).[77]

Subcutaneous administration is the preferred route for bortezomib based on the results of the MMY-3021 trial. The trial randomized patients (n = 222) to single-agent bortezomib administered either by the conventional IV route or by subcutaneous route.[78] The findings from the study demonstrate noninferior efficacy with subcutaneous versus IV bortezomib with regard to the primary endpoint (ORR after 4 cycles of single-agent bortezomib). The results showed no significant differences in terms of PFS or 1-year OS between groups.[78,79] However, patients receiving bortezomib subcutaneously had a significant reduction in peripheral neuropathy. Ixazomib may be substituted for bortezomib in select patients.

Carfilzomib can potentially cause cardiac, renal, and pulmonary toxicities. Careful assessment before initiating treatment with carfilzomib and close monitoring during treatment is recommended. Regarding dosing and administration, the Panel notes that carfilzomib may be used once or twice weekly and at different doses.

A randomized trial has compared two formulations of daratumumab as monotherapy. The subcutaneous formulation of daratumumab and hyaluronidase-fihj resulted in a similar ORR, PFS, and safety profile and fewer infusion-related reactions compared with IV daratumumab.[80] According to the NCCN Panel, daratumumab IV infusion or daratumumab and hyaluronidase-fihj, subcutaneous injection may be used in all daratumumab-containing regimens. Some patients may not be appropriate for subcutaneous treatment (eg, those with significant thrombocytopenia).

Prolonged use of steroids can be detrimental in patients >60 years of age who are frail. Therefore, the NCCN Panel recommends a tailored approach of reducing the dose to 20 mg weekly and discontinuing with either treatment response plateau or toxicity.[81,82]

By binding to CD38 expressed on the surface of RBCs, daratumumab and isatuximab-irfc may result in false-positive indirect antiglobulin test (indirect Coombs test). The binding of these antibodies to RBCs masks detection of antibodies to minor antigens in the patient's serum and interferes with serologic testing. Therefore, type and screen should be performed before using daratumumab or isatuximab-irfc.

Treatment with IgG kappa monoclonal antibodies can produce spurious positive immunofixation or M protein determinations. However, this is clinically relevant for patients with IgG kappa MM because the therapeutic monoclonal antibody band can be mistaken for the original M-protein, leading to a potential misinterpretation of immunofixation results as positive. In such instances, special interference testing or mass spectrometry-based assessment may be used to differentiate between the two

Bone disease, renal dysfunction, and other complications such as infections, hypercalcemia, hyperviscosity, and coagulation/thrombosis should be treated with appropriate adjunctive measures (see *Supportive Care for Multiple Myeloma* in this Discussion).

Preferred Primary Therapy Regimens for HCT Candidates

The preferred primary therapy options for patients who are HCT eligible include the following 4-drug regimens: daratumumab/bortezomib/lenalidomide/dexamethasone (Dara-VRd) and isatuximab/bortezomib/lenalidomide/dexamethasone (Isa-VRd).

Daratumumab/Bortezomib/Lenalidomide/Dexamethasone

The Phase 3, PERSEUS, trial evaluated the efficacy and safety of adding daratumumab to bortezomib/lenalidomide/dexamethasone (dara-VRd) in HCT-eligible patients (n = 709) with NDMM and demonstrated a clear benefit of adding the monoclonal antibody.[83] Patients were randomized to receive treatment induction with four cycles of dara-VRd or bortezomib/lenalidomide/dexamethasone (VRd), followed by autologous HCT, two cycles of consolidation with the same regimen as induction and then maintenance with lenalidomide in the VRd group and daratumumab plus lenalidomide as maintenance therapy for patients in the dara-VRd group. The addition of daratumumab to VRd backbone during induction, consolidation, and to lenalidomide maintenance significantly improved progression free survival (PFS). The median PFS was not reached in either arm, at 47.5 months, the PFS rates were 84.3% (95% CI, 79.5–88.1) in the Dara-VRd group compared to 67.7% (95% CI, 62.2–72.6) in the VRd group, with a HR of 0.42 (95% CI, 0.30–0.59; P < .001) indicating a 58% reduction in the risk of progression or death.[83] The dara-VRd group also demonstrated a higher rate of complete response (CR) or better (87.9% vs. 70.1%; P < .001).[83] A substantial proportion of patients in the dara-VRd group achieved MRD by NGS at the $<10^{-5}$ threshold (75.2% vs. 47.5%; P < .001), achieving deeper responses and potentially longer duration of remission. After at least 24 months of maintenance therapy, daratumumab maintenance therapy was discontinued in patients who had a CR or better and had sustained MRD–negative status for at least 12 months. These patients continued to receive maintenance lenalidomide until disease progression or unacceptable toxicity. The percentage of patients with MRD-negative status for at least 12 months was higher in the dara-VRd group compared with the VRd group (64.8% vs. 29.7%).[83] The percentage of patients with MRD-negative status assessed at a sensitivity threshold of 10^{-6} was 65.1% in the dara-VRd group and 32.2% in the VRd group.

The safety profile of dara-VRd was consistent with known profiles of the individual agents. Common grade 3 or 4 adverse events observed more frequently in the daratumumab arm compared to the control arm included neutropenia (62.1% vs. 51%) and thrombocytopenia (29.1% and 17.3%). Serious adverse events occurred in 57% who were treated with the Dara-VRd regimen compared with 49.3% treated with VRd.[83]

The final analysis of the GRIFFIN trial also supports the clinical benefit of adding daratumumab to VRd, demonstrating a significant improvement in depth of response (stringent CR of 67% vs. 48% with VRd) and a 4-year PFS rate of 87.2% versus 70% with VRd (HR, 0.45; 95% CI, 0.21–0.95).[84]

Lastly, a subgroup analysis comparing dara-VRD to VRD from a systematic review and meta-analysis of daratumumab-based quadruplet versus triplet induction regimens in HCT-eligible patients (n = 3327) with NDMM from three randomized controlled trials and one non-randomized controlled study, further confirmed significant improvements in both OS (pooled HR, 0.68; 95% CI, 0.48–0.97; *P* = .03) and PFS (pooled HR, 0.41; 95% CI, 0.31–0.54; *P* < .0000).[85]

Isatuximab/Bortezomib/Lenalidomide/Dexamethasone

In the phase 3 GMMG-HD7 trial, patients with NDMM (n = 660) eligible for autologous HCT were randomly assigned to receive three 42-day cycles of VRd induction therapy with or without isatuximab. An initial analysis found that Isa-VRd led to deeper responses and higher rates of MRD negativity (NGF x 10^{-5}) after induction (50% vs. 36%; overall response [OR], 1.82; 95% CI, 1.33–2.48). Addition of isatuximab resulted in higher incidences of grade 3 or 4 neutropenia (23% vs. 7%) and grade 3 or 4 infections (12% vs. 10%). There was one treatment-related death reported in the isatuximab group (septic shock) and four in the VRd group (one cardiac decompensation, one hepatic and renal failure, one cardiac arrest, and one drug-induced enteritis).[86] The NGF MRD negative rates continued to deepen, and after HCT, they were 66% with Isa-VRd versus 48% with VRd.[87] Importantly, at a median follow-up of 4 years from the first random assignment, the data shows that 18-week induction with Isa-VRd therapy significantly prolonged PFS compared with VRd, regardless of maintenance therapy (HR, 0.70; [95% CI, 0.52–0.95]; *P* = .0184), with PFS not reached yet in either arm.[87]

Other Recommended Primary Therapy Regimens for Newly Diagnosed HCT Candidates

Bortezomib/Lenalidomide/Dexamethasone

Phase II and III study results have shown that primary therapy with bortezomib/lenalidomide/dexamethasone is active and well tolerated in patients with NDMM, HCT eligible as well as HCT ineligible.[88,89,90]

Bortezomib/lenalidomide/dexamethasone (VRd) was compared to lenalidomide/dexamethasone (Rd) in the multicenter phase III SWOG S0777 trial.[91] Patients (n = 525) with previously untreated MM were randomly assigned to receive 6 months of induction therapy with either VRd (N = 264) or Rd (N = 261), each followed by maintenance therapy with Rd until progression or unacceptable. The triple-drug regimen group had longer PFS (43 vs. 30 months; HR, 0.712; 96% CI, 0.56–0.906) and improved median OS (75 vs. 64 months; HR, 0.709; 95% CI, 0.524–0.959).[91] As expected, ≥ grade 3 neuropathy was more frequent in the bortezomib-containing arm (24% vs. 5%; *P* < .0001) as bortezomib was administered twice weekly and intravenously in this study.[91]

With longer-term follow-up (median, 84 months), the benefits of adding bortezomib to lenalidomide and dexamethasone were seen to be maintained.[92] The PFS with VRd was 41 versus 29 months for Rd.[92] The OS was not yet reached (>84 months) with the bortezomib regimen versus 69 months for Rd.[92]

In order to minimize the toxicities seen with the standard dose of VRd, a phase II study evaluated the efficacy of dose-adjusted regimen, VRd-lite.[93] The VRd-lite regimen included subcutaneous bortezomib (1.3 mg/m^2) on days 1, 8, 15, and 22, and oral dexamethasone (20 mg) on the day of and the day after bortezomib administration. Lenalidomide was omitted on days 1, 8, and 15, which are the days of bortezomib administration. The ORR after 4 cycles of VRd-lite was 83%, including a

CR of 25%. The ORR and VGPR or better were further improved to 100% and 74%, among those who received autologous HCT.[93]

Based on the above results, the NCCN Panel included bortezomib/lenalidomide/dexamethasone as a category 1, other recommended option for primary treatment of HCT-eligible patients with MM.

Carfilzomib/Lenalidomide/Dexamethasone
Multiple trials have evaluated KRd in NDMM. A multicenter phase I/II trial evaluated the combination of KRd in NDMM.[94] In this trial, patients (n = 53) received carfilzomib with lenalidomide and low-dose dexamethasone. After 4 cycles, hematopoietic cells were collected from eligible patients.[94] Out of 35 patients from whom hematopoietic cells were collected, 7 proceeded to transplantation, and the remainder continued with KRd.[94] With a median follow-up of 13 months, 24-month PFS was estimated at 92%.The most common grade 3 and 4 toxicities in ≥10% of patients included hypophosphatemia (25%), hyperglycemia (23%), anemia (21%), thrombocytopenia (17%), and neutropenia (17%). Peripheral neuropathy was limited to grade 1/2 (23%).[94]

Another phase II trial also evaluated the same regimen (KRd) in patients with NDMM (n = 45). After 8 cycles of treatment, patients with at least stable disease (SD) received up to 24 cycles of lenalidomide 10 mg/day on days 1 to 21. Thirty-eight patients were evaluable for response and toxicity.[95] Thirty-eight patients were evaluable for response and toxicity. After a median follow-up of 10 months, PFS was 83.3%. Twenty-five patients completed eight cycles of the carfilzomib, lenalidomide, and dexamethasone regimen, of which 24 continued to lenalidomide therapy and one patient opted to exit the study after initial therapy. The most common non-hematologic and hematologic toxicities (≥ grade 3) in >10% of patients included electrolyte disturbances (18%), liver function test elevation (13%), rash/pruritus (11%), fatigue (11%), lymphopenia (63%), anemia (16%), leukopenia (13%), and thrombocytopenia (11%).[95] This study also assessed MRD. The PFS was found to be longer in patients with negative MRD.[96]

A separate phase II study examined patients (n = 76) with NDMM who received KRd primary therapy followed by HCT. The primary endpoint of sCR after 8 cycles of therapy was met, with a 60% sCR rate in the overall population. In a median follow-up of 56 months, median PFS and OS were not reached. The estimated 5-year PFS rate was 72%, and the estimated 5-year OS rate was 84% in the intention to treat population. Adverse events of grade 3 to 4 included neutropenia (34%), lymphopenia (32%), infection (22%), and cardiac events (3%).[97]
A randomized multicenter phase III trial (ENDURANCE E1A11) studied patients with NDMM (n = 1053) with MM treated with either VRd or carfilzomib/lenalidomide/dexamethasone (KRd) as induction therapy. Patients with high-risk features [except for patients with t(4;14)] were not included in this trial. After a median follow-up of 9 months, median PFS was 34.4 months with the bortezomib-regimen versus 34.6 months with the carfilzomib regimen.[98] A response of VGPR or better was seen in 65% of patients treated with VRd and 74% of patients treated with VRd (*P* = .0015). With respect to adverse events, the carfilzomib regimen was associated with less peripheral neuropathy but more cardiac, pulmonary, and renal toxicities.[98] The results of the phase III ENDURANCE trial[98] showed similar PFS with KRd versus VRd. However, patients deemed as high risk were not included. KRd was associated with less neuropathy but more dyspnea, hypertension, heart failure, and acute kidney injury compared with VRd.[98] Based on the data from the above studies, the NCCN Panel has included the KRd regimen as other recommended option for primary treatment of HCT-eligible patients with MM.

The NCCN Panel has included in a footnote that ixazomib may be substituted for carfilzomib in select patients receiving the KRd regimen especially in situations where the convenience of an oral agent is preferred by the patient. The data for this comes from the phase III TOURMALINE-MM2 trial, which evaluated ixazomib/lenalidomide/dexamethasone versus lenalidomide/dexamethasone in patients not eligible for HCT.[99] Higher rates of CR were reported with the addition of ixazomib to lenalidomide/dexamethasone (26% vs. 14%; OR, 2.10; $P < .001$). The median TTP was longer in the ixazomib arm (45.8 months vs. 26.8 months; HR, 0.738).[99] The primary endpoint was PFS, and median PFS was increased by 13.5 months with the addition of ixazomib (35.3 months vs. 21.8 months; HR, 0.830; $P = .073$).[99] This trial, however, did not meet its pre-specified primary endpoint of improved PFS as the data did not meet the threshold for statistical significance.[100]

Daratumumab/Carfilzomib/Lenalidomide/Dexamethasone

A non-randomized clinical trial (n = 41) examined adding daratumumab to the 3-drug combination regimen of carfilzomib/lenalidomide/dexamethasone (dara-KRd) without HCT. The primary endpoint of MRD rate was achieved in 71% of patients, with a PFS rate of 98% and an OS rate of 100% at a median of 11 months follow-up. The most common grade 3/4 adverse events included neutropenia (27%), rash (9%), lung infection (7%), and increased alanine aminotransferase levels (4%).[101] Extended primary therapy with 24 cycles of dara-KRd without HCT led to a 75% rate of sCR and/or MRD < 10^{-5} following 8 cycles. [102] The 3-year PFS was 85% and 3-year OS was 95%.[102]

The phase II MASTER trial studied dara-KRd in NDMM and used MRD status by NGS to determine whether additional consolidation with the induction regimen after autologous HCT is needed, depending on whether the subject had two consecutive MRD-negative results at a level of 10^{-5}. The most common serious adverse events included pneumonia and venous thromboembolism.[103] After a longer follow-up (median duration of 42.2 months), of the 123 patients in the trial, MRD was evaluable by NGS in 96%. Of these, MRD of less than of less than 10^{-5} was seen in 81% with no high-risk chromosomal abnormalities (HRCA), 86% of patients with one HRCA, and 79%, of patients with two or more HRCA. The 36-month PFS was 88% for patients with no HRCA, 79% for those with one HRCA, and 50% for those with two or more HRCAs and OS was 94% for patients with no HRCA, 92% for those with one HRCA, and 75% for those with two or more HRCA.[104]

Another phase II trial evaluated dara-KRd in combination with tandem HCT for patients with high-risk, NDMM. High-risk was defined by the presence of del(17p), t(4;14), and/or t(14;16). The ORR was 100% for patients completing a second HCT, including 81% complete response. MRD negativity rate (10^{-6}) was 94% before maintenance. After a median follow-up of 33 months, the PFS and OS were 80% and 91%, respectively.[105]

The NCCN Panel has included daratumumab/carfilzomib/lenalidomide/dexamethasone as a primary therapy option for HCT-eligible patients with MM under the category other recommended (category 2A) based on the above data.

Isatuximab/carfilzomib/lenalidomide/dexamethasone (Isa-KRd)

The phase III IsKia trial assessed the efficacy and safety of isatuximab-carfilzomib-lenalidomide-dexamethasone (Isa-KRd) compared to KRd in HCT-eligible patients with NDMM. The trial found that Isa-KRd significantly increased MRD negativity rates after induction and consolidation phases, with 77% of Isa-KRd patients achieving MRD negativity by NGS at the $<10^{-5}$ threshold after consolidation, compared to 67% for KRd. Isa-KRd also showed higher MRD negativity rates across all subgroups, including patients with high-risk MM.[106]

The GMMG-CONCEPT trial evaluated the combination of Isa-KRd for the treatment of patients with high-risk NDMM (n =125) who were HCT-eligible and HCT-ineligible. High-risk disease was defined by ISS Stage II or III and specific cytogenetic abnormalities such as del17p, t(4;14), t(14;16), or more than three copies of 1q21. The enrolled patients received induction therapy followed by consolidation with Isa-KRd plus maintenance therapy with Isa-KR. The patients eligible for HCT received high-dose melphalan followed by HCT, while those not eligible for HCT received two additional cycles of Isa-KRd after initial treatment. The primary endpoint was MRD negativity rate of $<10^{-5}$ by NGF after consolidation and the secondary endpoint was PFS. After consolidation, the MRD negativity rate was 67.7% in patients who were HCT eligible and 54.2% in patients not eligible for HCT. In total, 81.8% of patients who were HCT-eligible achieved MRD negativity, with 62.6% sustaining it for at least one year. The median PFS had not yet been reached in either group at the time of publication of the study.[107]

The MIDAS trial evaluated a MRD-driven consolidation and maintenance strategy after Isa-KRD in HCT-eligible patients with NDMM. Those with MRD-negative at 10^{-5} sensitivity by NGS after Isa-KRD did not derive additional benefit from autologous HCT compared with continued therapy using Isa-KRd.[108]

Regimens Useful In Certain Circumstances for HCT Candidates

Bortezomib/Cyclophosphamide/Dexamethasone

Data from three phase II studies involving NDMM have demonstrated high response rates with cyclophosphamide, bortezomib, and dexamethasone (CyBorD) as primary treatment.[90,109,110] The trial by Reeder et al carried out in the United States and Canada demonstrated an ORR of 88%, including a VGPR or greater of 61% and 39% CR/near CR with CyBorD as the primary regimen.[109] The depth of response seen after primary treatment was maintained after HCT in those who underwent transplantation (70% rates of CR/near CR; rate of at least VGPR or better was 74%).[109] According to the long-term follow-up analysis, the 5-year PFS and OS rates were 42% (95% CI, 31–57) and 70% (95% CI, 59–82).[111]

Analysis of the German DSMM XIa study also demonstrated high responses with CyBorD as primary treatment (ORR was 84%, with a 71.5% PR rate and 12.5% CR rate). High response rates were seen in patients with unfavorable cytogenetics.[110]

In the updated results of the phase II EVOLUTION study, primary treatment with CyBorD demonstrated an ORR of 75% (22% CR and 41% ≥ VGPR), and the 1-year PFS rate was 93%.[90]

Based on data from these and other phase II studies, the NCCN Multiple Myeloma Panel has now included the combination of bortezomib/cyclophosphamide/dexamethasone to the list of primary treatment available for HCT candidates. This is designated as a regimen that may be useful in certain circumstances especially in patients with acute renal insufficiency who do not have access to daratumumab. According to the NCCN Panel, one can consider switching to dara-VRd after renal function improves.

Carfilzomib/Cyclophosphamide/Dexamethasone

The carfilzomib/cyclophosphamide/dexamethasone regimen has been studied in phase I/II trials of transplant-ineligible patients with NDMM. Trials have investigated both once-weekly and twice-weekly carfilzomib dosing combined with fixed-dose cyclophosphamide and dexamethasone.[112,113] A pooled analysis of two phase I and II studies comparing two alternative schedules of carfilzomib, HCT-ineligible n patients with NDMM showed similar response rates in those treated with once-weekly carfilzomib at a dose of 70 mg/m^2 compared to those treated with twice-weekly carfilzomib at a dose of 36 mg/m^2. The PFS and OS were also similar. The median PFS was 35.7 months in the once-weekly group and 35.5 months in the twice-weekly group (HR, 1.39; *P* = .26). The 3-year OS was 70% and 72%, respectively (HR, 1.27; *P* = .5).[114]

Consistent with the above results, a phase Ib study, CHAMPION-2, evaluated the safety and tolerability of twice-weekly carfilzomib (three different doses) in combination with cyclophosphamide and dexamethasone for the treatment of NDMM. This study found that 56 mg/m^2 carfilzomib combined with weekly cyclophosphamide and dexamethasone was effective and with manageable toxicity.[115]

The NCCN Panel has included carfilzomib/cyclophosphamide/dexamethasone for both HCT and non-HCT settings as an option useful in certain circumstances such as those with renal insufficiency, peripheral neuropathy, and/or inability to take lenalidomide. The evidence for this comes from a phase I/II study for HCT-eligible candidates that evaluated the safety and efficacy of the all-oral combination of ixazomib with lenalidomide and dexamethasone in patients with NDMM treated with combination lenalidomide and dexamethasone.[116] Both tolerability and activity of this regimen in older patients (those ≥65 years of age) was similar to that in younger patients in this study.

Daratumumab/Bortezomib/Cyclophosphamide/Dexamethasone

Patients with NDMM (*n* = 87) and patients with relapsed MM (*n* = 14) received daratumumab/bortezomib/cyclophosphamide/dexamethasone.[117] In NDMM, after 4 cycles of induction therapy, VGPR or better was seen in 44.2% and the ORR observed was 79.1%.[117] The median PFS was not reached and the 12-month PFS rate was 87%. At the time of clinical cut-off, the 12-month OS rate was 98.8% (95% CI, 92.0%–99.8%).[117] Efficacy was also observed in patients with relapsed MM.

Based on the above results, NCCN Panel has included daratumumab/bortezomib/cyclophosphamide/dexamethasone for NDMM (HCT eligible and ineligible patients) as an option under useful in certain circumstances such as for those with acute renal insufficiency or who do not have access to dara-VRd. According to the NCCN Panel, one can consider switching to dara-VRd after renal function improves.

Bortezomib/Dexamethasone/Thalidomide/Cisplatin/Doxorubicin/Cyclophosphamide/Etoposide

The Total Therapy 3 (TT3) trial evaluated induction therapy with the multi-agent regimen, VTD-PACE (bortezomib, dexamethasone, thalidomide, cisplatin, doxorubicin, cyclophosphamide, and etoposide), prior to high-dose melphalan-based tandem auto-transplants and later as consolidation therapy.[118] This regimen is a potent combination of newer agents as well as traditional chemotherapy agents.

This regimen is listed under the category useful in certain circumstances. According to the NCCN Panel, VTD-PACE could be an option for NDMM with high-risk and aggressive extramedullary disease or plasma cell leukemia.

Preferred Primary Therapy Regimens if HCT is Deferred or if HCT is Not Indicated

Many of the regimen described above for HCT candidates are also options for non-HCT candidates. In general, in this setting, primary therapy is continued until progression with de-escalation of therapy (modification of dose and duration) as needed.

As in HCT-eligible patients, 4-drug regimens are preferred by the NCCN Panel as these regimens have been shown to induce higher response rates and depth of response in clinical trials. Based on functional status, patients may be started with a 2- or 3-drug regimen with additional drugs added as performance status improves.

Daratumumab/Lenalidomide/Dexamethasone

In HCT-ineligible patients (n = 737) with NDMM, a phase III trial (MAIA) compared daratumumab/lenalidomide/dexamethasone to lenalidomide/dexamethasone. The addition of daratumumab to lenalidomide/dexamethasone resulted in median PFS not being reached at a median follow-up of 56.2 months compared to a median PFS of 34.4 months in the control group (HR, 0.68; P < .0001). The rates of several adverse events were higher in the daratumumab group compared to the control group, including neutropenia (54% vs. 37%), pneumonia (19% vs. 11%), and lymphopenia (16% vs. 11%). Treatment-related deaths occurred in 4% of patients in the daratumumab group and 3% of patients in the control group. Based on the results of this study the FDA has approved the use of daratumumab/lenalidomide/dexamethasone in this setting.[119]

Results from an updated analysis of the efficacy and safety in MAIA at a median follow-up of 64.5 months reported median PFS with D-Rd of 61.9 months versus 34.4 months with Rd (HR, 0.55; 95% CI, 0.45–0.67; P < .0001). D-Rd arm also demonstrated higher rates of CR or better (≥CR; 51.1% vs. 30.1%), MRD negativity (32.1% vs. 11.1%), and sustained MRD negativity (≥18 months: 16.8% vs. 3.3%) versus Rd (all P < .0001). Median OS was not reached in the D-Rd group versus 65.5 months in the Rd group (HR, 0.66; 95% CI, 0.53–0.83; P = .0003). No new safety concerns were reported.[120]

The NCCN Panel has also included daratumumab/lenalidomide/dexamethasone as a category 1, preferred option for patients with NDMM who are HCT ineligible.

Of note, the phase III IFM2017-03 trial data has demonstrated that a "dexamethasone-sparing" regimen (daratumumab plus lenalidomide, with 2 cycles of dexamethasone) significantly improved outcomes in older, frail patients with NDMM compared to the standard lenalidomide plus continuous dexamethasone regimen.[121] The median PFS was significantly longer in the DR group at 53.4 versus 22.5 months in the Rd group (HR, 0.51; P < .0001). The median OS was also longer in the DR group (not reached) compared to 47.2 months in the Rd group (HR, 0.52; P = .0001).[121]

Daratumumab/Bortezomib/Lenalidomide/Dexamethasone (for patients <80 years of age who are not frail)

The phase III CEPHEUS study compared triplet bortezomib/lenalidomide/dexamethasone (VRd) with daratumumab-based triplet (Dara-VRd) in patients who were HCT ineligible or not planning to receive HCT (deferred).[122] The population of patients in this trial excluded patients assessed as frail and patients above 80 years.

Three hundred and ninety-five patients who were not planning to proceed to HCT were randomized to be treated with either Dara-VRd or VRd, both with the same VRd doses used in the SWOG-S0777 trial for 8 cycles. The primary endpoint was overall minimal residual disease (MRD)-negativity rate at 10^{-5} by NGS. Following induction subjects randomized to Dara-VRD proceeded to indefinite maintenance with Dara-R, while the control group received R alone. A median follow-up of 58.7 months revealed the improvement of treatment with addition of daratumumab into the VRd regimen. The MRD-negativity rate was 60.9% with dara-VRd versus 39.4% with VRd . Dara-VRd showed superior PFS compared with VRd, with HR value of 0.57 (95% CI, 0.41–0.79; P = .0005).[122] Importantly, median PFS data was not fully achieved for Dara-VRd whereas the median PFS for VRd was 52.6 months. The median PFS (54-month estimates) were 68.1% for dara-VRd (95% CI, 60.8–74.3) and 49.5% for VRd (95% CI, 41.8–56.8).

The NCCN Panel has also included dara-VRd as a category 1, preferred option for patients <80 years old who are not frail).

Isatuximab-irfc/Bortezomib/Lenalidomide/Dexamethasone (for patients <80 years old who are not frail) (category 1)

The registrational, phase III, open label IMROZ trial randomized patients (n = 446) ≤80 years with NDMM who were not eligible for ASCT, in four 6-week cycles to receive either Isatuximab-VRd or VRd, followed by continuous treatment in 4-week cycles of IsaRd versus Rd, respectively. At a median follow-up of 59.7 months, the estimated PFS with isatuximab-irfc/bortezomib/lenalidomide/dexamethasone was 63.2% compared with 45.2% in the VRd group with a 40% reduction in risk of disease progression or death (HR, 0.60; 95% CI, 0.44, 0.81; P = .0009). Complete response or better was significantly higher in the isatuximab-group (74.7% vs. 64.1%; P = .01) and the MRD-negative status ($<10^{-5}$ by NGS) and CR was also higher in the isatuximab-group (55.5% vs. 40.9%; P = .003).[123] It is of note that patients ≥ 65 years of age were deemed ineligible for HCT, and those ≥ 80 years of age were ineligible for the study. The incidence of peripheral sensory neuropathy was not higher with isatuximab-VRd than with VRd in this trial (all grade 54.4% vs. 60.8%, respectively), this high rate reflects the use of twice weekly bortezomib dosing.

The non-registrational phase III, BENEFIT/IFM2020-05 study demonstrated the efficacy and safety of isatuximab with weekly VRd compared with the triplet combination isatuximab with lenalidomide/dexamethasone in patients with NDMM who are not eligible for HCT.[124] The NGS MRD negativity rate at 10^{-5} at 18 months was significantly higher in the quadruplet arm, with 71 patients (53%; 95% CI, 44–61) compared with 35 patients (26%, 95% CI, 19–34) in the triplet arm. The odds ratio (OR) for MRD negativity in the quadruplet group compared with the triplet group was 3.16 (95% CI, 1.89–5.28; $P < .0001$)).[124] Bortezomib subcutaneously weekly (cycles 1–12) and bimonthly (cycles 13–18) schedule in the Isa-VRd arm led to higher rate of peripheral neuropathy: grade ≥2 occurred in 37 (27%) patients (4 patients had grade 3 neuropathy) in the Isa-VRd arm versus 13 (10%) patients (1 patient had grade 3 neuropathy) in the Isa-Rd arm, but as expected lower than with twice weekly dosing.

The NCCN Panel has also included Isa-VRd as a category 1, preferred option for patients <80 years of age who are not frail) and prefer weekly bortezomib dosing to minimize peripheral neuropathy.

Other Recommended Primary Therapy Regimens if HCT is Deferred or if HCT is Not Indicated

Bortezomib/Lenalidomide/Dexamethasone

Phase II study results (discussed in the HCT setting) have shown that primary therapy with bortezomib/lenalidomide/dexamethasone is active and well tolerated in all patients with NDMM regardless of autologous HCT status.[88]

The randomized phase III SWOG S0777 trial (discussed in the HCT setting), comparing bortezomib/lenalidomide/dexamethasone to lenalidomide/dexamethasone as induction therapy without an intent of immediate transplantation, reported superior results with the 3-drug regimen.[91,92]

The NCCN Panel included the bortezomib/lenalidomide/dexamethasone regimen as a category 1, other recommended option for patients with MM not eligible for HCT.

Carfilzomib/Lenalidomide/Dexamethasone

The results of a phase I/II trial demonstrated that the combination of carfilzomib/lenalidomide/dexamethasone (KRd) is well tolerated and is also effective in all patients with NMDD.[94] An updated follow-up analysis of the subset of 23 older patients (aged ≥65 years) showed that use of the carfilzomib, lenalidomide, and low-dose dexamethasone regimen for an extended period of time resulted in deep and durable responses. All patients achieved at least a PR. With a median follow-up of 30.5 months, the reported PFS rate was 79.6% (95% CI, 53.5–92.0) and OS was 100%.[125]

The phase II trial by Korde et al[96] also showed that treatment with the KRd regimen results in high rates of deep remission. The results were very similar across age groups, with the oldest patient on the trial being 88 years of age, and the regimen was found to be effective in individuals with high-risk disease.[96]

Based on the above phase II studies that did not exclude HCT-ineligible patients, the NCCN Panel has included carfilzomib/ lenalidomide/dexamethasone as an option for treatment of all patients with NDMM, including those who are not eligible for HCT.

The Panel has noted in a footnote that ixazomib may be substituted for carfilzomib in select patients. This is based on the data from a multicenter, phase II trial investigated the efficacy and toxicity of ixazomib, cyclophosphamide and low-dose dexamethasone as induction, followed by single-agent ixazomib maintenance, in older, HCT-ineligible, patients with NDMM.[126] The ORR after initial therapy with ixazomib/cyclophosphamide/dexamethasone was 73%. After a median follow-up of 26.1 months, the PFS was 23.5 months.[126]

Isatuximab-irfc/Lenalidomide/Dexamethasone
Based upon the IMROZ (Isa-VRd vs. VRd)[123] described above, and BENEFIT (Isa-VRd vs. Isa-Rd)[124] trials, Isa-VRd has been approved in those in whom HCT was deferred, who are not deemed as frail, populations. Both the IMROZ and BENEFIT trials excluded patients over 80 or a PS>2.

The BENEFIT trial investigated whether triplet or quadruplet therapy with isatuximab could improve MRD-negativity and response in patients ineligible for HCT or HCT-deferred. Interestingly, the response rates, whether very good partial response (VGPR) or complete response or better (≥CR) were superior for Isa-Rd at 18 months compared to Isa-VRd.[127] Rates for ≥CR for Isa-Rd and Isa-VRd were 58% and 31%, respectively (*P* < .0001).Though the time since trial treatment is short, estimated 24-month PFS are 80% (95% CI, 73.3–87.4) for Isa-Rd and 85.2% (95% CI, 79.2–91.7) for Isa-VRd. Although statistical differences between the two treatment groups for PFS and OS are marginal, the omission of bortezomib from the regimen may lead to less AEs as bortezomib induced AEs that led to temporary or permanent discontinuation. Addition of bortezomib to Isa-Rd led to 37 AEs of any grade (compared to Isa-Rd having 19), and 18 AEs grade 3 or above (compared to Isa-Rd having 8).[127] Isa-Rd is an option in the setting of peripheral neuropathy. This is category 2A, other recommended regimen.

Regimens Useful In Certain Circumstances if HCT is Deferred or if HCT is Not Indicated

Lenalidomide/Low-Dose Dexamethasone
The results of the SWOG SO232 trial[128] that included HCT-ineligible patients and the ECOG E4A03 trial[129] that included older patients with MM demonstrate that lenalidomide in combination with low-dose dexamethasone is a well-tolerated and effective regimen for these groups of patients. In the ECOG E4A03 trial the OS rate was significantly higher in the lenalidomide plus low-dose dexamethasone arm compared with the lenalidomide plus high-dose dexamethasone arm.[129] The inferior survival outcome seen with high-dose dexamethasone was greatest in patients aged ≥65 years. At 2 years, patients who did not proceed to HCT had an OS rate of 91% with lenalidomide and low-dose dexamethasone.[129]

The international, multicenter trial (FIRST trial) evaluated efficacy and safety of lenalidomide/dexamethasone given continuously for 72 weeks with melphalan/prednisone/thalidomide (MPT) in older (n = 1623) HCT-ineligible patients with NDMM.[130] The primary endpoint of this trial was PFS, and secondary endpoints were OS and adverse events, including the incidence of secondary malignancies. After a median of 37 months of follow-up, the risk of progression or death was reduced by 28% in patients receiving continuous lenalidomide/dexamethasone versus MPT (HR, 0.72; 95% CI, 0.61–0.85; *P* < .001).[130] Continuous lenalidomide/dexamethasone also reduced the risk of progression or death compared with 18 cycles of lenalidomide/dexamethasone (HR, 0.70; 95% CI, 0.89–1.20; *P* = .70). In the interim analysis, an OS benefit was seen in the lenalidomide/dexamethasone arm versus MPT (HR, 0.78; CI, 0.64–0.96; *P* = .02).[130]

In an analysis based on renal function of patients enrolled in the FIRST trial, continuous lenalidomide/low-dose dexamethasone compared with MPT reduced the risk of progression or death in patients with normal, mild, and moderate renal impairment by 33%, 30%, and 35%, respectively.[131]

Lenalidomide/low-dose dexamethasone is considered an option useful in certain circumstances (category 1) by the NCCN Multiple Myeloma Panel for HCT-ineligible patients with MM. The Panel recommends appropriate thromboprophylaxis for patients receiving this therapy. Based on the results of the FIRST trial,[130,132] the NCCN Panel recommends considering treatment with continuous lenalidomide/dexamethasone until disease progression for select patients who are not eligible for HCT.

Bortezomib/Cyclophosphamide/Dexamethasone

The role of bortezomib/cyclophosphamide/dexamethasone as initial therapy for patients with MM ineligible for HCT was studied in a small phase II trial (n = 20).[133] The median age of patients in this study was 76 years (range, 66–90 years). After a median of five cycles, the ORR was 95% with 70% of patients achieving VGPR or better response. With respect to toxicity, six patients experienced non-hematologic grade 3 or 4 adverse events (20%), including muscle weakness, sepsis, and pneumonia. Neutropenia and thrombocytopenia were seen in 2 patients (10%).[133]

Based on the above *and* the results from the EVOLUTION trial[90] (described earlier) that had included HCT-ineligible patients and the above phase II trial results,[133] the NCCN Panel has included bortezomib/cyclophosphamide/dexamethasone as an option useful in certain circumstances for non-HCT candidates. This option may be considered especially in patients with acute renal insufficiency or for patients who have no access to a PI in combination with lenalidomide/dexamethasone. According to the NCCN Panel, one can consider switching to a quadruplet therapy with Dara-RVd or to RVd, if antiCD38 is not available after renal function improves.

Bortezomib/Dexamethasone

A U.S. community-based, randomized, open-label, multicenter, phase IIIb UPFRONT trial compared the safety and efficacy of three highly active bortezomib-based regimens in previously untreated older patients with MM ineligible for HCT.[134] The patients with symptomatic, measurable MM were randomized (1:1:1) to one of the following regimens: bortezomib/dexamethasone (n = 168); bortezomib/thalidomide/dexamethasone (n = 167); or melphalan/prednisone/bortezomib (n = 167) followed by maintenance therapy with bortezomib. The primary endpoint was PFS; secondary endpoints included ORR, CR/near CR and VGPR rates, OS, and safety. All three induction regimens exhibited substantial activity, with an ORR of 73% (bortezomib/dexamethasone), 80% (bortezomib/thalidomide/dexamethasone), and 70% (melphalan/prednisone/bortezomib) during the treatment period.[135] After a median follow-up of 42.7 months, the median PFS and OS were not significantly different between the three treatment arms.[134] Response rates, including CR and greater than or equal to VGPR, improved after bortezomib maintenance, with no concomitant increase in the incidence of peripheral neuropathy.

The NCCN Multiple Myeloma Panel has included bortezomib/dexamethasone as a primary therapy as an option that is useful in certain circumstances for patients with MM who are ineligible for HCT.

Bortezomib/Lenalidomide/Dexamethasone (VRD-lite) for patients assessed as being frail

In HCT-ineligible patients with NDMM, a phase II study with the dose-adjusted VRd-lite regimen showed that the dose-adjusted regimen had comparable efficacy and better tolerability than the standard-dose regimen. The VRd-lite dosage included lenalidomide 15 mg orally on days 1–21; bortezomib 1.3 mg/m^2 subcutaneously on days 1, 8, 15, and 22; and dexamethasone 20 mg orally on the day of and the day after bortezomib for 9 cycles followed by 6 cycles of consolidation with lenalidomide and bortezomib. The ORR after 4 cycles of VRd-lite was 86%, with 66% achieving a VGPR or better.[136]

The NCCN Panel included VRD-lite regimen[136] as an option useful in certain circumstances for patients who are frail and not eligible for HCT. `

Carfilzomib/Cyclophosphamide/Dexamethasone

A phase II study examined the safety and efficacy of carfilzomib/cyclophosphamide/dexamethasone in patients ≥65 years of age with NDMM and ineligible for autologous HCT.[112] Out of 55 patients, 52 (95%) had at least a PR, 39 of 55 (71%) patients had at least a VGPR, 27 of 55 (49%) patients had a near CR or CR, and 11 of 55 (20%) patients had an sCR. After a median follow-up of 18 months, the 2-year PFS and OS rates were 76% and 87%, respectively.[112] Frequently reported grade 3 to 5 toxicities were neutropenia (20%), anemia (11%), and cardiopulmonary events (7%). Peripheral neuropathy was limited to grades 1 and 2 (9%).

The NCCN Panel has included carfilzomib/cyclophosphamide/dexamethasone as an option useful in certain circumstances for treatment of patients with NDMM not eligible for HCT with renal insufficiency and/or peripheral neuropathy.

Daratumumab/Cyclophosphamide/Bortezomib/Dexamethasone

Daratumumab has effectively enhanced patient treatment outcomes in patients with myeloma whether as a standalone treatment or in combination with previously established treatments.[122,137] A prospective multicenter phase II trial compared the response rates, MRD induction, PFS, and OS in patients with myeloma who are HCT ineligible of bortezomib/cyclophosphamide/dexamethasone (VCD) with daratumumab added to VCD (dara-VCD) induction, followed by daratumumab maintenance.[138] Patients (n = 121) were randomized into treatment groups receiving either VCD (n = 57) or dara-VCD (n = 64). The median PFS for VCD and dara-VCD were 16.8 months (95% CI, 15.3–21.7) and 25.8 months (95% CI, 19.9–33.5), respectively, resulting in HR of 0.67 (95% CI, 0.438–-1.030; *P* = .066). Dara-VCD yielded superior OR (PR or better) than VCD (85.94 vs. 64.91; *P* = .007). MRD-negativity induction rates were affected by the COVID-19 pandemic, leading to imperfect comparisons of the treatment regimen groups.

The NCCN Panel included daratumumab/cyclophosphamide/bortezomib/dexamethasone regimen as an option useful in certain circumstances for patients with NDMM not eligible for HCT. It is used as a treatment option for patients with renal insufficiency or acute kidney injury (AKI). Though randomized controlled efforts are still lacking, mounting evidence highlights the feasibility and importance of early initiation of daratumumab-based treatment in MM patients who present with AKI, to induce rapid and significant reductions in disease burden, improve renal outcomes, and provide a plasmapheresis-free approach.[139] According to the NCCN Panel, one can consider switching to dara-RVd after renal function improves.

Isatuximab-irfc/Carfilzomib/Lenalidomide/Dexamethasone
Isa-KRd was added to the list of regimens “Useful in Certain Circumstances” for HCT-ineligible patients with high-risk cytogenetics based on the results of GMMG-CONCEPT trial data (discussed above).[107] However, since the GMMG-CONCEPT trial included very few HCT-ineligible patients, the data for inclusion for non-HCT candidates is based more on extrapolation from the data from HCT-eligible population, therefore there was no uniform consensus for use of this regimen in this setting given other available options, therefore it is category 2B.

Ixazomib/Lenalidomide/Dexamethasone
The TOURMALINE-MM2 randomized trial evaluated ixazomib/lenalidomide/dexamethasone (I-Rd) in 705 HCT-ineligible patients with NDMM. Patients received 18 cycles of IRd or placebo-Rd, followed by a lower dose of lenalidomide with lower-dose ixazomib or placebo until progression or toxicity. IRd resulted in higher numeric median PFS, however statistical significance was not reached (median PFS, 35 vs. 22 months; HR, 0.83; 95% CI, 0.68–1.02). This benefit was seen in all prespecified subgroups, including the 280 patients with high-risk cytogenetics (median PFS, 24 vs. 18 months; HR, 0.69; 95% CI, 0.51–0.94). An OS difference has not been demonstrated (HR, 0.998; 95% CI, 0.79–1.26).

The PFS benefit seen with IRd in this trial appears to be less than the PFS benefit seen with bortezomib/lenalidomide/dexamethasone in the SWOG S0777 trial or with daratumumab/lenalidomide/dexamethasone in the MAIA trial. Therefore, considering this, NCCN Panel has added this as an all-oral option that would be useful in certain circumstances such as for patients with constraints to travel to treatment site.

Ixazomib-containing regimens are currently FDA indicated for the treatment of patients with MM who have received at least one prior therapy. The NCCN Panel has noted in a footnote that in patients with NDMM receiving regimens that have bortezomib or carfilzomib, these drugs may be substituted with ixazomib only in select patients in cases of intolerance to bortezomib/carfilzomib[140] or for logistical reasons (eg, patients who are assessed as very frail with many comorbid conditions; or unable to travel to a treatment site; or during public health emergencies).[141]

Lenalidomide/Cyclophosphamide/Dexamethasone
The efficacy and tolerability of cyclophosphamide/lenalidomide/dexamethasone in NDMM was demonstrated in a phase II study. Of the fifty-three patients enrolled in the trial, 85% had a PR or better including VGPR in 47%. The median PFS was 28 months (95% CI, 22.7–32.6) and at 2 years the OS was 87% (95% CI, 78–96).[142,143] The Myeloma XI trial compared responses to cyclophosphamide/lenalidomide/dexamethasone with cyclophosphamide/thalidomide/dexamethasone.[144] The results reported that the combination of cyclophosphamide/lenalidomide/dexamethasone was associated with significantly longer PFS (median, 36 vs. 33 months; P = .0116) and OS at 3 years (82.9% vs. 77.0%; P = .0072).[144]

The NCCN Panel included lenalidomide/cyclophosphamide/dexamethasone as a primary therapy option for patients with MM who are not eligible for HCT under the category useful in certain circumstances (category 2A).

Monitoring After Primary Myeloma Therapy

Response Criteria

Assessing the response to treatment is a key determinant of MM treatment. Patients on treatment should be monitored for response to therapy and for symptoms related to disease and/or treatment.

The updated IMWG response criteria definitions[9,145,146] for CR, sCR, immunophenotypic CR, molecular CR, VGPR, PR, minimal response (MR) for relapsed/refractory MM, SD, and progressive disease (PD) are outlined in *Response Criteria for Multiple Myeloma* in the algorithm. This has been updated to include measures of MRD assessments. It is recommended that the IMWG uniform response criteria should be used in all clinical trials.[147] According to the NCCN Panel, response should be assessed using the IMWG criteria.[9]

The same imaging modality used during the initial workup should ideally be used for the follow-up assessments. Follow-up tests after primary MM therapy include those used for initial diagnosis: a CBC with differential and platelet counts, blood glucose, electrolytes, and metabolic panel.

Assessing changes in levels of various proteins, particularly the M protein, helps track disease progression and response to treatment. SPEP is used to track for quantitative immunoglobulins and 24-hour urine UPEP helps track total protein.

The FLCr is required for documenting sCR according to the IMWG Uniform Response Criteria.[9] The serum FLC assay cannot replace the 24-hour UPEP for monitoring patients with measurable urinary M protein and can also be affected by renal function.

Bone marrow aspiration and biopsy with FISH should be performed as clinically indicated, especially at relapse.

Follow-up with advanced whole-body imaging (ie, FDG-PET/CT, low-dose CT, whole-body MRI without contrast) is recommended as needed. Residual focal lesions detected by either FDG-PET/CT or MRI have been shown to be of adverse prognostic significance.[148-151] Zamagni et al reported PFS of 44 months in patients with residual focal lesions on FDG-PET/CT versus 84 months for those without residual focal lesions on FDG-PET/CT after systemic treatment (P = .0009).[150] In the IMAJEM trial, both PFS and OS were significantly better in patients with negative FDG-PET/CT results before initiation of maintenance therapy (P = .011 and P = .033, respectively).[151] An analysis by Walker et al showed that conventional MRI normalizes over a prolonged period of time making FDG-PET/CT superior in this regard.[148] However, in small cohorts, functional imaging sequence for MRI called diffusion-weighted imaging was shown to have superior sensitivity to detect residual disease compared with FDG-PET/CT.[152-154] Furthermore, unlike FDG-PET/CT, MRI does not expose the patient to radiation.

A meta-analysis of 14 studies has shown that MRD negativity predicts improved PFS and OS, including in those who achieved CR.[155] Therefore, the NCCN Panel recommends consideration of MRD testing as indicated for prognostication after shared decision-making.

Hematopoietic Cell Transplantation

The NCCN Panel recommends considering collection of hematopoietic stem cells prior to prolonged exposure to lenalidomide and/or daratumumab (ie, optimally four or fewer induction cycles) in patients for whom HCT is being considered. Collecting enough hematopoietic stem cells for two HCTs (depending on the intended number of HCTs and age) in anticipation of a tandem HCT, boost after cellular therapy, or a second HCT as subsequent therapy is recommended. Alternatively, all patients may consider continuation of primary therapy until the best response is reached. The optimal duration of primary therapy after achieving maximal response is unknown; hence, maintenance therapy (see section on

Maintenance Therapy) or observation can be considered beyond maximal response.

HCT Eligibility

All patients are assessed to determine eligibility for HCT. The NCCN Panel recommends that all patients eligible for HCT should be referred for evaluation by HCT center and hematopoietic stem cells (for at least two HCTs, in younger patients) should be harvested.

High-dose therapy with HCT support is a critical component in the treatment plan of eligible patients with NDMM. The types of HCT may be single autologous HCT, a tandem HCT (a planned second course of high-dose therapy and HCT within 6 months of the first course), or an allogeneic HCT.

The NCCN Guidelines® for Multiple Myeloma indicate that all types of HCT are appropriate in different clinical settings; these indications are discussed further below. In general, all candidates for high-dose chemotherapy must have sufficient hepatic, renal, pulmonary, and cardiac function. However, renal dysfunction is not an absolute contraindication to HCT.

Autologous HCT

Autologous HCT results in high response rates and remains the standard of care after primary therapy for eligible patients. In 1996, results of the first randomized trial were reported; this trial demonstrated that autologous HCT is associated with statistically significantly higher response rates and increased OS and event-free survival (EFS) when compared with the response of similar patients treated with conventional therapy.[156] In 2003, results of a second trial comparing high-dose therapy to standard therapy showed an increase in the CR rate and an improvement in OS (54 months in the high-dose group compared to 42 months for standard therapy).[157] Barlogie and colleagues reported on the results of an American trial that randomized 510 patients to receive high-dose therapy with autologous HCT or standard therapy.[158] With a median follow-up of 76 months, there were no differences in response rates, PFS, or OS between the two groups. The reason for the discrepant results is not clear but may be related to differences in the specific high-dose and conventional regimens between the American and French study. For example, the American study included total body irradiation (TBI) as part of the high-dose regimen; TBI has subsequently been found to be inferior to high-dose melphalan.[159]

Another trial included 190 patients 55 to 65 years of age randomized to standard or high-dose therapy.[160] This study was specifically designed to include older patients, since the median age of the participants in other trials ranged from 54 to 57 years and the median age in this trial was 61 years. After 120 months of follow-up, there was no significant difference in OS, although there was a trend toward improved EFS in the high-dose group (P = .7). Additionally, the period without symptoms, treatment, or treatment toxicity was significantly longer in the high-dose group. The study concluded that the equivalent survival suggests that the treatment choice between high-dose and conventional-dose chemotherapy should be based on personal choice in older patients. For example, an early HCT may be favored because patients can enjoy a longer interval of symptom-free time.

A phase III study compared high-dose melphalan followed by autologous HCT with MPR (melphalan, prednisone, and lenalidomide) consolidation after induction. Patients (n = 402) were randomly assigned (in a 1:1:1:1 ratio) to one of the four groups: high-dose therapy and autologous HCT followed by maintenance with lenalidomide; high-dose therapy and HCT alone; primary therapy with MPR followed by lenalidomide; and primary therapy with lenalidomide alone.[161] At a median follow-up of 51 months, HCT resulted in longer median PFS (43 vs. 22 months; HR, 0.44; 95% CI, 0.32–0.61) and OS (82% vs. 65% at 4 years; HR 0.55; 95% CI, 0.32–0.93).[161]

Results from the IFM 2005/01 study of patients with symptomatic MM receiving primary therapy with bortezomib and dexamethasone versus VAD showed a marked improvement in ORR with bortezomib and dexamethasone over VAD.[162] Responses were evaluated after primary treatment and post-autologous HCT. After the first autologous HCT, CR/near-CR rates were 35.0% in the bortezomib plus dexamethasone arm, compared with 18.4% in the VAD arm.[162] The VGPR rates were 54.3% versus 37.2%. Median PFS was 36.0 versus 29.7 months (*P* = .064) with bortezomib plus dexamethasone versus VAD after a median follow-up of 32.2 months.[162] Also, PFS was also significantly longer in patients achieving greater than or equal to VGPR after primary treatment than in patients achieving less than VGPR (median, 36 vs. 29.7 months).[162]

In another study, 474 patients were randomized to primary therapy with bortezomib/dexamethasone/thalidomide (n = 236) or thalidomide/dexamethasone (n = 238) before double autologous HCT and as consolidation therapy after HCT.[163] The 3-drug regimen yielded high response rates compared with the 2-drug regimen, with a CR rate of 19% (vs. 5%) and greater than or equal to VGPR of 62% (vs. 31%). After HCT, improved incremental responses were still seen with bortezomib/dexamethasone/thalidomide compared with thalidomide plus dexamethasone.[163] The IFM 2009 phase III trial compared the efficacy and safety of bortezomib/lenalidomide/dexamethasone (RVD) alone versus (RVD) plus autologous HCT for the treatment of patients ≤65 years with NDMM.[164] The reported CR rate was 48% in the group that received induction therapy alone versus 59% in the HCT group (*P* = .03). No MRD was detected in 65% of the patients who received (RVD) alone versus no MRD in 79% of the patients who received induction therapy plus autologous HCT (*P* < .001).[164] There was a clear improvement in PFS with HCT (50 vs. 36 months). These results clearly show the benefit of autologous HCT, with higher rates of durable responses in those with no MRD after initial therapy.[164] Taken together, the studies suggest that improved responses with the primary regimen result in improved outcomes after HCT even for patients receiving an IMiD and PI-based triplet regimen.

The OS of patients in the IFM 2009 phase III trial was high in both groups, the one that received autologous HCT and the one that did not.[164] Although autologous HCT improved PFS it did not improve OS.

The randomized phase 3 DETERMINATION trial investigated the role of autologous HCT in patients with NDMM. Patients (n = 722) were randomly assigned to receive 3 cycles of lenalidomide, bortezomib, and dexamethasone (RVD) followed by stem cell collection and 5 more cycles of RVD or 3 cycles of RVD, stem cell collection, autologous HCT and an additional 2 more cycles of RVD.[165] Both groups (those who received HCT and those who did not) received indefinite lenalidomide maintenance.[165] Primary end point was PFS; secondary end points included response rates, duration of response, time to progression, OS, safety, and quality of life. At a median 76-month follow-up, median PFS was improved by 21 months with HCT, however no OS benefit was seen.[165] The percentage of patients with a partial response or better was 95.0% in the RVD-alone group and 97.5% in the HCT group (*P* = .55); 42.0% and 46.8%, respectively, had a complete response or better (*P* = .99). Treatment-related adverse events of grade 3 or higher occurred in 78.2% and 94.2%, respectively; 5-year survival was 79.2% and 80.7% (HR for death, 1.10; 95% CI, 0.73–1.65).

The results of IFM 2009 and DETERMINATION trials suggest that delaying HCT is an option and is not associated with negative effects on OS.

In an exploratory subgroup analysis of DETERMINATION, patients who achieved negative MRD at 10^{-5} sensitivity reported similar PFS in the VRd-alone arm versus early HCT (5-year PFS, 59% vs. 54%; HR, 0.91; 95% CI, 0.46–1.79).[165] Subsequently, the phase III IFM2020-02-MIDAS trial evaluated a MRD-driven consolidation and maintenance strategy after induction with isatuximab, carfilzomib, lenalidomide, and dexamethasone (Isa-KRD) in HCT-eligible patients with NDMM. Those MRD-negative at 10^{-5} sensitivity by NGS after Isa-KRD did not derive additional benefit from autologous HCT compared with continued therapy using Isa-KRd. The results of the MIDAS trial suggest that autologous HCT may not offer additional benefit in terms of deep molecular response for patients who are MRD-negative after induction with Isa-KRd.[108]

According to the NCCN Guidelines, for HCT-eligible patients autologous HCT is the preferred option after primary induction therapy while a delayed HCT after early stem cell collection and storage is appropriate as well (category 1). Repeat HCT can be considered for treatment of progressive/refractory disease after primary treatment in patients with prolonged response to initial HCT.

Tandem HCT

Tandem HCT refers to a planned second course of high-dose therapy and HCT within 6 months of the first course. Planned tandem HCTs have been studied in several randomized trials. The IFM94 trial reported by Attal et al randomized patients with NDMM to single or tandem autologous HCTs.[166] A total of 78% of patients assigned to the tandem HCT group received the second HCT at a median time of 2.5 months after the first. A variety of options for therapy of relapsed disease were provided. For example, relapsing patients in either group underwent either no therapy, additional conventional therapy, or another HCT. The probability of EFS for 7 years after the diagnosis was 10% in the single HCT group compared to 20% in the double HCT group. In a subset analysis, those patients who did not achieve a complete CR or VGPR within 3 months after the first HCT appeared to benefit the most from a second HCT. The investigators of the IFM94 study have suggested that the improvement in projected survival associated with tandem HCT is related not to improved response rates, but to longer durations of response. Four other randomized trials have compared single versus tandem HCT.[160,167-169] None of these trials showed a significant improvement in OS. However, since the median follow-up in these trials ranged from 42 to 53 months, the lack of significant improvement is not surprising. The trial by Cavo et al[167] found that patients not in CR or near CR after the first HCT benefited the most from a second HCT. This confirms the observations of the IFM94 trial using non-TBI–based high-dose regimens. In both the French and Italian trials, the benefit of a second autologous HCT was seen in patients who do not achieve a CR or VGPR (>90% reduction in M protein level) with the first procedure. These two studies were not adequately powered to evaluate the equivalence of one versus two HCTs in patients achieving a CR or VGPR after the first HCT.

A review of long-term outcomes of several trials of autologous HCT found that tandem HCTs were superior to both single transplantations and standard therapies.[170] Also, post-relapse survival was longer when EFS was sustained for at least 3.5 years after tandem transplantation.[170,171] Results of the multicenter, phase III study (EMN02/HO95 MM trial) suggested that tandem autologous HCT for NDMM may be superior in extending PFS compared with single autologous HCT after induction therapy with a bortezomib-based regimen.[172] In another more recent study, after initial HCT patients were randomly assigned to receive a second HCT followed by lenalidomide maintenance; or 4 cycles of bortezomib, lenalidomide, and dexamethasone followed by lenalidomide maintenance; or lenalidomide maintenance alone.[173] At 38 months, all three arms showed similar PFS and OS.[173] In the MIDAS trial, among patients who were MRD-positive at 10^{-5} sensitivity after induction, the %

with a pre-maintenance MRD-negative status at 10^{-6} sensitivity was similar with tandem HCT compared with single HCT.[108]

The NCCN Panel recommends collecting enough hematopoietic stem cells for at least one HCT in *all* eligible patients, and for two transplants in the younger patients if tandem HCT or repeat HCT would be considered. According to the NCCN Panel, a tandem HCT with or without maintenance therapy can be considered for all patients who are candidates for HCT and is an option for patients who do not achieve at least a VGPR after the first autologous HCT and those with high-risk features. The support for use of maintenance therapy after tandem HCT comes from the study by Palumbo et al,[161] which addressed the role of maintenance therapy with lenalidomide after autologous HCT.[161] Although associated with more frequent grade 3 or 4 neutropenia and infections, maintenance therapy with lenalidomide was found to significantly reduce risk of disease progression or death (HR, 0.47) after both single and tandem HCT compared with no maintenance.[161]

A second autologous HCT can be considered at the time of disease relapse. A retrospective case-matched control analysis was performed comparing patients who underwent a second autologous HCT to those treated with conventional chemotherapy for relapsed MM.[174] Similar to previously published smaller studies,[175-177] this retrospective analysis demonstrated that a second autologous HCT is associated with superior relapse-associated mortality compared with conventional chemotherapy (68% vs. 78%), along with improved OS (32% vs. 22%) at 4 years. In this analysis, factors associated with improved OS and PFS included younger age (<55 years), beta-2 microglobulin <2.5 mg/L at diagnosis, a remission duration of >9 months, and a greater than PR to their first autologous HCT. This analysis indicates that a second autologous HCT, for relapsed or progressive MM, may be an option for carefully selected patients. Some of these patients can achieve durable complete or partial remission.[177,178]

A multicenter, randomized phase III trial compared treatment with high-dose melphalan plus second autologous HCT with cyclophosphamide in patients with relapsed MM who had received autologous HCT as primary treatment.[179] The patients included in the study were >18 years of age and needed treatment for progressive or relapsed disease at least 18 months after a previous autologous HCT. All patients first received bortezomib/doxorubicin/dexamethasone induction therapy. Patients with adequately harvested hematopoietic stem cells were then randomized to high-dose melphalan plus second autologous HCT (n = 89) or oral cyclophosphamide (n = 85). The primary endpoint was time to disease progression.[179] After a median follow-up of 31 months, median TTP in patients who underwent second autologous HCT after induction therapy was 19 months versus 11 months for those treated with cyclophosphamide (HR, 0.36; 95% CI, 0.25–0.53; *P* < .0001). Grade 3–4 neutropenia (76% vs. 13%) and thrombocytopenia (51% vs. 5%) were higher in the group that underwent autologous HCT versus cyclophosphamide.[179] Median OS in the HCT group was 67 months versus 52 months in the cyclophosphamide maintenance group.[180]

According to the NCCN Panel, repeat autologous HCT for relapsed disease may be considered either on or off clinical trial depending on the time interval between the preceding HCT and documented progression.

The prognosis of patients who relapse after autologous HCT appears to differ depending on the timing of the relapse.[181-185] Data from retrospective studies[186-189] suggest 2 to 3 years as the minimum length of remission for consideration of second autologous HCT for relapsed disease.

Allogeneic HCT

Allogeneic HCT includes either myeloablative or nonmyeloablative (ie, “mini” HCT) HCTs. Allogeneic HCT has been investigated as an alternative to autologous HCT to avoid the contamination of reinfused autologous tumor cells, but also to take advantage of the beneficial graft-versus-tumor effect associated with allogeneic HCTs. However, lack of a suitable donor and increased morbidity has limited this approach, particularly for the typical older MM population. Non-myeloablative HCTs are designed to decrease the morbidity of the high-dose chemotherapy but preserve the beneficial graft-versus-tumor effect. Therefore, the principal difference between myeloablative and nonmyeloablative HCTs relates to the chemotherapy regimen used. Specific preparatory regimens have not been a focus of the NCCN Guidelines, and therefore these guidelines do not make a distinction between these approaches.

Given the small candidate pool, it is not surprising that there have been no randomized clinical trials comparing myeloablative allogeneic to autologous HCT, but multiple case series have been published describing allogeneic HCT as an initial therapy or as therapy for relapsed/refractory MM. In a 1999 review, Kyle et al reported a mortality rate of 25% within 100 days and overall HCT-related mortality of approximately 40% and few patients were cured.[190] Other reviews have also reported increased morbidity without convincing proof of improved survival.[191,192] However, there are intriguing data from the SWOG randomized trial of autologous HCT versus conventional chemotherapy.[158] The original trial had an ablative, allogeneic HCT group consisting of patients with HLA identical siblings. Thirty-six patients received allografts, and due to the high 6-month mortality of 45%, the allogeneic arm was closed. After 7 years of follow-up the OS of the conventional chemotherapy, autologous, and allogeneic arms were all identical at 39%. The autologous and conventional chemotherapy arms do not demonstrate a plateau, whereas the allogenic curve was flat at 39%. This suggests that a proportion of these patients are long-term survivors. Thus, there is ongoing interest in myeloablative allogeneic HCT, particularly given the lack of a significant cure rate for single or tandem autologous HCT.

Patients whose disease either does not respond to or relapses after allogeneic hematopoietic cell grafting may receive donor lymphocyte infusions to stimulate a beneficial graft-versus-myeloma effect[193-200] or other myeloma therapies on or off a clinical trial.

Follow-up After HCT

Follow-up tests after HCT are like those done after primary myeloma therapy. MRD assessment is indicated for prognostication. Studies have shown that MRD negativity after autologous HCT translates to significantly improved PFS and OS rates.[201-203] Similar results have also been reported in the allogeneic HCT setting where the presence of MRD after allogeneic HCT has been associated with a significantly adverse PFS and OS.[204] The NCCN Panel recommends assessing for MRD during follow-up as indicated prognostication after shared decision-making with patient.[147] The Panel recommends a whole-body FDG-PET/CT scan around day 100 following autologous HCT.

Maintenance Therapy

Preferred Regimen

Lenalidomide as Maintenance

Lenalidomide as maintenance therapy after autologous HCT has been evaluated in two independent randomized phase III studies.[205,206]

In the CALGB 100104 trial, patients were randomized to maintenance therapy with lenalidomide (n = 231) versus placebo (n = 229) after autologous HCT.[206] At a median follow-up of 34 months, 37% of the patients who received lenalidomide versus 58% who received placebo had disease progression or died. The median TTP in the lenalidomide group was 46 months versus 27 months in the placebo group (P < .001). Second primary cancers occurred in 18 patients who received lenalidomide (8%) and in 6 patients who received placebo (3%).[206]

Data from the international, randomized, double-blind phase III IFM 2005-02 trial (n = 614) show that patients treated with lenalidomide as consolidation therapy after an autologous HCT followed by lenalidomide as maintenance therapy had upgraded responses. Of the 614 patients enrolled in the trial, 307 were randomly assigned to lenalidomide maintenance therapy and 307 to placebo. Maintenance treatment was continued until the patient withdrew consent, the disease progressed, or unacceptable toxic effects occurred. The final analysis of the IFM 2005-02 trial was performed after a median follow-up of 30 months and 264 patients had disease progression (104 in the lenalidomide group and 160 in the placebo group). The median PFS was 41 months in the lenalidomide group, compared with 23 months in the placebo group (HR, 0.50; P < .001; median follow-up period was 30 months). The probability of surviving without progression for 3 years after randomization was 59% of those treated with lenalidomide and 35% in those who received the placebo. The benefit of lenalidomide maintenance therapy, evidenced by rate of PFS at 3 years after randomization, was higher in all patients who received lenalidomide maintenance therapy compared with those who received placebo. This benefit was observed in patients who had a VGPR at randomization (64% vs. 49%; P = .006) and those who did not (51% vs. 18%; P < .001).[205] An increased incidence of second primary cancers was observed in the lenalidomide group (32 had second primary cancers in the lenalidomide group and 12 in the placebo group).[205] The updated survival analysis of the same study after 91 months for follow-up reported median TTP of 57.3 months (95% CI, 44.2–73.3) with lenalidomide and 28.9 months (23.0–36.3) with placebo (HR, 0.57; 95% CI, 0.46–0.71; P < .0001).[207] The most common grade 3–4 adverse events in the lenalidomide group compared to placebo were neutropenia (50% vs. 18%) and thrombocytopenia (15% vs. 5%). An increased rate of second primary malignancies (hematologic plus solid tumor) were diagnosed in the lenalidomide group compared with placebo (14% vs. 4%).[207]

The study by Palumbo et al[161] (discussed in *Autologous HCT*) showed that although maintenance therapy with lenalidomide is associated with more frequent grade 3 or 4 neutropenia and infections, it significantly reduced risk of disease progression or death (HR, 0.47) compared with no maintenance.[161]

The benefit of lenalidomide maintenance was studied in a meta-analysis of data from 1209 patients enrolled in the trials discussed above randomized to maintenance with lenalidomide or placebo.[208] The study showed improved median PFS with lenalidomide maintenance (52.8 vs. 23.5 months; HR 0.48; 95% CI, 0.42–0.55). At 7 years, the OS was 62% in the group receiving lenalidomide maintenance versus 50% in the group receiving placebo. In those with high-risk cytogenetics, a PFS benefit, but not an OS benefit was seen with lenalidomide maintenance versus placebo.

The lenalidomide group had higher rates of second primary malignancy occurring before progression, and the rates of PD were higher in the group receiving placebo. There are several reports showing higher incidences of second malignancies when lenalidomide is used as a maintenance therapy post-HCT or in a melphalan-containing regimen.[205,206,209,210] In the FIRST trial, the overall incidence of second malignancies, including hematologic malignancies, was lower in the continuous lenalidomide/dexamethasone arm. The overall rates of second primary cancers were 3.0% in the continuous lenalidomide/dexamethasone arm, 6.0% in the arm receiving 18 cycles of lenalidomide/dexamethasone, and 5.0% in the MPT arm.[130]

A report from the HOVON 76 trial indicates that lenalidomide maintenance may not be a feasible option after mini-allogeneic HCT.[211] However, another recently reported study has shown the feasibility of maintenance therapy with low-dose lenalidomide after allogeneic HCT in patients with high-risk MM.[212]

Data from the phase III MM-015 study show that lenalidomide maintenance after primary therapy with melphalan/prednisone/lenalidomide (MPL) significantly reduced the risk of disease progression and also increased PFS.[213] In this study, patients with NDMM (n = 459) aged ≥65 years were randomized to receive MP followed by placebo, MPL, or MPL followed by lenalidomide until progression. Maintenance with lenalidomide significantly prolonged PFS. The PFS of patients treated with MPL followed by maintenance lenalidomide was significantly prolonged (n = 152; median, 31 months) compared with the other two arms: MPL (n = 153; median, 14 months; HR, 0.49; $P < .001$) or MP (n = 154; median, 13 months; HR, 0.40; $P < .001$). Lenalidomide maintenance therapy improved PFS by 66% compared with placebo, regardless of age.[213] In the FIRST trial, use of lenalidomide indefinitely until progression was associated with a superior PFS compared with a fixed duration of 18 months.

The GMMG-MM5 trial (n = 502) examined maintenance therapy with lenalidomide following induction therapy and HCT in NDMM. Patients received lenalidomide for a fixed duration of 2 years or until either CR. The primary endpoint was PFS, which was not significantly different between arms after a median follow-up of 60.1 months. Higher OS was seen with lenalidomide given for 2 years than for lenalidomide given until CR (HR, 1.42; 95% CI, 1.04–1.93; $P = .03$). Serious adverse events were more common in the group that received lenalidomide for 2 years versus until CR (77.6% vs. 58.2%) and included infections, cardiac disorders, neuropathy, and thromboembolic events.[214]

Based on the evidence from the phase III trials,[205,206,213] the NCCN Panel lists single-agent lenalidomide as one of the preferred maintenance regimens (category1) Lenalidomide lacks the neurologic toxicity seen with thalidomide. However, there seems to be an increased risk for second cancers, especially post-HCT,[205,206,209] or after a melphalan-containing regimen.[210]

A meta-analysis of randomized controlled trials examined patients treated with lenalidomide maintenance versus patients with no maintenance or placebo in the HCT setting.[208] The analysis showed that patients treated with lenalidomide maintenance had significantly improved PFS (HR, 0.48; 95% CI, 0.41–0.55) and a trend toward OS with lenalidomide maintenance (HR, 0.75; 95% CI, 0.63–0.90; $P = .001$) versus no maintenance or placebo.[208]

The benefits of improved PFS with lenalidomide maintenance must be weighed against the increased rate of severe (grade 3 and 4) neutropenia, risk of second cancers, and other toxicities.[215] The NCCN Panel has noted that the benefits and risks of maintenance therapy with lenalidomide versus second cancers should be discussed with patients.

Other Recommended Regimen for Maintenance Therapy

Carfilzomib/Lenalidomide as Maintenance

Patients enrolled in the FORTE trial (described above) were randomized to receive carfilzomib/lenalidomide maintenance therapy or lenalidomide alone as maintenance therapy.[216] After a median duration of follow-up of 37 months, 3-year PFS was 75% in the carfilzomib/lenalidomide group versus 65% in the lenalidomide alone group (HR, 0.64; 95% CI, 0.44–0.94; $P = .023$). Most common grade 3–4 adverse events in the carfilzomib/lenalidomide group versus the lenalidomide group included neutropenia (20% vs. 23%), infections (5% vs. 7%), and vascular events (7% vs. 1%), with one treatment-related adverse event leading to death in the carfilzomib/lenalidomide group.[216]

The randomized phase III ATLAS trial evaluated carfilzomib/lenalidomide/dexamethasone compared with single agent lenalidomide as maintenance therapy following autologous HCT in patients with NDMM. After a median follow-up of 33.8 months, the carfilzomib/lenalidomide/dexamethasone group demonstrated a median PFS of 59.1 months, compared to 41.6 months in the lenalidomide group, representing a 49% reduction in the risk of disease progression or death. The NCCN Panel has included carfilzomib/lenalidomide maintenance therapy as an option useful in certain circumstances for HCT-eligible candidates.

Daratumumab/Lenalidomide as Maintenance Therapy

In the randomized, phase II GRIFFIN study described in a section above, daratumumab in combination with lenalidomide, bortezomib, and dexamethasone, followed by maintenance therapy with daratumumab and lenalidomide was compared to lenalidomide, bortezomib, and dexamethasone followed by maintenance therapy with lenalidomide alone, in newly diagnosed, HCT-eligible patients with MM. The addition of daratumumab to primary therapy, followed by daratumumab/lenalidomide maintenance therapy, led to deep and durable responses including high rates of sCR and MRD negativity.[217]

The PERSEUS (discussed in a section above) evaluated the efficacy and safety of Dara-VRd versus VRd The trial was not designed to isolate the effect of daratumumab or DR in the maintenance phase of treatment.[83]

Since both PERSUES and GRIFFIN trials did not randomize the maintenance therapy after induction plus consolidation, it is not clear whether the MRD negativity and PFS benefit seen with daratumumab was from the induction followed by consolidation therapy with the daratumumab-containing regimen, or whether adding daratumumab to the maintenance regimen had additional benefit.

A post hoc analysis of the PERSEUS trial was performed to look into high-risk patients, defined as patients experiencing relapse or progression within 18 months of treatment initiation to determine whether Dara-VRd followed by DR maintenance reduces risk and secondly to explore the impact of sustained MRD negativity (10^{-5}) ≥CR on PFS. Results from this analysis showed that Dara-VRd with DR maintenance leads to deeper and durable responses and is associated with improved PFS. The rate of patients with high-risk disease was reduced by 50% with Dara-VRd (3.1%; n = 11) versus VRd (6.8%; n = 24).[218] In addition, when considering pre-progression deaths, the rate of patients who progressed or who died within 18 months of treatment initiation, was also lower with Dara-VRd (5.4%; n = 19) versus VRd (11.0%; n = 39). The number of patients with sustained MRD negativity (10^{-5}) ≥CR was more than doubled with Dara-VRd versus VRd for ≥12 months (64.8% vs. 29.7%; OR. 4.42; 95% CI, 3.22–6.08; $P < .0001$) and 24 months (55.8% vs. 22.6%; OR, 4.36; 95% CI, 3.15–6.05; $P < .0001$). Among patients with sustained MRD-negativity rates at ≥12 months (10^{-5}), the PFS rates at 48 months were 95.3% for Dara-VRd and 94.2% for VRd. Among patients with sustained MRD-negativity rates at ≥24 months, these 48 months PFS rates increased even more to 97.8% for Dara-VRd and 98.8% for VRd, respectively.[218]

The phase 3 AURIGA study evaluated DR maintenance in patients with MRD-positivity after HCT versus lenalidomide alone.[219] At median follow-up 32.3 months, the MRD-negative rates were 50.5% for DR versus 18.8% for lenalidomide alone, reaching the primary endpoint. As a secondary endpoint, the PFS was 82.7% for DR compared to 66.4 for lenalidomide. The Grade 3 and Grade 4 cytopenias and infections were slightly higher with DR compared to lenalidomide alone (54.2% vs. 46.9%) and (18.8% vs.13.3%), respectively.

Based on the above trial data, NCCN Panel has included daratumumab/lenalidomide maintenance therapy as an option useful in certain circumstances for HCT-eligible candidates.

Maintenance Regimens Useful in Certain Circumstances

Bortezomib/Lenalidomide

Patients with high-risk cytogenetics have been shown to benefit from a combination of PI and IMiD as maintenance therapy.[220,221] The study by Joseph et al of patients (n = 1000) with NDMM treated with lenalidomide, bortezomib, and dexamethasone induction therapy and risk-adapted maintenance showed that early HCT and maintenance with doublet therapy (lenalidomide plus bortezomib) established excellent outcomes for those with high-risk myeloma.[221]

Bortezomib as Maintenance Therapy

The results from the HOVON study show that maintenance with single-agent bortezomib after autologous HCT is well tolerated and is associated with improvement of ORR.[222] Patients in the HOVON trial were randomly assigned to one of the two arms consisting of either primary treatment with VAD followed by autologous HCT and maintenance with thalidomide or with bortezomib/doxorubicin/dexamethasone followed by autologous HCT and bortezomib as maintenance therapy for 2 years. The study reported high near-CR/CR rates after primary treatment with the bortezomib-based regimen. Bortezomib as maintenance therapy was well tolerated and associated with additional improvement of response rates.[222] (see *Other Recommended Primary Therapy Regimens for HCT Candidates).*

A multicenter phase III trial in patients with NDMM showed that consolidation with bortezomib after autologous HCT improved PFS only in patients not achieving at least VGPR after autologous HCT.[223] There was no difference in PFS in patients with greater than or equal to VGPR after autologous HCT.

The results of the phase III UPFRONT study also show that maintenance with single-agent bortezomib is well tolerated when administered after treatment with bortezomib-based primary therapy.[134] Patients with NDMM ineligible for high-dose therapy and HCT enrolled in the UPFRONT trial were randomized (1:1:1) and treated with one of the following bortezomib-based primary regimens: bortezomib and dexamethasone; bortezomib in combination with thalidomide and dexamethasone; or bortezomib with melphalan and prednisone followed by maintenance treatment with bortezomib. The results show that the response rates, including CR and greater than or equal to VGPR, improved after bortezomib maintenance in all arms, with no concomitant increase in the incidence of peripheral neuropathy.[134]

Bortezomib monotherapy is included as maintenance therapy based on the results of the above-noted clinical trials. The isolated benefit of bortezomib monotherapy in the maintenance setting is unclear. Bortezomib monotherapy as a maintenance therapy option under “Useful in Certain Circumstances,” used in instances when a patient is unable to tolerate lenalidomide due to toxicity.

Daratumumab as Maintenance Therapy

CASSIOPEIA trial was an open-label, phase 3 trial of patients with NDMM eligible for. Patients were randomized to induction and post-HCT consolidation with dara-VTd or VTd. The second randomization compared daratumumab maintenance for up to 2 years with observation.[224] The most common grade 3 or 4 adverse events in the daratumumab maintenance group included lymphopenia (4%), hypertension (3%), and neutropenia (2%). Serious adverse events were more common in the daratumumab group (23% vs. 19%), with two treatment-related deaths in the daratumumab group.[224] At a median follow-up of 7 years from first randomization, dara-VTd induction/consolidation significantly reduced the risk of disease progression or death by 39% compared to VTd induction/consolidation alone, regardless of second randomization. Median PFS was approximately 2.5 years longer with the addition of daratumumab to VTd. Dara-VTd also significantly reduced the risk of death by 45% compared to VTd. With a median follow-up of approximately 6 years from second randomization, daratumumab maintenance significantly reduced the risk of disease progression or death by 51%, with a 20% higher 72-month progression-free survival rate compared to observation (57·1% versus 36·5%, respectively).[225]

The NCCN Panel has added daratumumab alone (based on the CASSIOPEIA trial data) as an option useful in certain circumstances after autologous HCT.

Ixazomib as Maintenance Therapy
The TOURMALINE-MM3 trial studied 2 years of maintenance with ixazomib versus placebo in patients who had achieved at least a PR following induction therapy and a single autologous HCT. After a median follow-up of 31 months, a 28% reduction in median PFS was observed with ixazomib versus placebo (median PFS was 26.5 vs. 21.3 months; HR, 0.72; 95% CI, 0.58–0.89; P = .0023).[226] A subsequent analysis of the same study demonstrated a higher rate of deepening responses with ixazomib versus placebo maintenance therefore prolonging PFS.[227] The median PFS in patients with VGPR/PR was 26.2 with ixazomib maintenance versus 18.5 months with placebo (HR, 0.636; P < .001).[227]

The TOURMALINE-MM4 trial is like TOURMALINE-MM3 but examined patients who were not receiving HCT. The study randomized patients who were not undergoing HCT and had achieved at least a PR following standard induction therapy to receive either ixazomib maintenance therapy or placebo. The primary endpoint was PFS since time of randomization. The primary endpoint was met, with a median PFS of 17.4 months in the ixazomib group versus 9.4 months in the placebo group (HR, 0.659; 95% CI, 0.542–0.801; P < .001). Grade 3 treatment-related adverse events were more common with ixazomib (36.6%) than placebo (23.2%). The most common any grade adverse events included nausea, vomiting, and diarrhea.[228] While both TOURMALINE-MM3 and TOURMALINE-MM4 have demonstrated statistically significant improvement in PFS, the OS data for MM3 and MM4 have not demonstrated a statistically significant difference.[229] Based on the above data, the NCCN Panel has included ixazomib as a category 2B recommendation useful in certain circumstances maintenance option for both HCT-eligible and HCT-ineligible patients.

Therapy for Previously Treated Multiple Myeloma

A variety of therapies are available for previously treated or relapsed/refractory MM. The choice of appropriate therapy for a specific patient depends on the context of clinical relapses such as prior treatment and duration of response.

The therapeutic options for previously treated MM include systemic therapy; autologous HCT for eligible patients who did not receive HCT as part of their initial treatment; or clinical trial. For those who had autologous HCT as part of initial treatment and had a durable response or had stable disease, consideration may be given to a second HCT at the time of relapse/disease progression.

As a general principle, if the relapse occurs at greater than 6 months after completion of the initial primary therapy, patients may be retreated with the same primary regimen. This, however, does not apply to HCT where a longer remission would be needed to justify another HCT. If using new triplet regimen, it should preferably include drugs or drug classes in patients who have not been exposed to or not exposed to for at least 6 months. Clinical trials with these triplet regimens primarily included patients who were naïve or sensitive to the novel drug in the doublet comparator arm. Clinical trials with these triplet regimens primarily included patients who were naïve or sensitive to the novel drug in the doublet comparator arm. Patients with disease refractory to the novel drug in the doublet backbone should be considered for triplet therapy that does not contain the drug that MM is progressing on.

Preferred Regimens for Previously Treated Multiple Myeloma – After One to Three Prior Therapies

For patients that are still sensitive to bortezomib and/or lenalidomide, any of the regimens listed below may be appropriate. Since, Anti CD-38- bortezomib- or lenalidomide-containing regimens are often given as induction therapy, and it is likely that at relapse the disease is refractory to one of these agents, especially if relapse is well within 6 months of primary treatment completion other combinations are preferred.

The NCCN Panel has provided a list of regimens for Anti CD-38 - refractory, bortezomib-refractory, and lenalidomide-refractory disease after one to three prior therapies.

Preferred Regimen(s) for Anti CD-38, Bortezomib and Lenalidomide-Refractory Disease

Carfilzomib/Pomalidomide/Dexamethasone

A phase II trial investigated carfilzomib/pomalidomide/dexamethasone followed by continuous pomalidomide/dexamethasone as second line therapy for relapsed/refractory MM in patients who had progression during lenalidomide maintenance therapy. Patients who were eligible for HCT and had not received it previously received HCT. On this regimen, 75% of patients had a VGPR, and 37% displayed CR. At 40-months of follow up, the median PFS was 26 months for patients who received therapy with HCT, and 17 months for patients who received carfilzomib/pomalidomide/dexamethasone therapy without HCT. The median OS was 67 months, with the most common grade 3 and 4 adverse events related to treatment including hematologic toxicity (41%), cardiovascular (6%) and respiratory (3%) events, and infections (17%).[230] Based on these data, the NCCN Panel has included carfilzomib/pomalidomide/dexamethasone as a preferred regimen option for relapsed or refractory MM.

Elotuzumab/Pomalidomide/Dexamethasone

In a phase II study, patients (n = 117) with refractory/relapsed MM and refractory to lenalidomide and a PI were randomized to receive pomalidomide/dexamethasone or elotuzumab/pomalidomide/dexamethasone.[231] After a follow-up of 9.1 months, the median PFS and ORR were both more than double with elotuzumab (PFS, 10.3 vs. 4.7 months; ORR, 53% vs. 26%).[231] Median OS was also significantly improved with elotuzumab/pomalidomide/dexamethasone compared with pomalidomide/dexamethasone (29.8 vs. 17.4 months; HR, 0.59) (95% CI, 0.37–0.93; *P* = .0217).[232]

The NCCN Panel has included the combination of pomalidomide/dexamethasone/elotuzumab as an option for patients who have received at least two prior therapies including an IMiD and a PI.

Preferred Regimens for Anti CD-38 or Lenalidomide-Refractory Disease

Ixazomib/Pomalidomide/Dexamethasone

In the phase I/II Alliance A061202 study (n = 29), patients with lenalidomide/PI refractory MM were treated with ixazomib/pomalidomide/dexamethasone- with 51.7% of patients having a PR or better, a median PFS of 4.4 months, a median response duration of 16.8 months, and a median OS of 34.3 months. Common adverse events included hematologic toxicity, and gastrointestinal events.[233]

Another phase I/II study studied the safety and efficacy of ixazomib/pomalidomide/dexamethasone in patients who had multiple prior therapies, were refractory to lenalidomide alone, or were refractory to lenalidomide and bortezomib, or lenalidomide, bortezomib, and carfilzomib.[234] The ORR was 33% and 40% with two different doses of ixazomib.[234]

Considering promising preliminary response rates, especially in patients refractory to both lenalidomide and a PI, the NCCN Panel has included ixazomib/pomalidomide/dexamethasone as a treatment option for patients with relapsed/refractory MM who have received at least two prior therapies including an IMiD and a PI and have demonstrated disease progression on or within 60 days of completion of the last therapy.

Pomalidomide/Bortezomib/Dexamethasone

A phase 3 open-label, multicenter, randomized, trial (OPTIMISMM) evaluated pomalidomide/bortezomib/dexamethasone (n = 281) versus bortezomib/dexamethasone in patients (n = 278)
with relapsed or refractory MM who previously received lenalidomide.[235] After a median follow-up of 15.9 months, a significantly improved PFS was seen in the pomalidomide arm (median, 11.20 vs. 7.10 months; HR, 0.61; 95% CI, 0.49–0.77; P < .0001). The most common grade 3/4 treatment-related adverse events in the pomalidomide arm reported in this trial were neutropenia, infections, and thrombocytopenia.[235] A post-hoc subgroup analysis of the OPTIMISSM trial evaluated outcomes in 226 patients at first relapse that had only received one prior line of therapy. Analyses were conducted by lenalidomide-refractory status, prior bortezomib exposure, and prior HCT. There were statistically significant improvements in PFS in both lenalidomide refractory (17.8 vs. 9.5 months; P = .0276) and lenalidomide non-refractory (22.0 vs. 12.0 months; P = .0491) patients. There were also statistically significant improvements in PFS in patients who had received prior bortezomib (17.8 vs. 12 months), and patients with (22 vs. 13.8 months) and without (16.5 vs. 9.5 months) prior HCT.[236]

Based on the above data, the NCCN Panel has included pomalidomide/bortezomib/dexamethasone as a category 1, preferred option preferred option for the treatment of patients with relapsed/refractory MM.

Preferred Regimens for Anti CD-38 or bortezomib -Refractory Disease

Carfilzomib/Lenalidomide/Dexamethasone

The randomized, multicenter, phase III ASPIRE trial studied the combination of lenalidomide and dexamethasone with or without carfilzomib in patients (n = 792) with relapsed/refractory MM who had received one to three prior lines of therapy and were sensitive to both lenalidomide and dexamethasone. The primary endpoint of the study was PFS. The results showed that addition of carfilzomib to lenalidomide and dexamethasone significantly improved PFS by 8.7 months (26.3 months for the carfilzomib arm vs. 17.6 months for lenalidomide and low-dose dexamethasone; HR for progression or death, 0.69; 95% CI, 0.57–0.83; P = .0001). The median duration of treatment was longer in the carfilzomib group (88 vs. 57 weeks). The incidence of peripheral neuropathy was identical in both arms (17%). Non-hematologic adverse effects (≥ grade 3) that were higher in the carfilzomib group compared with lenalidomide and dexamethasone included dyspnea (2.8% vs. 1.8%), cardiac failure (3.8% vs. 1.8%), and hypertension (4.3% and 1.8%). There were fewer discontinuations due to side effects in the carfilzomib arm (15.3% vs. 17.7%). Patients in the carfilzomib arm reported superior health-related quality of life than those who received lenalidomide and dexamethasone alone.[237]

Based on the above data, the NCCN Multiple Myeloma Panel has included the combination of carfilzomib with lenalidomide and dexamethasone as a category 1, preferred option for patients with relapsed/refractory MM.

In addition to the regimens listed in the section above, the following two lenalidomide-containing regimens may be used for lenalidomide-sensitive or naïve and bortezomib-refractory disease.

Preferred Regimens for Bortezomib or Lenalidomide-Refractory Disease

Daratumumab/Carfilzomib/Dexamethasone

A phase 1b, open-label, non-randomized, multicenter trial first studied this regimen in patients (n = 82) with relapsed or refractory MM. At a median follow-up of 16 months, the ORR was 84%. In the overall treatment population, while the median PFS was not reached, the 12- and 18-month PFS rates were 74% and 66%, respectively.[238] In a multicenter, open-label phase 3 trial (CANDOR), the addition of daratumumab to carfilzomib plus dexamethasone showed deeper responses and improved PFS.[239] This response has been shown to be maintained with longer follow up analyses of about 27 months. PFS was 28.6 months in the daratumumab group versus 15.2 months in the carfilzomib alone group (HR, 0.59; 95 CI, 0.45–0.78; P < .0001).[240] Based on the above data and the FDA approval, the NCCN Panel has included this regimen as a category 1, preferred option for relapsed/refractory MM, for patients with relapsed or refractory MM.

Isatuximab-irfc/Carfilzomib/Dexamethasone

A prospective, randomized, open label, phase 3 study (IKEMA) examined the utility of isatuximab/carfilzomib/dexamethasone versus carfilzomib/dexamethasone in 302 patients with relapsed/refractory MM who had received one to three prior lines of therapy (median two prior lines of therapy). Treatment was continued until disease progression or unacceptable toxicity, with the primary endpoint being PFS. Median PFS was 35.7 months in the isatuximab/carfilzomib/dexamethasone group versus a median PFS of 19.15 months in the carfilzomib/dexamethasone group (HR, 0.53; 99% CI, 0.32–0.89; P = .0007). Grade 3 or higher treatment related adverse events occurred in 77% of patients in the isatuximab group versus 67% of patients in the control group.[241] Based on this data, the NCCN Panel has included isatuximab-irfc/carfilzomib/dexamethasone as a category 1, preferred regimen option for relapsed or refractory MM.

Daratumumab/Pomalidomide/Dexamethasone

The combination of daratumumab/pomalidomide/dexamethasone was evaluated in an open-label, multicenter, phase 1b study (MMY1001). This study included patients (n = 103 patients) who had received at least two prior lines of therapy (excluding daratumumab or pomalidomide).[242] At a median follow-up of 13.1 months, the ORR was 60%. The median PFS and OS were 8.8 and 17.5 months, respectively, and estimated survival at 1 year was 66%.[242] Toxicities reported were similar to those seen in other trials of pomalidomide and daratumumab, except for increase in neutropenia.[242]

The open label phase III APOLLO trial randomly assigned patients with relapsed/refractory disease and at least one previous line of therapy (n = 304) to receive pomalidomide/dexamethasone or daratumumab/pomalidomide/dexamethasone. With a median follow up time of 16.9 months, there was a statistically significant improvement in the primary endpoint of PFS for the added daratumumab group (12.4 vs. 6.9 months; P = .0018). Serious adverse events occurred in 50% of patients in the daratumumab group compared to 39% of patients in the pomalidomide/dexamethasone group, the most common being pneumonia and lower respiratory tract infections.[137]

The MM-014 study evaluated 112 patients with relapsed/refractory MM who had previously been treated with lenalidomide and assigned them to a regimen containing daratumumab/pomalidomide/dexamethasone. The primary endpoint was ORR which was achieved in 77.7% of patients in a median follow up of 17.2 months (median PFS was not reached at time of follow up). The most common adverse event of grade 3 or higher was infection, which developed in 31.3% of patients (13.4% with grade 3 or higher pneumonia).[243]

Based on the above data, the NCCN Panel has included daratumumab/pomalidomide/dexamethasone as a treatment option for patients with relapsed/refractory MM who have received one prior therapy including an IMiD and a PI.

Daratumumab/Teclistamab-cqyv:

Content Development for this regimen is in progress

Isatuximab-irfc/pomalidomide/dexamethasone

In an open-label, multicenter, phase III trial (ICARIA-MM), patients (n= 307) with MM who had received at least two lines of prior therapy, including lenalidomide and a PI were randomized to receive isatuximab-irfc with pomalidomide/dexamethasone (Isa-Pd) or pomalidomide/dexamethasone (Pd).[244] After a median follow-up of 12 months, a higher ORR (60% vs. 35%) and improved PFS (median, 11.5 vs. 6.5 months; HR, 0.6; 95% CI, 0.44–0.81) were reported in the Isa-Pd arm. In a prespecified subgroup analysis of this study, the addition of isatuximab-irfc showed improved ORR and PFS in patients with renal impairment.[245]

The final analysis after median follow-up duration was 52.4 months reported a median OS of 24.6 months with Isa-Pd and 17.7 months with Pd.[246] Daratumumab was used in the Pd group and therefore had a potential benefit on PFS in the first subsequent therapy line. Despite this, median PFS 2 was significantly longer with Isa-Pd versus Pd (17.5 vs. 12.9 months).[246]

The NCCN Panel has included isatuximab-irfc/pomalidomide/dexamethasone as a category 1, preferred option for the treatment of patients with relapsed/refractory MM after two prior therapies including lenalidomide and a PI.

Additional Preferred Regimens for Bortezomib-Refractory Disease:

Daratumumab/Lenalidomide/Dexamethasone

In a multicenter, open-label phase 3 trial (POLLUX), patients (n = 569) with relapsed/refractory MM were randomized to receive lenalidomide/dexamethasone with or without daratumumab until disease progression or unacceptable toxicity.[247]

After a median follow-up of 13.5 months, daratumumab in combination with lenalidomide and dexamethasone was associated with better PFS and ORR compared with lenalidomide/dexamethasone alone. After a median follow-up of 25.4 months, a subsequent analysis reported that the higher ORR (92.9% vs. 76.4%; *P* < .001), and PFS (83% vs. 60% at 12 months; 68% vs. 41% at 24 months; HR, 0.41; 95% CI, 0.31–0.53) was maintained in those who had received daratumumab.[247]

The most common adverse events of grade 3 or 4 in patients treated with the daratumumab regimen versus lenalidomide/dexamethasone were neutropenia (51.9 vs. 37.0%), thrombocytopenia (12.7% vs. 13.5%), and anemia (12.4% vs. 19.6%). Daratumumab-associated infusion-related reactions (mostly grade 1 or 2) were reported in 47.7% of patients.

With an extended follow-up of 3.5 years, the improvements in PFS and ORR continued to be maintained in patients treated with the daratumumab regimen (PFS, 16.7 vs. 7.1 months; HR, 0.31; 95%; CI, 0.25–0.40; *P* < .0001). In a subgroup of patients with one prior line of therapy, the median PFS was 27.0 months with daratumumab versus 7.9 months with daratumumab and lenalidomide (HR, 0.22; 95% CI, 0.15–0.32; *P* <

.0001). The ORR rates for patients with one prior line of therapy for those receiving the daratumumab-regimen was 92% compared with 74% in those receiving daratumumab/dexamethasone.[248]

Based on the above data, the NCCN Panel has added daratumumab/lenalidomide/dexamethasone as a category 1, preferred option for the treatment of patients with relapsed/refractory MM.

Additional Preferred Regimens for Lenalidomide refractory Disease

In addition to the regimens listed in the section for bortezomib- and lenalidomide refractory disease, the following bortezomib-containing regimens may be used for bortezomib-sensitive or naïve and lenalidomide-refractory disease.

Daratumumab/Bortezomib/Dexamethasone

A phase III trial showed that adding daratumumab to bortezomib and dexamethasone markedly improved outcomes for patients with recurrent/refractory MM.[249] Patients (n = 498) were randomized to receive daratumumab/bortezomib/dexamethasone or bortezomib/dexamethasone. The ORR in the daratumumab arm was 82.9% compared to 63.2% in the control arm (P < .001).[249] The rates of VGPR and CR were double in the daratumumab arm compared to the control arm (59.2% vs. 29.1%; P < .001 and 19.2% vs. 9.0%; P = .001, respectively). The 12-month estimated rate of PFS was significantly higher in the daratumumab arm compared to the control arm (60.7% vs. 26.9%).[249] The most common grade 3 or 4 adverse events reported in the daratumumab and control groups were thrombocytopenia (45.3% and 32.9%, respectively), anemia (14.4% and 16.0%, respectively), and neutropenia (12.8% and 4.2%, respectively).[249] Grade 1 or 2 infusion-related reactions associated with daratumumab were reported in 45.3% of the patients in the daratumumab group and grade 3 in 8.6% of the patients. These infusion-related reaction rates are consistent with findings from previous trials of daratumumab.[250,251]

After a median follow-up of 40 months, patients receiving the daratumumab containing regimen demonstrated a 69% reduction in the risk of disease progression or death (median PFS, 16.7 vs. 7.1 months; HR, 0.31; 95% CI, 0.25–0.40; P < .0001); showed significantly better ORR (85% vs. 63%; P < .0001).[252] Patients who received one prior line of therapy demonstrated the greatest benefit with daratumumab (median PFS, 27.0 vs. 7.9 months; HR, 0.22; 95% CI, 0.15–0.32; P < .0001).

Based on the above phase III data, the NCCN Panel has added daratumumab/bortezomib/dexamethasone as a category 1, preferred option for the treatment of patients with relapsed/refractory MM.

Preferred CAR T-Cell Therapies for Relapse After One to Three Prior Therapies

Ciltacabtagene autoleucel: In the phase 3 CARTITUDE-4 study patients treated with ciltacabtagene autoleucel (cilta-cel) therapy (n = 208) were compared to those treated with two standard regimens, either pomalidomide /daratumumab/dexamethasone or pomalidomide/bortezomib/dexamethasone (n = 211).[253] In the cilta-cel arm the rate of CR or better was 73.1% compared to 21.8% with standard regimens (odds ratio [OR], 10.3; 95% CI, 6.5–16.4). The ORRs in the cilta-cel and standard-regimen arms were 84.6% and 67.3%, respectively (OR, 3.0; 95% CI, 1.8–5.0). MRD negativity (<10^{-5} determined by NGS) was higher in the cilta-cel arm, 60.6% compared to 15.6% in the standard-regimen arm (OR, 8.7; 95% CI, 5.4–13.9). At a median follow-up of 15.9 months (range, 0.1–27.3), the median PFS was not yet reached with cilta-cel and was 11.8 months with standard regimens (HR, 0.26; 95% [CI, 0.18–0.38; P < .001).[253] Based on these data, the Panel included cilta-cel as an option for patients after one prior *line* of therapy including an IMiD and a PI, and refractoriness to lenalidomide. Importantly, after median follow-up 33.6 months, cilta-cel significantly improved OS at 30 months 76.4% versus 63.8% (HR [95% CI]: 0.55 [0.39–0.79]; P = .0009), with consistent cilta-cel OS benefit across prespecified subgroups.[254] The peer-reviewed publication of this key follow-up analysis is awaited.

Idecabtagene vicleucel: In the phase 3 KarMMa-3 trial patients treated with Idecabtagene vicleucel (ide-cel) therapy (n = 254) were compared to those treated with standard regimens (n = 132).[255] In this trial, 66% of the patients had triple-class–refractory disease, and 95% had daratumumab-refractory disease. At a median follow-up of 18.6 months, the median PFS was 13.3 months (95% CI, 11.8–16.1) in the ide-cel arm compared with 4.4 months (95% CI, 3.4–5.9) in the standard-regimen arm leading to a 51% reduction in the risk of disease progression or death (HR for disease progression or death, 0.49; 95% confidence interval, 0.38 to 0.65; P < .001). Ide-cel also demonstrated an improvement in ORR versus standard regimens, with 71% (95% CI, 66%–77%) of patients achieving a response compared with 42% (95% CI, 33%–50%) in those who received standard regimens (P < .0001).[255] The Panel included ide-cel as an option for patients after two prior *lines* of therapy including an IMiD, an anti-CD38 monoclonal antibody and a PI.

Bridge Therapy: Since CAR-T therapy is being used in earlier lines of treatment and it takes time to manufacture the individualized CAR-T therapy, bridging therapy may be considered in order to slow disease progression, reduce tumor burden, and maintain CAR-T eligibility, especially in patients with rapid disease progression and with high-risk factors. Bridging therapy prior to CAR-T infusion helps reduces risk of CRS, immune effector cell–associated neurotoxicity syndrome (ICANS), prolonged cytopenias, and improves outcomes. The type of bridging therapy used needs to be individualized.[256] The panel recommends that talquetamab-tgvs may be considered as a bridge to BCMA CAR-T therapy in relapsed and/or refractory MM and agreed that BCMA targeted CAR-T that immediately follows BCMA targeted bispecific use may not represent optimal sequencing.[257]

Other Recommended Regimens for Relapse After One to Three Prior Therapies

Elotuzumab/Lenalidomide/Dexamethasone

The FDA has approved elotuzumab in combination with lenalidomide and dexamethasone for the treatment of patients with MM who have received one to three prior therapies. This is based on the results of the phase III trial, ELOQUENT-2. The trial randomized 646 patients (1:1) to receive either elotuzumab in combination with lenalidomide and dexamethasone or lenalidomide and dexamethasone alone.[258]

The rates of PFS at the end of 1 and 2 years were higher for those receiving the elotuzumab-containing regimen (68% at 1 year and 41% at 2 years) compared with those receiving lenalidomide and dexamethasone alone (57% at 1 year and 27% at 2 years).[258] Median PFS in the group receiving the elotuzumab-containing regimen was 19.4 months versus 14.9 months in those receiving lenalidomide and dexamethasone alone (HR for progression or death in the elotuzumab group, 0.70; 95% CI, 0.57–0.85; P < .001) indicating a relative reduction of 30% in the risk of disease progression or death.[258] Common grade 3 or 4 adverse events in both arms of the trial were lymphocytopenia, neutropenia, fatigue, and pneumonia. Infusion reactions occurred in 33 patients (10%) in the elotuzumab group and were grade 1 or 2 in 29 patients.[258]

Consistent with the above findings, a subset analysis of 3-year follow-up reported a reduced risk of progression by 27% with the elotuzumab/lenalidomide/dexamethasone combination compared with lenalidomide/dexamethasone.[259]

The final results of the ELOQUENT-2 study have demonstrated that the addition of elotuzumab to lenalidomide/dexamethasone improved OS in patients with MM who received one to three prior lines of therapy (48.3 vs. 39.6 months).[260]

Based on the above data and FDA approval the NCCN Panel has included elotuzumab in combination with lenalidomide and dexamethasone as a category 1 option for previously treated MM.

Ixazomib/Lenalidomide/Dexamethasone

A double-blind, randomized, placebo-controlled, phase III TOURMALINE MM1 trial randomized 722 patients with relapsed and/or refractory MM to a combination of ixazomib plus lenalidomide and dexamethasone or lenalidomide and dexamethasone alone (control group). This trial was designed based on the promising results of a phase I/II study (discussed under *Other Recommended Primary Therapy Regimens for HCT Candidates*).[116]

The results of the TOURMALINE MM1 trial show a significant improvement in PFS with the ixazomib-containing regimen. After a median follow-up of almost 15 months, a 35% improvement in PFS was seen in the group treated with the ixazomib regimen compared with the control group (HR, 0.74; P = .01).[261] Median PFS was 20.6 months in the ixazomib-treated group versus 14.7 months in the group receiving lenalidomide and dexamethasone alone. In the ixazomib-treated group versus the control group, the ORR (78% vs. 72%; P = .035) and CR (11.7% vs. 6.6%; P = .019) were also improved. Of note, patients with high-risk cytogenetics enrolled in the trial receiving ixazomib had a similar HR for PFS as the entire study population (HR, 0.596 and 0.543, respectively).[261] Grade ≥3 adverse events were reported in 74% and 69% of patients in the ixazomib-treated and control groups, respectively. These included anemia (9% with ixazomib/lenalidomide/dexamethasone vs. 13% with lenalidomide/dexamethasone), thrombocytopenia (19% vs. 9%), and neutropenia (23% vs. 24%).[261] The addition of the ixazomib/lenalidomide/dexamethasone group had a slightly higher rate of peripheral neuropathy compared to lenalidomide/dexamethasone (27% vs. 22%).

Based on the results of the phase III TOURMALINE MM1 trial[261] the NCCN Panel has included ixazomib/lenalidomide/dexamethasone as a category 1, preferred regimen option for previously treated MM after one to three prior therapies.

Selinexor/Bortezomib/Dexamethasone

An ongoing phase 3, randomized open label trial (BOSTON) compared selinexor/bortezomib/dexamethasone with bortezomib/dexamethasone in patients with previously treated MM (one to three prior lines of therapy, including PIs). Four hundred two patients were randomized to the selinexor/bortezomib/dexamethasone and 206 to the bortezomib/dexamethasone group. After a median follow up duration of 13.2 months in the selinexor/bortezomib/dexamethasone group, the median PFS was 13.93 months compared to a median follow up duration of 16.5 months and median PFS of 9.46 months in the bortezomib/dexamethasone group (HR, 0.70; 95% CI, 0.53–0.93; P = .0075). The most frequent adverse events of grade 3-4 that were more common in the selinexor/bortezomib/dexamethasone group were thrombocytopenia (39% vs. 17%), fatigue (13% vs. 1%), and anemia (16% vs. 10%).[262]

Based on the above data the NCCN Panel has included once weekly selinexor in combination with bortezomib and dexamethasone as a category 1, other recommended regimen option for previously treated MM.

Bortezomib/Cyclophosphamide/Dexamethasone

The effects of adding an alkylating agent (such as cyclophosphamide) and a novel agent (such as lenalidomide or bortezomib) to dexamethasone have been investigated for patients with relapsed/refractory MM. The combination of bortezomib, dexamethasone, and cyclophosphamide was found to be effective in patients with relapsed/refractory MM with an acceptable toxicity profile.[263,264] The NCCN Multiple Myeloma Panel members have included bortezomib/cyclophosphamide/dexamethasone as other recommended regimen for relapsed/refractory MM after one to three prior therapies.

Bortezomib/Lenalidomide/Dexamethasone

Data from preclinical studies showed lenalidomide sensitizes myeloma cells to bortezomib and dexamethasone. The results of phase I and phase II studies show that bortezomib/lenalidomide/dexamethasone is well tolerated and active, with durable responses in heavily pretreated patients with relapsed and/or refractory MM, including patients who have had prior lenalidomide, bortezomib, thalidomide, and HCT.[265,266] After a median follow-up of 44 months, the median PFS was 9.5 months and median OS was 30 months (95% CI, 24–37).[266] The NCCN Multiple Myeloma Panel members have included bortezomib/lenalidomide/dexamethasone as other recommended regimen for relapsed/refractory MM after one to three prior therapies.

Carfilzomib/Cyclophosphamide/Dexamethasone

Carfilzomib/cyclophosphamide/dexamethasone has been shown to be well tolerated with the toxicity profile of carfilzomib being similar to that seen in other trials.[267]

A phase II trial (MUKfive) compared the safety and toxicity of carfilzomib/cyclophosphamide/dexamethasone with bortezomib/cyclophosphamide/dexamethasone in patients with relapsed/refractory MM, who had received one prior regimen.[267] A higher proportion of patients receiving carfilzomib achieved VGPR or better and was noninferior to bortezomib. Carfilzomib/cyclophosphamide/dexamethasone was well tolerated with the toxicity profile of carfilzomib being similar to that seen in other trials.[267] This study also included a maintenance phase and demonstrated a median PFS of 11.9 versus 5.6 months in favor of carfilzomib maintenance versus observation.

Another phase II trial compared treatment with cyclophosphamide plus carfilzomib and dexamethasone to treatment with carfilzomib and dexamethasone in patients (n = 197) with relapsed/refractory MM after one to three prior lines.[268] After a median follow-up of 37 months, median PFS was 19.1 with the 3-drug regimen compared to 16.6 months with the 2-drug regimen (*P* = .577).[268] The combination of cyclophosphamide with carfilzomib and dexamethasone did not improve outcomes significantly compared with carfilzomib and dexamethasone alone in the overall population. However, in a sub-group analysis of the lenalidomide-refractory population, the addition of cyclophosphamide to carfilzomib and dexamethasone resulted in a PFS benefit of 18.4 versus 11.3 months (HR, 1.7; 95% CI, 1.1–2.7; *P* = .043).[268]

The NCCN Panel has included carfilzomib/cyclophosphamide/dexamethasone as treatment as other recommended regimen for relapsed/refractory MM after one to three prior therapies.

Daratumumab/Cyclophosphamide/Bortezomib/Dexamethasone

In the LYRA study,[117] among the small cohort of patients with relapsed MM (*n* = 14), after 4 cycles of induction therapy ORR was 12.3% and VGPR or better was seen in 57.1% of patients.[117] The ORR after 4 induction cycles was 71.4%. The median PFS was 13.3 months (95% CI, 6.8–13.3). At 12-months, the OS rate was 54.5% (95% CI, 8.6%–86.1%).[117]

Based on this, the NCCN Panel has included daratumumab/bortezomib/cyclophosphamide/dexamethasone as other recommended treatment option for relapsed/refractory MM.

Daratumumab/Carfilzomib/Pomalidomide/Dexamethasone

This 4-drug regimen is being studied in phase II trials in relapsed/refractory setting. The initial analysis (NCT04176718) reported on twenty-four patients who received the 4-drug regimen. The median age of the patients was 65 (range, 42–73), and median number of prior regimens received was 2 (range, 1–3). All patients were refractory to their last line of therapy and had prior lenalidomide and bortezomib and were daratumumab-naïve. Median PFS seen at 12 months was 86.2% (95% CI, 70%–100%).[269] This trial is ongoing and currently recruiting patients (NCT04176718).

Based on the initial promising report, the NCCN Panel has included daratumumab/carfilzomib/pomalidomide/dexamethasone as other recommended treatment option for relapsed/refractory MM.

Elotuzumab/Bortezomib/Dexamethasone

Multiple randomized trials have shown that three-drug combinations are consistently more effective than 2-drug combinations for the treatment of MM. A phase II trial studied the effect of addition of elotuzumab to bortezomib/dexamethasone in patients with relapsed/refractory MM.[270]

Interim analysis results demonstrated a 28% reduction in risk of disease progression or death for patients in the elotuzumab-containing triple-drug arm compared to patients treated with bortezomib/dexamethasone (HR, 0.72; 70% CI, 0.59–0.88). Median PFS was significantly higher in the elotuzumab-containing arm (9.7 vs. 6.9 months). After 2 years the addition of elotuzumab continued to show an efficacy benefit compared to bortezomib/dexamethasone alone with a 24% relative risk reduction in PFS (HR, 0.76; 70% CI, 0.63–0.91).[270]

Based on the above phase II trial data, the NCCN Panel has included elotuzumab/bortezomib/dexamethasone as other recommended regimen for relapsed/refractory MM after one to three prior therapies.

Ixazomib/Cyclophosphamide/Dexamethasone

This regimen has been shown to be tolerable and efficacious in patients with NDMM.[126,271] A phase II study evaluated this regimen in the relapsed/refractory setting in patients with a median age of 63.5 years and found that it is well tolerated. At a median follow-up of 15.2 months in the phase II study, median PFS was 14.2 months. The PFS trend with this regimen was better in patients aged 65 and older compared with those less than 65 years (median, 18.7 vs. 12.0 months; HR, 0.62; P = .14).[272] The NCCN Panel has included this all oral regimen under other recommended regimen for relapsed/refractory MM after one to three prior therapies.

Lenalidomide/Cyclophosphamide/Dexamethasone

A retrospective analysis to assess the efficacy of lenalidomide in combination with cyclophosphamide and dexamethasone showed that this regimen is effective in heavily pre-treated patients with manageable adverse effects.[273] The NCCN Panel has included cyclophosphamide/lenalidomide/dexamethasone treatment as other recommended regimen for relapsed/refractory MM after one to three prior therapies.

Pomalidomide/Cyclophosphamide/Dexamethasone

A phase II study compared the combination of pomalidomide/cyclophosphamide/dexamethasone to pomalidomide/dexamethasone in patients (n = 70) with relapsed/refractory MM who had received more than two prior therapies.[274]

The triple-drug combination significantly improved the ORR (≥PR, 64.7% vs. 38.9%; P = .0355). The median PFS reported was 9.5 versus 4.4 months. There were no significant differences in adverse event reports between the treatment arms; grade 3 and 4 anemia, neutropenia, and thrombocytopenia, respectively, were reported in 11%, 31%, and 6% of patients treated with pomalidomide/dexamethasone and 24%, 52%, and 15% of patients treated with the triplet regimen.[274] Similar results were reported by a single-center retrospective study of patients (n = 20) with relapsed/refractory MM who received pomalidomide/cyclophosphamide/dexamethasone until HCT or disease progression.[275] Response to the triple-drug regimen was 63%, with nearly half of patients (42%) after 1 cycle with a median time to response of 3 cycles. One-year median PFS was 80.7% and 65% of patients were relapse-free.[275]

Based on the above phase II trial data, the NCCN Panel has included pomalidomide/cyclophosphamide/dexamethasone as other recommended treatment option for patients with relapsed/refractory MM who have received two prior therapies, including an IMiD and a PI and disease progression on/within 60 days of completion of last therapy.

Regimens Useful In Certain Circumstances for Previously Treated MM – Relapsed/Refractory Disease (one to three prior therapies)

Bortezomib/Dexamethasone

The addition of dexamethasone to bortezomib in patients with relapsed/refractory MM who had PD during bortezomib monotherapy resulted in improvement of response in 18% to 34% of patients.[276-278] The NCCN Multiple Myeloma Panel members have included the bortezomib and dexamethasone regimen as an option that is useful in certain circumstances for patients with relapsed/refractory MM (category 1).

Bortezomib/Liposomal Doxorubicin/Dexamethasone

Bortezomib with liposomal doxorubicin (PLD) was approved by the FDA as a treatment option for patients with MM who have not previously received bortezomib and have received at least one prior therapy. The approval was based on a priority review of data from an international phase III trial (n = 646) showing that use of the combination significantly extended the median time to disease progression compared with bortezomib alone (9.3 vs. 6.5 months).[279] Median duration of response was increased from 7.0 to 10.2 months with the combination therapy. Based on these results, the NCCN Multiple Myeloma Panel considers bortezomib with the PLD regimen as a category 1 option that is useful in certain circumstances for patients with relapsed/refractory MM.

Lenalidomide/Dexamethasone

Lenalidomide combined with dexamethasone received approval from the FDA as a treatment option for patients with MM who had received at least one prior treatment. This was based on the results of two studies of a total of 692 patients randomized to receive dexamethasone either with or without lenalidomide. The primary efficacy endpoint in both studies was TTP. A pre-planned interim analysis of both studies reported that the median TTP was significantly longer in the lenalidomide arm compared to the control group.[280,281] The updated clinical data from the pivotal North American phase III trial (MM-009) in 353 previously treated patients with MM reported increased OS and median time to disease progression in patients receiving lenalidomide plus dexamethasone compared to patients receiving dexamethasone plus placebo.[281] Similar results were seen in the international trial MM-010.[280] Patients in both of these trials had been heavily treated before enrollment. Many had three or more prior lines of therapies with other agents and more than 50% of patients had undergone HCT.[280,281] Most adverse events and grade 3/4 adverse events were more frequent in patients with MM who received the combination of lenalidomide/dexamethasone compared to placebo and dexamethasone. Thrombocytopenia (61.5%) and neutropenia (58.8%) were the most frequently reported adverse events observed. The NCCN Multiple Myeloma Panel now considers this regimen as a category 1 option that is useful in certain circumstances for patients with relapsed/refractory MM.

Carfilzomib/Cyclophosphamide/Thalidomide/Dexamethasone

The results of the phase I/II trial showed that this 4-drug regimen is efficacious with an ORR of 91%, with 59% achieving VGPR or greater after 4 cycles in patients with MM.[282] The PFS and OS at 24 months (median, 17.5 months) was 76% and 96%, respectively.[282] This regimen has now been included under the list of regimens "useful in certain circumstances" for relapsed/refractory MM.

Carfilzomib/Dexamethasone

The results of the phase III ENDEAVOR trial in patients with relapsed/refractory MM treated with multiple prior lines of therapy showed a two-fold improvement in median PFS with carfilzomib/dexamethasone compared to bortezomib/dexamethasone (18.7 vs. 9.4 months; HR, 0.53; P < .0001).[283] ORR was 77% in the carfilzomib group versus 63% in the bortezomib group; rates of CR or better were 13% and 6% and rates of VGPR were 42% and 22%, respectively. The median duration of response was 21.3 months in the carfilzomib group and 10.4 months in the bortezomib group. Adverse events (grade 3 or higher) in the carfilzomib arm compared to the bortezomib arm included hypertension (6% vs. 3%), anemia (12% vs. 9%), thrombocytopenia (10% vs. 14%), and dyspnea (5% vs. 2%). Rate of grade ≥2 peripheral neuropathy was 6% in the carfilzomib group and 32% in the bortezomib group.[283]

The OS analysis showed that those treated with carfilzomib/ dexamethasone lived 7.6 months longer (median OS was 47.6 months in the carfilzomib group vs. 40 months in the bortezomib group; HR, 0.791 [95% CI, 0.648–0.964]; P = .010).[284] The most frequent grade 3 or worse adverse events in the carfilzomib arm compared to the bortezomib arm included hypertension (15% vs. 3%), anemia (16 % vs. 10%), dyspnea (6% vs. 2%), decreased lymphocyte count (6% vs. 2%), diarrhea (4% vs. 9%), and peripheral neuropathy (1% vs. 6%).[284] Rates of thrombocytopenia, pneumonia, and fatigue were similar in both groups.[284]

In the phase III A.R.R.O.W. trial, patients (n = 578) with relapsed and refractory MM previously treated with two or three treatments, including PI and IMiD were randomly assigned (1:1) to receive carfilzomib once a week (70 mg/m^2) or twice a week (27 mg/m^2). All patients received dexamethasone. The media PFS was higher in the once weekly (11.2 months) compared with those who received twice weekly carfilzomib (7. 6 months; HR, 0.69; 95% CI, 0.54–0.83; P = .0029). The overall safety was comparable between the two groups.[285] The NCCN Panel has included this combination on the list of regimens "useful in certain circumstances" for relapsed/refractory MM.

Based on the above data, the NCCN Multiple Myeloma Panel has included the combination of carfilzomib and dexamethasone as a category 1, useful in certain circumstances for patients with relapsed/refractory MM.[285]

Selinexor/Carfilzomib/Dexamethasone

In a study of thirty-two patients who had received a median of four prior therapies were assigned to receive once weekly selinexor, carfilzomib, and dexamethasone. The ORR was 78% with a median PFS of 15 months. The most common grade 3 or higher treatment-related adverse events were thrombocytopenia (47%), nausea (6%), anemia (19%), and fatigue (9%).[286]

Another analysis of a subset of this patient population that had triple class refractory MM also showed an ORR of 66.7% with a median PFS of 13.8 months, and median OS of 33 months.[287]

This regimen has now been included under the list of regimens "useful in certain circumstances" for relapsed/refractory MM.

Selinexor/Daratumumab/Dexamethasone

A phase Ib/II trial assessed the safety and efficacy of adding daratumumab to selinexor dexamethasone. Patients (n = 34) enrolled in the trial had received three or more prior lines of therapy, including a PI and an IMiD. In daratumumab naïve patients, the ORR was 73%, with 11 VGPR, and 11 PR. The median PFS was 12.5 months. This regimen has been included under the list of regimens useful in certain circumstances for relapsed/refractory MM.

Venetoclax/Dexamethasone with or without Daratumumab or PI for t(11;14) Patients

A phase I study of patients (n = 66) with relapsed/refractory MM who received a median of 5 prior lines of therapy studied venetoclax monotherapy and reported an ORR in 21% of patients with the response rate being higher in patients (n = 30) with t(11;14) compared with those without t(11:14) (40% vs. 6%).[288] Similar higher response rates have been found in patients with t(11:14) in real-world experience as well.[289] An open label phase I/II study examined venetoclax/dexamethasone in heavily pretreated t(11;14) patients. In this phase II part of the study, patients had received a median of five prior lines of therapy. At a median follow-up of 9.2 months, the ORR was 48%, with a median TTP of 10.8 months.[290]

Several prospective trials have reported on the efficacy and tolerability of venetoclax/dexamethasone containing combination regimens in relapsed t(11;14) MM. A phase I study found that venetoclax/dexamethasone in combination with daratumumab with or without bortezomib produced high rates of durable responses in patients with relapsed or refractory MM with t(11;14) translocation.[291]

In patients with no prior treatment with carfilzomib, venetoclax/dexamethasone plus carfilzomib was found to be safe and efficacious especially in those with t(11;14) translocations.[292] This finding has been supported by case studies.[293]

The NCCN Panel had included venetoclax/dexamethasone with or without daratumumab or a PI as options for patients with t(11:14) translocation.

Pomalidomide/Dexamethasone

Pomalidomide, like lenalidomide, is an analogue of thalidomide. It possesses potent immunomodulatory and significant anti-myeloma properties.[294]

A phase III, multicenter, randomized, open-label study (MM-003) conducted in Europe compared the efficacy and safety of pomalidomide and low-dose dexamethasone (n = 302) versus high-dose dexamethasone (n = 153) in patients with relapsed MM who were refractory to both lenalidomide and bortezomib.[295] After a median follow-up of 10 months, PFS, the primary endpoint of the study, was significantly longer in patients who received pomalidomide and low-dose dexamethasone compared with those who received high-dose dexamethasone (4 vs. 1.9 months; HR, 0.45; $P < .0001$).[295] The median OS was significantly longer in the patients who received pomalidomide and low-dose dexamethasone as well (12.7 vs. 8.1 months; HR, 0.74; $P = .0285$).[295] The most common hematologic grade 3 and 4 adverse effects found to be higher with the low-dose dexamethasone compared with the high-dose dexamethasone were neutropenia and pneumonia.[295] Other phase III studies of pomalidomide plus low-dose dexamethasone in combination with other agents (eg, bortezomib) are currently ongoing (Clinical Trial ID: NCT01734928). A European multicenter, single-arm, open-label, phase IIIb trial evaluated the safety and efficacy of pomalidomide and low-dose dexamethasone in a large patient population (N = 604).[296] The median PFS reported was 4.2 months and OS was 11.9 months. Whether the patients received prior lenalidomide or bortezomib, the PFS, OS, and ORR reported were similar.[296] The results of this trial are consistent with those observed in the pivotal MM-003 trial.[295]

In addition, several complementary phase II studies have been published evaluating the use of pomalidomide and dexamethasone in patients with MM relapsed/refractory to lenalidomide and/or bortezomib. A phase II study investigated two different dose regimens of pomalidomide and dexamethasone in 84 patients with advanced MM. Pomalidomide (4 mg) was given orally on days 1 to 21 or continuously over a 28-day cycle, and dexamethasone (40 mg) was given orally once weekly.[297] ORR was 35% and 34% for patients in the 21-day and 28-day groups, respectively. With a median follow-up of 23 months, median duration of response, PFS, and OS were 7.3, 4.6, and 14.9 months across both groups, respectively. All patients experienced similar adverse events in both groups. The adverse events were primarily due to myelosuppression.[297] Another phase II trial evaluated two doses of pomalidomide 2 or 4 mg/day with dexamethasone 40 mg weekly in heavily pre-treated patients (n = 35).[298] The ORR in the 2-mg cohort was 49% versus 43% in the 4-mg cohort. OS at 6 months was 78% and 67% in the 2- and 4-mg cohort, respectively. Myelosuppression was the most common toxicity.[298]

The FDA has approved pomalidomide for patients with MM who have received at least two prior therapies including lenalidomide and bortezomib and have demonstrated disease progression on or within 60 days of completion of the last therapy. The FDA-recommended dose and schedule of pomalidomide is 4 mg orally on days 1 to 21 of repeated 28-day cycles with cycles repeated until disease progression along with the recommendation to monitor patients for hematologic toxicities, especially neutropenia.

Based on the above data, the NCCN Panel has included pomalidomide plus dexamethasone as a therapeutic option in patients who have received at least two prior therapies, including an IMiD and a PI, and have demonstrated disease progression on or within 60 days of completion of the last therapy (category 1).

Selinexor/Pomalidomide/Dexamethasone

An abstract presented at the 2021 ASCO Annual Meeting presented data from an ongoing phase I/II clinical trial that contains one arm evaluating the regimen of selinexor/pomalidomide/dexamethasone (SPd) (NCT02343042). Sixty-five patients were enrolled initially in phase I with a median of three prior lines of therapy. After determining a recommended phase II dose, it was administered to twenty patients. Among these patients, the ORR was 65% and the median PFS was not reached in a median follow up of 3.9 months.[299]

Subsequently an analysis was performed to identify the optimal dose of SPd by comparing the safety and efficacy of SPd - 60 mg of selinexor (from STOMP Phase 1/2) to that of SPd-40 mg dose of selinexor (from STOMP Phase 2 and XPORT-MM-028). Weekly SPd in patients with RRMM was found to be tolerable and the lower weekly selinexor dosing (SPd-40) showed an improved adverse effects profile compared to SPd-60.[300]

Based on the above data, the NCCN Panel has included SPd as a therapeutic option in patients who have received at least two prior therapies, including an IMiD and bortezomib, and have demonstrated disease progression on or within 60 days of completion of the last therapy (category 1).

Daratumumab

Daratumumab is a human IgG kappa monoclonal antibody that targets the CD38 surface protein on myeloma cells.[250] In a phase I/II study, patients who had received more than three lines of therapy including an IMiD and a PI or were double refractory to PI and IMiD were randomized to two different doses of daratumumab (8 vs. 16 mg/kg). Findings from 106 patients who received 16 mg/kg noted an ORR of 29.2% in 31 patients (3 sCR, 10 VGPR, and 18 PR). The median duration of response was 7.4 months and median TTP was 3.7 months. The estimated 1-year OS rate was 65%.[251] Adverse events reported were fatigue (39.6%), anemia (33.0%), nausea (29.2%), and thrombocytopenia (25.5%). Grade 1 and 2 infusion-related reactions were seen in 42.5% of patients during first infusion. No patients discontinued the study due to infusion-related reactions.[251]

Based on the above phase II results and FDA approval, the Panel has added daratumumab as an option for the treatment of patients with MM who have received at least three prior lines of therapy including a PI and an IMiD or who are double refractory to a PI and IMiD.

DCEP and VTD-PACE for Aggressive MM
Patients with an aggressive relapse may need multi-drug combinations such as DCEP,[301-303] TD-PACE (thalidomide, dexamethasone, cisplatin, doxorubicin, high-dose cyclophosphamide, and etoposide),[304,305] and VTD-PACE (bortezomib, thalidomide, dexamethasone, cisplatin, doxorubicin, cyclophosphamide, and etoposide)[306-308] for effective disease control.

Preferred Regimens for Relapse After Three Prior Therapies
Currently there are three bispecific antibodies (elranatamab-bcmm, talquetamab-tgvs, and teclistamab-cqyv) and two CAR T-cell therapies (idecabtagene vicleucel and ciltacabtagene autoleucel) approved by the FDA and included as preferred options by the NCCN Panel for relapsed/refractory MM after at least three prior therapies including an anti-CD38 monoclonal antibody, a PI, and an IMiD.

CAR T-Cell Therapies
Idecabtagene vicleucel: Idecabtagene vicleucel is a BCMA-directed CAR-T cell therapy. In a phase II study (n = 128), patients with relapsed and refractory MM who had received at least three prior regimens (including a PI, an IMiD, and an anti-CD38 antibody) received idecabtagene vicleucel. Patients had received a median of six previous regimens for MM and 94% had received HCT. In this population of heavily pretreated patients, after a median 13-month follow-up, 73% of patients demonstrated response, with 33% having a CR or better. The median time to response was 1 month and median time to a CR or better was 2.8 months. High response rates (>50%) were found in several examined subgroups including older patients, patients with high-risk cytogenetic abnormalities, penta-refractory disease, and high tumor burden. Adverse events (grade 3 or 4) were common and included neutropenia (91%), anemia (70%), and thrombocytopenia (63%). Twenty-eight patients were retreated with idecabtagene vicleucel following disease progression and 21% demonstrated a second response. Grade 3 or 4 adverse events were common and were reported in 99% of patients. The most common adverse events were related to hematologic toxicity such as neutropenia (89%), anemia (60%), and thrombocytopenia (52%). Infections (69%) and cytokine release syndrome (84%) were also common treatment related adverse events, although the incidence of grade 3 or higher cytokine release syndrome was lower (5%).[309] The NCCN Panel has included idecabtagene vicleucel as an option for patients who have received at least four prior therapies, including an anti-CD38 monoclonal antibody, a PI, and an IMiD.

Ciltacabtagene autoleucel: Ciltacabtagene autoleucel is another BCMA directed CAR T-cell therapy. The CARTITUDE-1 trial (n = 97) was an open label phase Ib/II study that looked to assess the safety and efficacy of ciltacabtagene autoleucel in patients with relapsed or refractory MM who had received three or more previous lines of therapy (including an IMiD, PI, and andti-CD38 antibody).[310] The median amount of prior therapies was six. After a median 12.4 months of follow-up, the ORR was 97%, with 67% of patients achieving sCR. The PFS rate was 77% with an 89% OS rate. Adverse events included neutropenia in 95% of patients and anemia in 68%. Other common adverse events included thrombocytopenia (60%), leukopenia (61%), and lymphopenia (50%). Cytokine release syndrome also occurred in 95% of patients. There were six deaths due to treatment related adverse events.[310] A follow up analysis at 18 months showed that responses were durable; 18-month PFS and OS rates were 66.0% and 80.9% respectively, with no new observed safety signals.[311]

A post hoc analysis of the CARTITUDE-1 study at a median follow-up of 61.3 months, 33% (n = 32) of patients remained progression free with no additional treatment for MM, of which 96.9% (31 of the 32 patients) achieved stringent CR.[312]

Bispecific Antibodies:

Elranatamab-bcmm: Elranatamab is a bispecific T-cell engager (BiTE) that binds CD3 on T cells and to B-cell maturation antigen (BCMA) on myeloma cells. In the phase II, MagnetisMM-3 trial, patients with relapsed/refractory MM received subcutaneous elranatamab once weekly.[313] The primary population in whom the efficacy was seen were patients (n = 123) without prior BCMA-directed therapy. The findings indicated an ORR of 61.0% (75/123) and 35.0% greater than or equal to CR. Fifty responders switched to biweekly dosing, and 40 of those (80.0%) improved or maintained their response for ≥6 months. Common adverse events reported were infections, CRS, anemia, and neutropenia. With biweekly dosing, grade 3–4 adverse events decreased from 58.6% to 46.6%.[313] Elranatamab is included as an option for patients with multiple myeloma who have relapsed after 4 prior lines of therapy.

Linvoseltamab-gcpt: Linvoseltamab is a bispecific antibody therapy which targets BCMA and CD3. In the phase II LINKER-MM1 study, 117 patients with relapsed/refractory myeloma, whose disease progressed after 3 lines of therapy or triplet therapy (all consisting of a PI, IMiD and a CD38-targeted antibody, received 200 mg linvoseltamab-gcpt for 14 weeks.[314] After a median follow-up time of 14.3 months, the ORR was 71%, with half of those achieving a CR or better.[314] The estimated median duration of response was 29.4 months. Grade 3 or 4 hematologic and infectious toxicities being the most common (in 74% of the patients).[314] Grade 1 or 2 CRS occurred in 46% of patients. ICANS (all grades) occurred in 7.7% (9 patients). All ICANS events were concurrent with CRS or infusion-related reactions. Linvoseltamab is included as an option for patients with multiple myeloma who have relapsed after 4 prior lines of therapy.

Talquetamab-tgvs: Talquetamab is a T cell redirecting bispecific antibody targeting both GPRC5D and CD3 on T cells. In the single-arm, open-label, multicenter trial, MMY1001 (MonumenTAL-1) patients (n = 187) who had previously received at least four prior systemic therapies received talquetamab-tgvs subcutaneously weekly or talquetamab-tgvs biweekly until disease progression or unacceptable toxicity.[315]The most common adverse reactions reported with weekly and biweekly dosing were CRS (in 77% and 80% of the patients, respectively), skin-related events (in 67% and 70%), and dysgeusia (in 63% and 57%).[315] Talquetamab is included as an option for patients with multiple myeloma who have relapsed after 4 prior lines of therapy

Teclistamab-cqyv: Teclistamab is a BiTE that binds to CD3 on T cells and BCMA on myeloma cells. A phase I/II study examined the T-cell-redirecting bispecific antibody teclistamab-cqyv in 165 patients who had triple class refractory disease, with a median of five prior lines of therapy.[316] After a median follow up of 14.1 months, the ORR was 63% and 39.4% of patients demonstrated a CR or better. The median PFS was 11.3 months, with a median response duration of 18.4 months. Common adverse events included cytokine release syndrome in 72.1% of patients (0.6% grade 3) and grade 3 or 4 hematologic toxicity including neutropenia (64.2%), anemia (37%), and thrombocytopenia (21.2%). Infections were also common, with grade 3 or 4 infections occurring in 44.8% of patients. Teclistamab is included as an option for patients with multiple myeloma who have relapsed after 4 prior lines of therapy.

Other Recommended Regimens for Relapse After Three Prior Therapies

High-Dose or Fractionated Cyclophosphamide

Studies have reported that high-dose cyclophosphamide or hyperfractionated cyclophosphamide is efficacious particularly in patients needing immediate disease control who have received multiple prior treatments.[317,318] Therefore the NCCN Panel has included high dose or fractionated cyclophosphamide as an option for relapsed/refractory MM.

Selinexor/Dexamethasone:
Selinexor in combination with dexamethasone was studied in a phase IIb trial (STORM) in patients with relapsed/refractory MM.[319] The patients in the trial had multiple prior therapies and were refractory to IMIDs (lenalidomide and pomalidomide), PIs (bortezomib and carfilzomib), and the CD38 antibody daratumumab. A total of 122 patients were included in the intent-to-treat population. PR or better was observed in 26% of patients (95% CI, 19–35 with sCR in 2%, VGPR in 5%, and PR in 20% of the patients.

The most common adverse events reported during treatment were thrombocytopenia in 73% of the patients, fatigue in 73%, nausea in 72%, and anemia in 67%.

Based on the above results, the NCCN Panel has included selinexor/dexamethasone under other recommended options for patients with relapsed/refractory MM who have received at least four prior therapies and whose disease is refractory to at least two PI, at least two immunomodulatory agents, and an anti-CD38 monoclonal antibody.[286]

Belantamab mafodotin/bortezomib/dexamethasone:
Belantamab mafodotin-blmf is a BCMA antibody, conjugated to a microtubule disrupting agent— monomethyl auristatin—via a stable, protease resistant linker. The DREAMM-7 trial randomized patients (N = 494) with relapsed/refractory MM to belantamab mafodotin plus bortezomib and dexamethasone (BVd, n = 243) or to daratumumab plus bortezomib and dexamethasone (DVd, n = 251).[320] The median patient age was 64.5 years and patients had to have received at least one line of therapy with disease progression during or after the most recent therapy. Patients were excluded from the trial if they had disease that was refractory to anti-CD38 therapy or had had exposure to anti-BCMA therapy. At a median follow-up of 28.2 months, median PFS in the BVd arm was 36.6 versus 13.4 months in the DVd arm (HR, 0.41; 95% CI, 0.31–0.53; $P < .00001$).[321] After a median follow-up of 39.4 months, the OS was prolonged with BVd (HR, 0.58; 95% CI, 0.43–0.79; $P = .0002$).[320,321] BVd also led to greater than double the MRD-negativity rates in patients with a CR or better (25% vs. 10%; $P < .0001$).[320] The BVd combination was associated with a higher incidence of grade 3 or higher adverse events compared to DVd. Of note, the ocular events were notably more common in the BVd group than in the DVd group (79% vs. 29%).[320]

Based on the DREAMM-7 trial results, the NCCN Panel has BVd as a category 1, other recommended option for patients whose disease is refractory to at least two prior therapies including an IMiD and a PI.

Regimens Useful in Certain Circumstances for Relapse After Three Prior Therapies

Bendamustine
In a trial by Knop and colleagues, 31 patients who had experienced relapse after autologous HCT were enrolled to receive increasing doses of bendamustine.[322] The ORR was 55%, with a median PFS of 26 weeks for all patients and 36 weeks for patients who received higher doses of bendamustine (90–100 mg/m^2). The toxicity was mild and hematologic. A retrospective analysis of 39 patients has reported that bendamustine is effective and tolerable in patients with advanced progressive MM, with an ORR of 36%.[323]

The ECOG studied treatment with high-dose cyclophosphamide in patients with poor-risk features who had disease that was refractory to prior chemotherapy.[324] The ORR reported was 43% (29% response rate in patients refractory to prior therapy with cyclophosphamide).[324] Bendamustine is currently a treatment option for relapsed/refractory MM.

Bendamustine/Bortezomib/Dexamethasone

A phase II study evaluated bendamustine/bortezomib/dexamethasone administered over six 28-day cycles and then every 56 days for 6 more cycles in patients (n = 75; median age 68 years) with relapsed/refractory MM treated with multiple prior therapies and *not* refractory to bortezomib. The PR rate was 71.5% (16% CR, 18.5% VGPR, 37% partial remission). At 12-month follow-up, median TTP was 16.5 months, and 1-year OS was 78%.[325]

Bendamustine/carfilzomib/dexamethasone

A multicenter trial evaluated combination therapy with bendamustine/carfilzomib/and dexamethasone in 63 patients with relapsed/refractory MM (with at least two lines of prior therapy). Fifty-two percent of patients achieved a PR or better and 32% achieved a VGPR or better. After a median follow-up of 22 months, the median PFS was 11.6 months with a median OS of 30.4 months. The most common adverse events of grade 3 or higher included lymphopenia (29%), neutropenia (25%), and thrombocytopenia (22%).[326] The NCCN Panel has included carfilzomib in combination with bendamustine and dexamethasone as a treatment option for relapsed/refractory MM.

Bendamustine/Lenalidomide/Dexamethasone

A multicenter phase I/II trial investigated the combination of bendamustine, lenalidomide, and dexamethasone as treatment for patients (n = 29) with relapsed/refractory MM.[327] PR was seen in 52% (n = 13) of patients, with VGPR in 24% (n = 6) of patients. The median PFS in the trial was 6.1 months (95% CI, 3.7–9.4 months), and the 1-year PFS rate was 20% (95% CI, 6%–41%).[327] The NCCN Panel has included lenalidomide in combination with bendamustine and dexamethasone as a treatment option for relapsed/refractory MM.

Belantamab Mafodotin-blmf

In the open-label phase II trial (DREAMM-2), belantamab mafodotin was evaluated in patients whose MM was refractory to multiple agents. Responses were seen in approximately one-third of patients.[328] In the final analysis of the DREAMM-2 study, after a median follow-up of 12.5 and 13.8 months, the ORR was 32% and 35%, median PFS of 2.8 and 3.9 months, and median OS of 15.3 and 14.0 months in the 2.5 mg/kg and 3.4 mg/kg cohorts, respectively. Median duration of response was 12.5 and 6.2 months. The common Grade 3 and 4 adverse events were keratopathy (29% vs. 25%), thrombocytopenia (22% vs. 29%), and anemia (21% vs. 28%).[329]

Based on this study, single agent belantamab mafodotin-blmf is included this as an option useful in certain circumstance for those after 4 prior therapies (including a PI, an IMiD, and an anti-CD38 monoclonal antibody).

Talquetamab + Teclistamab: Talquetamab and teclistamab were used in combination in patients with relapsed/refractory MM in the RedirecTT-1 phase Ib/IIa study.[330] Patients enrolled (n = 94) were treated with both agents in one of multiple dose regimens. Talquetamab doses ranged from 0.2 mg/kg to 0.8 mg/kg, and teclistamab doses ranged from 0.75 mg/kg to 3.0 mg/ kg and were given either weekly (lower doses) or biweekly (higher doses). A median follow-up period of 20.3 months revealed a response in 80% of patients receiving the recommended phase 2 regimen (0.8 mg/kg talquetamab, 3.0 mg/kg teclistamab), while a response was seen in 78% of patients across all dose levels.[330] Finally, patients were estimated to have an 86% probability of maintaining a response after 18 months.

The NCCN Panel has included talquetamab and teclistamab as a treatment option for relapsed/refractory MM noting that prophylactic tocilizumab may be considered prior to first dose to reduce the risk of CRS.

Supportive Care for Multiple Myeloma

Important advances have been made in adjunctive treatment/supportive care of patients with MM. This involves careful patient education about the probable side effects of each drug, the drug combinations being used, and the supportive care measures required. Supportive care can be categorized into those measures required for all patients and those that address specific drugs.

Bony manifestations in the form of diffuse osteopenia and/or osteolytic lesions develop in 85% of patients with MM. Related complications are the major cause of limitations in quality of life and performance status in patients with MM. A large, double-blind, randomized trial has shown that monthly use of IV pamidronate (a bisphosphonate) can decrease pain and bone-related complications, improve performance status, and, importantly, preserve quality of life in patients with Durie-Salmon stage III MM and at least one lytic lesion.[331,332] Zoledronic acid has equivalent benefits.[333] Results from the study conducted by Zervas et al[334] show a 9.5-fold greater risk for the development of osteonecrosis of the jaw (ONJ) with zoledronic acid compared to pamidronate. Patients who are on bisphosphonates should have their renal function monitored. They should have a dental exam prior to the start of bisphosphonate therapy and should be monitored for ONJ.

The Medical Research Council (MRC) Myeloma IX study examined effects of zoledronic acid versus clodronate (a bisphosphonate not currently FDA approved) in patients with MM initiating chemotherapy regardless of bone disease. The patients were randomized to receive zoledronic acid (n = 981) or clodronic acid (n = 979). Zoledronic acid was reported to reduce mortality and significantly improve PFS.[335] Patients on clodronate and zoledronic acid had similar occurrence of acute renal failure and treatment-related serious adverse events. Zoledronic acid was associated with higher rates of ONJ than was clodronic acid.[336] An extended follow-up (median, 5.9 years) of the MRC Myeloma IX showed significant improvement in OS (52 vs. 46 months; HR, 0.86; *P* = .01) compared with clodronic acid.[337] The long-term rates of ONJ were also observed to be higher with zoledronic acid compared with clodronate (3.7% vs. 0.5%; *P* = .0001).[337]

A recent meta-analysis of 20 randomized controlled trials comparing bisphosphonates with either placebo or a different bisphosphonate as a comparator concluded that adding bisphosphonates to the treatment of MM reduces vertebral fractures and probably reduces pain.[338] It did not find a particular bisphosphonate to be superior to another.[338] In a multicenter trial (CALGB 70604), patients with MM or bone metastases from a solid malignancy were randomly assigned to zoledronic acid either monthly or every 3 months for 2 years.[339] The rates of skeletal-related events (SRE) were similar in both arms. Among the 278 patients with MM, rates of SRE were 26% in those receiving monthly versus 21% in those receiving treatment every 3 months.[339]

A large, placebo-controlled, randomized trial compared denosumab with zoledronic acid in patients (n = 1718) with NDMM with bone lesions. Time to first SRE and OS were similar in both arms. The denosumab arm had lower rates of renal toxicity and higher rates of hypocalcemia. ONJ was slightly higher in the denosumab arm (3% vs. 2%) but not statistically significant.[340]

The NCCN Guidelines for Multiple Myeloma recommend bisphosphonates (category 1) or denosumab for all patients receiving therapy for symptomatic MM regardless of documented bone disease. Denosumab is preferred by the NCCN Panel in patients with renal disease. The NCCN Panel recommends a baseline dental exam and monitoring for ONJ in all patients receiving a bone-modifying agent and monitoring for renal dysfunction with use of bisphosphonate therapy. With respect to duration of therapy, the Panel also recommends continuing bone-targeting treatment (bisphosphonates or denosumab) for up to 2 years and continuing beyond 2 years would be based on clinical judgement. The frequency of dosing (monthly vs. every 3 months) would depend on the individual patient criteria and response to therapy.

Excess bone resorption from bone disease can lead to excessive release of calcium into the blood, contributing to hypercalcemia. Symptoms include polyuria and gastrointestinal disturbances, with progressive dehydration and decreases in glomerular filtration rate. Hypercalcemia should be treated with hydration, bisphosphonates, denosumab,[340] steroids, and/or calcitonin. Among the bisphosphonates (zoledronic acid, pamidronate, and ibandronate), the NCCN Multiple Myeloma Panel members prefer zoledronic acid for treatment of hypercalcemia.[333,341,342]

The NCCN Panel has provided general principles of palliative RT for patients with MM. Careful planning of the radiation field and radiation technique is important to minimize toxicity to the spinal cord, brain, bone marrow, and adjacent OARs as patients with MM may be treated multiple times during their disease course. The Panel has noted that initiation of systemic therapy should not be delayed for RT and can often be given concurrently and that patients should be carefully monitored for toxicities.[343,344] In the setting of CAR-T cell therapy, bridging radiation treatment may be considered to treat symptomatic/at-risk sites of disease. Apply the above principles of radiation planning including careful planning of radiation fields, technique, and doses to limit toxicity. If clinically feasible, bridging radiation should be delivered after apheresis and completed prior to CAR-T cell infusion. Low-dose RT (8 Gy x 1 fraction) or 20–30 Gy in 5–10 total fractions can be used as palliative treatment for uncontrolled pain, for impending pathologic fracture, or for impending cord compression. Moderately fractionated courses of 20–25 Gy in 8–10 fractions are generally preferred over higher doses (30 Gy) absent extenuating circumstances (eg, severe symptomatic cord compression) to limit unnecessary toxicity and reduce risk of future treatment of adjacent or overlapping OARs (eg, spinal cord).[345] For RT dose constraint suggestions regarding bone marrow and other OARs, see NCCN Guidelines for Hodgkin Lymphoma (available at www.NCCN.org).

Plasmapheresis should be used as adjunctive therapy for symptomatic hyperviscosity.[346] Institutions differ in their use of plasmapheresis for adjunctive treatment of renal dysfunction.

Erythropoietin therapy may be considered for anemic patients, especially those with renal failure. Measuring endogenous erythropoietin levels may also be helpful in treatment planning[347,348] (see NCCN Guidelines for Prevention and Treatment of Cancer-Related Infections, available at www.NCCN.org). Daratumumab can interfere with cross-matching and RBC antibody screening. The NCCN Panel recommends performing type and screen prior to receiving daratumumab to inform future matching.

The highest risk for venous thromboembolism (VTE) is in the first 6 months following new diagnosis of MM. The NCCN Panel has outlined management of VTE, risk stratification, and VTE prophylaxis in a separate section in the NCCN Guidelines for Multiple Myeloma (MYEL-K pages 1-3) and VTE prophylaxis is administered to all patients, assuming there are no contraindications to anticoagulation agents or anti-platelets (see NCCN Guidelines for Cancer-Associated Venous Thromboembolic Disease, available at www.NCCN.org).To prevent infections in patients with MM, the Panel recommends referring to the NCCN Guidelines for Prevention and Treatment of Cancer-Related Infections (available at www.NCCN.org), the CDC recommendations for CDC for Use of COVID-19 Vaccines in the US, and in the most recent update, the Panel has outlined recommendations for prophylaxis and management of infections in patients undergoing treatment with CAR T-cell and bispecific antibody.

Management of Renal Disease in Multiple Myeloma

In patients with MM and monoclonal gammopathies, renal disease usually results from the production of monoclonal immunoglobulin or light/heavy chains by a clonal proliferation of plasma cells or B cells. Renal disease is seen in 20% to 50% of patients with MM and has been observed to negatively affect outcomes.[349-351] The NCCN Panel has added a new page outlining management of renal disease in MM.

Renal insufficiency defined as elevated serum creatinine greater than 2 mg/dL or established glomerular filtration rate (eGFR) less than 60 mL/min/1.73 m^2 in patients with MM is usually due to light chain cast nephropathy, but other etiologies need to be considered including hypercalcemia, volume depletion, and hyperuricemia as well as nephrotoxic medications or IV contrast. In addition, concomitant amyloidosis and monoclonal immunoglobulin deposition should be suspected when renal insufficiency or albuminuria is present without high levels of light chains.

Diagnostic Tests

According to the NCCN Panel, diagnostic workup of patients with symptomatic MM should include serum creatinine, electrolyte measurements, eGFR, electrophoresis of a sample from a 24-hour urine collection, serum electrophoresis, and serum FLC measurement. If proteinuria consists of light chains with high serum levels of FLC, and the cause of renal insufficiency can be attributed to MM, a renal biopsy may not be necessary. However, patients without a clear and complete explanation for their renal insufficiency should undergo renal biopsy to look for other pathophysiology such as monoclonal immunoglobulin deposition disease (MIDD) or membranoproliferative glomerulonephritis (MPGN).

Treatment Options

The initial treatment of cast nephropathy includes initiating appropriate MM therapy and providing adequate supportive care.

Myeloma Therapy: Bortezomib and/or a daratumumab-containing regimen can be administered in patients with severe renal impairment and also those on dialysis and does not require renal dose adjustment.[352-356] Once the renal function improves or stabilizes, one can switch to other regimens.

Agents used in myeloma therapy should be used with caution and with dose adjustments based on the degree of renal function impairment as recommended by the IMWG.[357] A retrospective study evaluated lenalidomide and dexamethasone based on two phase III trials of lenalidomide/low-dose dexamethasone in patients with relapsed/refractory MM with a serum creatinine of <2.5 mg/dL. Patients grouped by creatinine clearance >60 mL/min (n = 243), 30–60 mL/min (n = 82), and <30 mL/min (n = 16) showed no difference in response rates to lenalidomide/low-dose dexamethasone.[358] Patients with renal insufficiency had higher rates of thrombocytopenia and lenalidomide discontinuation than seen in patients without renal insufficiency. The NCCN Panel has outlined recommendations for lenalidomide dosing based on the degree of renal function in patients with MM and renal impairment. While prospective data to define optimal dosing are often lacking, pomalidomide has been studied in patients with relapsed MM in three different categories of renal insufficiency (eGFR 30–40 mL/min/1.73 sqm, eGFR <30 mL/min/1.73 sqm, and those requiring dialysis) and full-dose pomalidomide of 4 mg daily was found to be safe in all three groups.[359]

Supportive Care: IV fluids should be started promptly to decrease the renal tubular light chain concentration with a goal urine output of 100 to 150 cc per hour. Careful assessment of the fluid status is critical to avoid hypervolemia, especially in patients with oliguria renal failure.

In addition, nephrotoxic medications should be discontinued, and other metabolic abnormalities such as hypercalcemia and hyperuricemia should be corrected. Hydration, bisphosphonates or denosumab, and calcitonin are recommended to reduce calcium levels in the case of hypercalcemia. In patients with renal disease, pamidronate and zoledronic acid should be used with caution. The NCCN Panel has provided the recommended dosing of these agents in those who have renal impairment.

Dialysis may be required in selected patients in addition to prompt institution of anti-myeloma therapy. Mechanical removal of light chains may be considered on a case-by-case basis. While the benefit of mechanical removal of FLCs has not been established, there is limited evidence for the use of plasmapheresis or high-cutoff dialysis to reduce pathogenic light chains.

Management of CNS Disease in Patients with Multiple Myeloma

The content for this section is under development

Monoclonal Gammopathy of Clinical Significance (MGCS)

Monoclonal gammopathy of undetermined significance (MGUS) is defined by the absence of MM-defining events, presence of monoclonal gammopathy of <3 g/dL, and clonal population of BMPCs less than 10%. The prevalence of MGUS in the general population is about 0.7%, and it increases with age.

MGCS refers to the potentially organ-toxic properties of M protein. Typically, the M protein in MGCS does not meet the diagnostic criteria for MM and Waldenström macroglobulinemia (WM). Previously MGCS were all grouped under MGUS. Monoclonal gammopathy affects the renal function, and it is referred to as MGRS. Peripheral neuropathy mediated by an M-protein in the serum and urine without any evidence of MM or WM is now defined as monoclonal gammopathy of neurological significance (MGNS).

Monoclonal Gammopathy of Renal Significance (MGRS)

The term "MGRS" was proposed by the International Kidney and Monoclonal Gammopathy Research Group to collectively describe patients who meet the criteria for MGUS but demonstrate renal injury attributable to the underlying M protein.[360]

When the presence of monoclonal gammopathy affects the renal function, it is referred to as MGRS. Renal damage in the setting of symptomatic MM is not considered MGRS.

Initial Workup

In patients suspected of having MGRS, kidney biopsy is performed. A kidney biopsy is essential in demonstrating the nephrotoxicity of the M protein. The biopsy may be deferred if the eGFR is stable, the urinalysis is normal, or there is no evidence of proteinuria (it is not always light chain proteinuria).

The presence of monoclonal immunoglobulin deposits in the kidney indicates the existence of a plasma cell, B cell, or lymphoplasmacytic clone that is responsible for the production of the M protein.

M protein must be detected by electrophoresis and immunofixation in the urine and serum and must be correlated with the one found in biopsy. Immunofluorescence staining should be performed with the biopsy sample for IgG subclasses, IgA and IgM, and kappa and lambda.

Imaging by PET/CT, low-dose CT, or whole-body MRI should be performed as clinically indicated. Bone marrow biopsy is carried out if suspected to have MM or WM.

Additional workup for appropriate diagnosis of suspected WM, CLL/SLL, or systemic light chain amyloidosis maybe carried out as outlined in the respective NCCN Guidelines (see NCCN Guidelines for Waldenström Macroglobulinemia/Lymphoplasmacytic Lymphoma, NCCN Guidelines for Chronic Lymphocytic Leukemia/Small Lymphocytic Lymphoma, and NCCN Guidelines for Systemic Light Chain Amyloidosis).

Treatment

The treatment of MGRS is directed at the underlying plasma cell or B-cell clones to improve or prevent further kidney damage in these patients. For IgG, IgA, and FLC MGRS, use the management algorithms for MM; for IgM MGRS, see NCCN Guidelines for Waldenström Macroglobulinemia/Lymphoplasmacytic Lymphoma. For any MGRS with monoclonal B-cell lymphocytosis (MBL) features, see NCCN Guidelines for Chronic Lymphocytic Leukemia/Small Lymphocytic Lymphoma.

The response assessment in patients with MGRS who are being actively treated is as per the NCCN Guidelines listed above and includes SPEP and immunofixation; 24-hour urine collection for total protein, protein electrophoresis, and immunofixation; serum FLC assay; and serum creatinine.

Monoclonal Immunoglobulin Deposition Disease (MIDD)

The content for this section is under development

Monoclonal Gammopathy of Neurological Significance (MGNS)

MGNS is a subset of MGCS that is characterized by neurologic symptoms (such as peripheral neuropathy) and the presence of M protein without evidence of active MM or WM.

Patients with MGNS may have varying electrophysiologic features. A single-center retrospective analysis suggests that approximately 69% of patients with MGNS showed demyelinating features while 26% showed axonal features.[361] As a precursor state to numerous B-cell disorders, suspected MGNS does not always necessitate pharmacotherapy, but does warrant additional workup and evaluation to rule out other causes of neuropathy and inform clinical decision-making.

Initial Workup

MGNS may be the result of other comorbidities or conditions such as diabetes, cobalamin deficiency, thyroid dysfunction, Lyme disease, HIV infection, syphilis, autoimmune disease, or cryoglobulinemia. A comprehensive assessment of other causes of neuropathy is needed, including an evaluation for light chain amyloidosis, WM, or POEMS (Polyneuropathy, Organomegaly, Endocrinopathy, Monoclonal protein, Skin changes) syndrome. Additional components of an initial workup include anti-MAG antibodies, ganglioside antibody panel, nerve conduction study (NCS) or electromyogram (EMG), neurology consult, and chest/abdominal/pelvic CT with contrast (if possible). Sural nerve biopsy may be considered in certain circumstances. A nerve biopsy is a less desirable diagnostic method due to the association with pain and potential permanent sensory or motor deficits.[362]

MYD88 L265P allele-specific PCR testing of bone marrow is included in the initial workup of a patient with suspected MGNS. A study analyzed the correlation of *MYD88* L265P mutation in a large number of tumor samples across a variety of B-cell disorders and reported higher levels of IgM ($P <$ .0001) and higher frequency of proteinuria ($P = .002$) in patients with the mutation compared to patients with wild-type *MYD88*. The incidence of *MYD88* L265P mutation was reported in 100% of patients with WM, 47% of patients with MGUS, 6% with splenic marginal zone lymphoma, and 4% of patients with B-cell chronic lymphoproliferative disorders.[363] Subsequent studies have also reported the strong correlation of *MYD88* mutation status in MGNS and WM.[364-366] It should be noted that wild-type *MYD88* occurs in less than 10% of patients and should not be used to exclude diagnosis of WM if other criteria are met.

CXCR4 gene mutation testing may be considered in certain circumstances due to its prevalence in MGNS. Several studies reported *CXCR4* gene mutation as a common somatic mutation that has been linked with the *MYD88* L265P mutation.[365-369] Notably, however, a study reported that *CXCR4* mutation status had no impact on risk of death.[366] Therefore, *CXCR4* gene mutation testing may be included in the initial workup of a patient with suspected MGNS but should be interpreted in the context of other clinical findings to confirm diagnosis.

Clinical Findings

If there is either low suspicion or intermediate suspicion of MGNS that does not affect activities of daily living (ADLs), observation and follow-up as clinically indicated is appropriate. A low suspicion of MGNS may be determined if the following clinical findings are present: motor/pain predominant, non-length dependent, rapid progression (weeks to months), unilateral or asymmetrical, antibodies not present, and no demyelination confirmed by EMG/NCS.

If there is either intermediate suspicion that affects ADLs or high suspicion of MGNS, please refer to the NCCN Guidelines for Waldenström Macroglobulinemia/Lymphoplasmacytic Lymphoma. A high suspicion of MGNS may be determined if the following clinical findings are present: sensory predominant, length dependent, slow progression (years), bilateral and symmetrical, antibodies present, and demyelination confirmed by EMG/NCS. Presence of the *MYD88* L265P and/or *CXCR4* mutations should also be considered in the further diagnosis of WM.

POEMS Syndrome

POEMS syndrome is a rare plasma cell disorder that is characterized by the dominant presence of demyelinating peripheral neuropathy and confirmed clonal plasma cell proliferation. The etiology is not well understood, but it is believed to be correlated with inflammatory cytokines, such as vascular endothelial growth factor (VEGF). Despite the individual components of the acronym, not all of the features are necessary to make a diagnosis of POEMS syndrome. There are also other notable features that are not included in the acronym. Because of its rarity, POEMS syndrome may be underdiagnosed and mistaken for other chronic inflammatory syndromes.[370-372]

The diagnosis of POEMS syndrome is confirmed when both mandatory major criteria, one of the other three major criteria, and one of the six minor criteria are present. The mandatory major criteria for diagnosis of POEMS syndrome include the presence of polyneuropathy (typical demyelinating) and monoclonal plasma cell proliferative disorder [almost always λ (lambda)]. Additionally, at least one of the following major criteria is required: Castleman disease, sclerotic bone lesions, and/or VEGF elevation. Minor criteria for diagnosis that may be present include organomegaly (ie, splenomegaly, hepatomegaly, lymphadenopathy), extravascular volume overload (eg, edema, pleural effusion, ascites), endocrinopathy (ie, adrenal, thyroid, pituitary, gonadal, parathyroid, pancreatic), skin changes (ie, hyperpigmentation, hypertrichosis, glomeruloid hemangiomata, plethora, acrocyanosis, flushing, white nails), papilledema, and/or thrombocytosis/polycythemia. Other signs and symptoms may include clubbing, weight loss, hyperhidrosis, pulmonary hypertension, restrictive lung disease, thrombotic diatheses, diarrhea, and/or low vitamin B12 levels.[371]

Initial Workup

If POEMS syndrome is suspected, a complete history and physical examination is warranted. The clinical findings should be evaluated for any potential correlation to organomegaly (including splenomegaly, hepatomegaly, and lymphadenopathy). Presentation of symptoms such as hyperhidrosis, diarrhea, weight loss, and menstrual/sexual dysfunction should be accounted for in the initial workup. Fundoscopic, skin, and neurologic examinations may be performed to identify abnormalities. [371]

Recommended initial testing beyond the initial workup includes electrophysiologic (nerve conduction) studies, CT of the chest/abdomen/pelvis (for lymphadenopathy, organomegaly, ascites, pleural effusion, and/or edema), echocardiography (for right ventricular systolic and pulmonary artery pressures), and CT body bone windows and/or FDG-PET/CT (for sclerotic bone lesions). Laboratory testing should include a CBC, complete metabolic panel, fasting glucose, serum immunoglobulins (ie, IgG, IgA, IgM), electrophoresis and immunofixation, serum FLC, 24-hour urine total protein, VEGF, interleukin-6 (IL-6), and numerous hormones (ie, testosterone, estradiol, thyroid-stimulating hormone, parathyroid hormone, prolactin, serum cortisol, luteinizing hormone). A bone marrow aspirate and biopsy, FISH panel, and PCR should be done to assess for MM.[371]

Additional testing may be considered, if indicated. These additional tests include sural nerve biopsy, follicle-stimulating hormone, adrenocorticotropin hormone, cosyntropin stimulation test, biopsy of bone lesion (if needed), and excisional lymph node biopsy (if Castleman disease or other B-cell lymphomas suspected).[371]

Treatment

For a confirmed diagnosis of POEMS syndrome, treatment options may include RT, autologous HCT, and/or systemic therapy. RT alone is recommended for isolated bone lesions (defined as <3 sites) in patients without clonal BMPCs.

Autologous HCT is recommended in patients who are eligible as sole therapy or as consolidation after induction therapy. Patients who are not eligible for HCT should receive induction therapy as treatment. Induction therapy options regardless of HCT eligibility include lenalidomide/dexamethasone with or without daratumumab, bortezomib/dexamethasone, melphalan/dexamethasone, cyclophosphamide/dexamethasone, or pomalidomide/dexamethasone.

Patients who progress after initial therapy should receive individualized treatment based on response and toxicity of prior therapy, with consideration of the patient’s performance status at the time of progression.

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